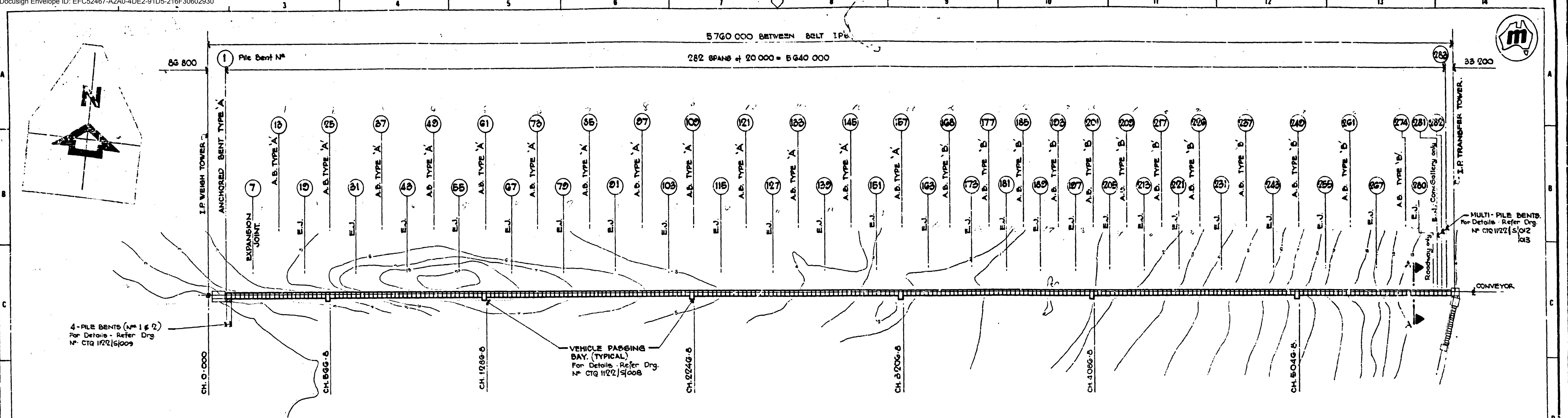


APPENDIX K JETTY DRAWINGS

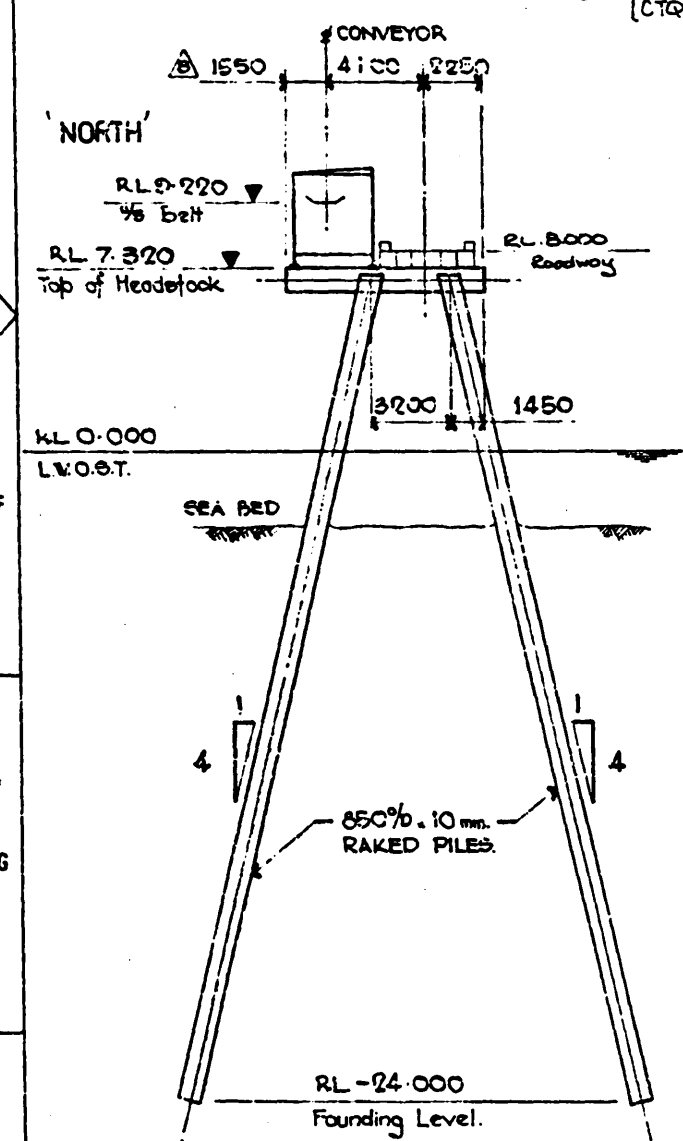
Drawing number	Description
CTQ-1122-S-001	Approach jetty - General plan and sections
CTQ-1122-S-002	Approach jetty - Pile schedule
CTQ-1122-S-003	Approach jetty - Sheet 1 superstructure layout
CTQ-1122-S-004	Approach jetty - Sheet 2 superstructure layout
CTQ-1122-S-005	Approach jetty - Typical cross section
CTQ-1122-S-006	Approach jetty - Type 'A' anchor bent
CTQ-1122-S-007	Approach jetty - Type 'B' anchor bent
CTQ-1122-S-008	Approach jetty - Passing bay details
CTQ-1122-S-010	Approach jetty - Pilebents no.s 275-279
CTQ-1122-S-012	Approach jetty - Pilebent no. 281
CTQ-1122-S-015	Approach jetty - Miscellaneous details
CTQ-1122-S-018	Approach jetty - Standard conveyor gallery
CTQ-1122-S-020	Approach jetty - Standard span cladding
CTQ-1122-S-021	Approach jetty - Gallery flashing details
CTQ-1122-S-022	Approach jetty - Conveyor stringer details
CTQ-1122-S-023	Approach jetty - Gallery tolerances
CTQ-1122-S-024	Approach jetty - Variations to standard gallery
CTQ-1122-S-025	Approach jetty - Variations to cladding
CTQ-1122-S-026	Approach jetty - Cross-under details
CTQ-1122-S-030	Approach jetty - Pre-cast roadway GA
CTQ-1122-S-031	Approach jetty - Standard roadway girder details
CTQ-1122-S-032	Approach jetty - Expansion joint details
CTQ-1122-S-033	Approach jetty - Pre-cast roadway girder arrangement and details
CTQ-1122-S-034	Approach jetty - Pre-cast roadway girder details sht 2
CTQ-1122-S-038	Approach jetty - As-constructed piles
CTQ-1122-S-039	Approach jetty - As-constructed piles



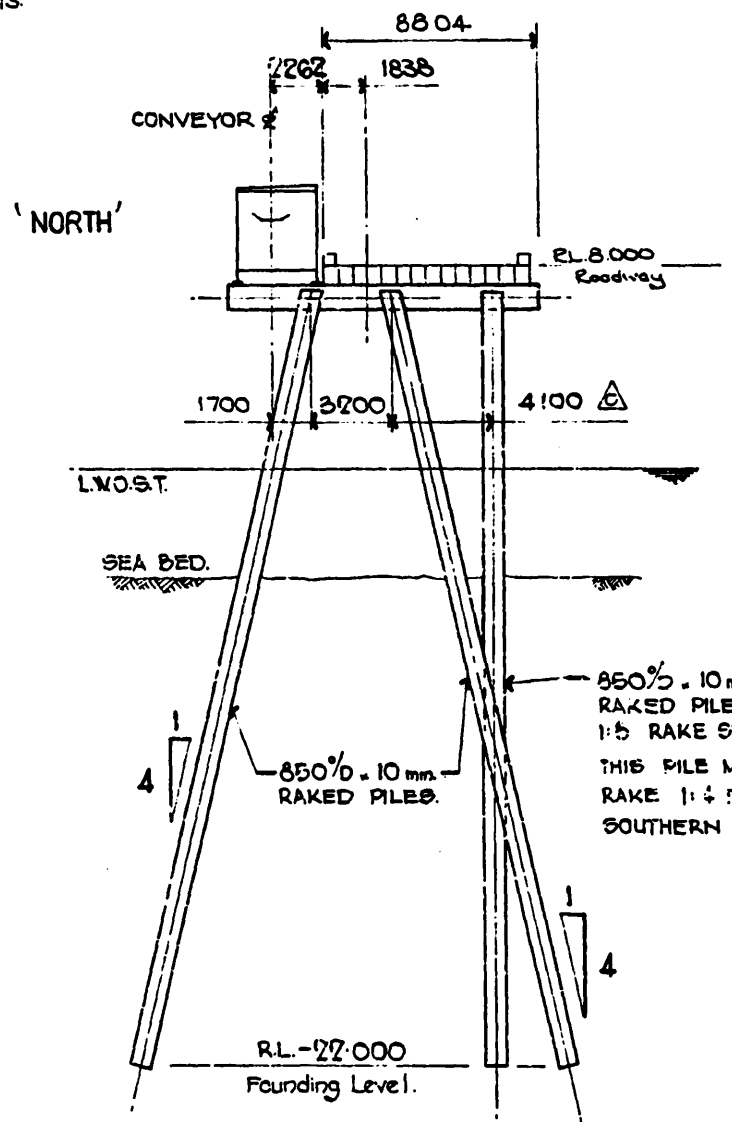
NOTE:
 A.B. = ANCHORED BENT.
 ANCHORED BENTS TYPE 'A' = 1 ANCHOR PILE - Refer Drg. No. CTQ 1122/S/006 for Details.
 ANCHORED BENTS TYPE 'B' = 2 ANCHOR PILES - Refer Drg. No. CTQ 1122/S/007 for Details.
 E.J. = EXPANSION JOINT. - Refer Drg. No. CTQ 1122/S/003 for Details.
 [CTQ 1122/S/032]

PLAN - APPROACH JETTY.

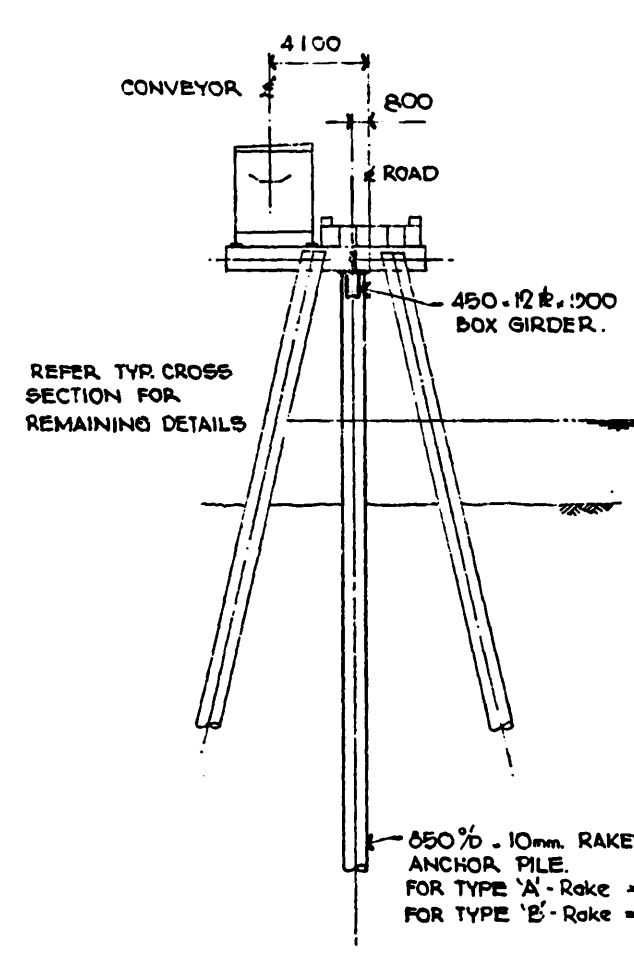
SCALE: 1:7,500.



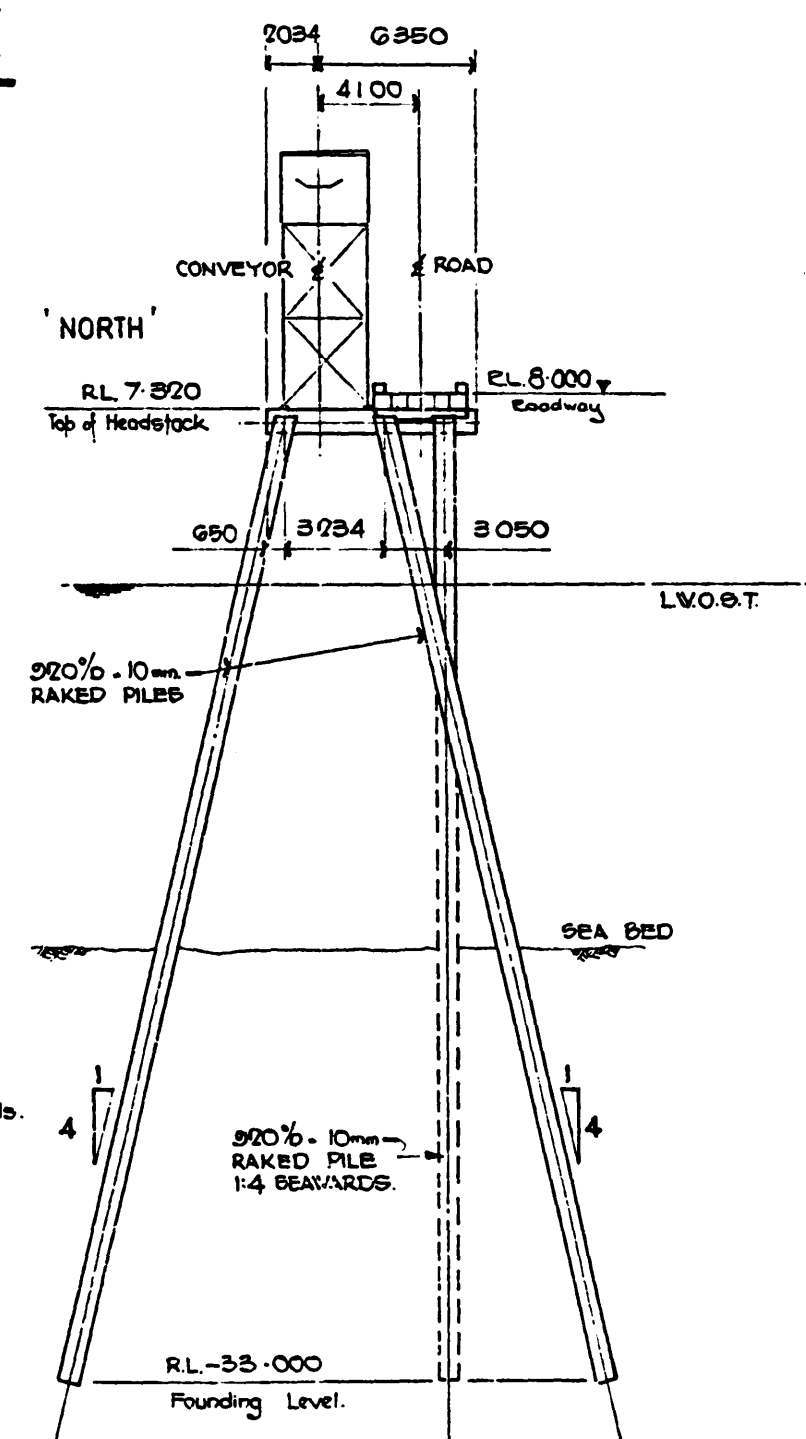
TYP. CROSS-SECTION.
 PILE BENT No. 10 SHOWN.



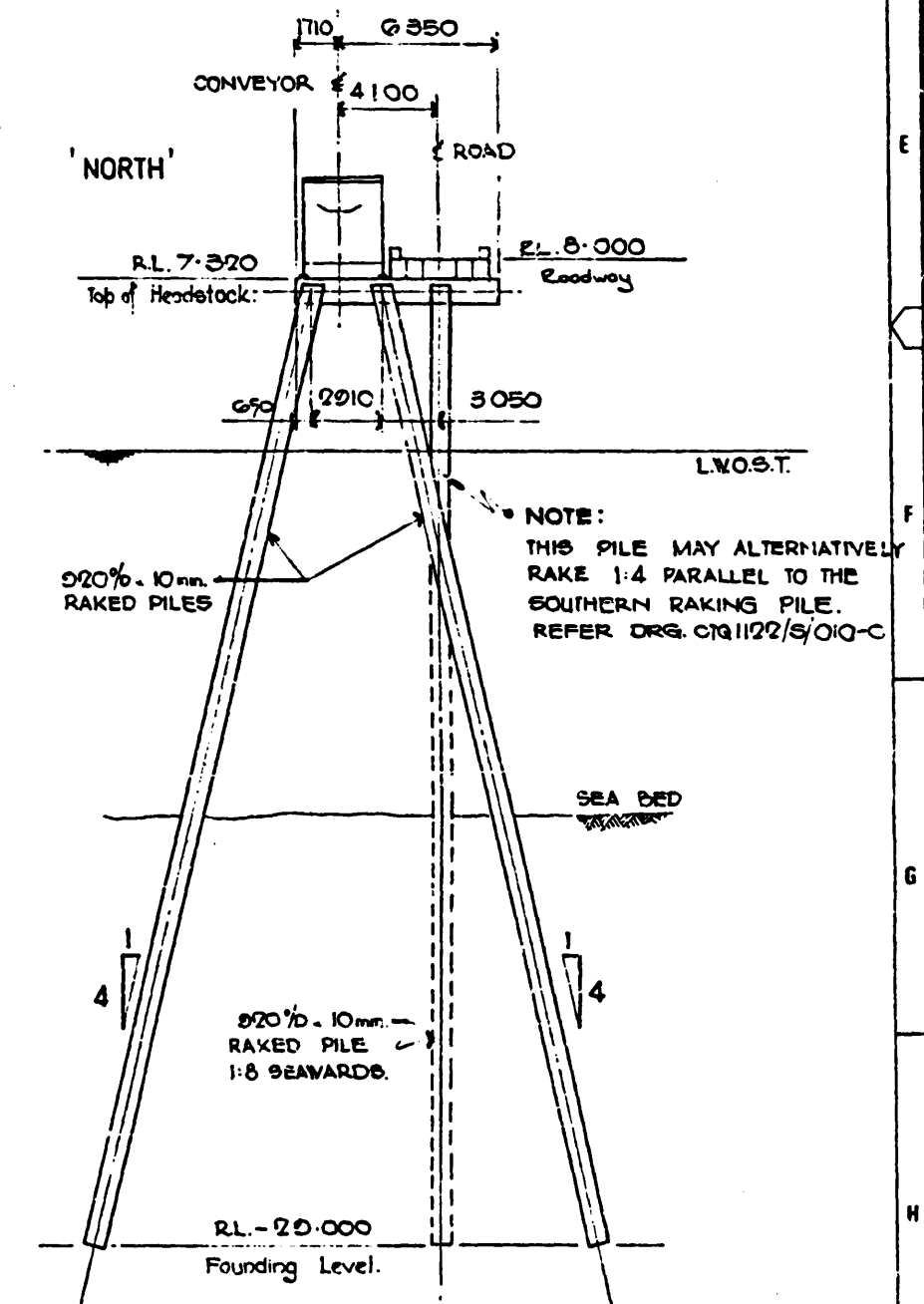
TYP. PASSING-BAY SECTION.
 BAY AT CHS. 500.0 SHOWN.



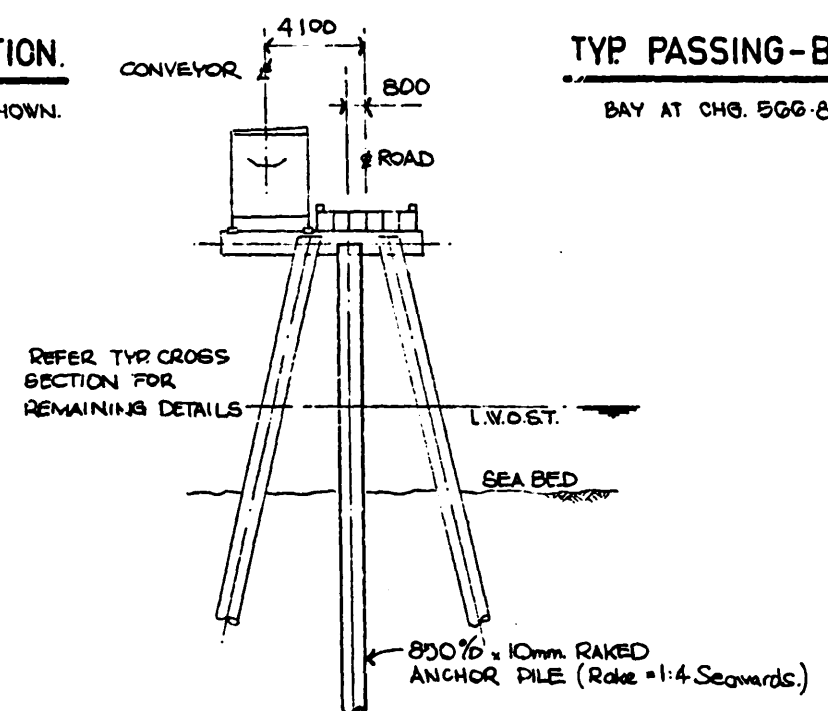
TYP. ANCHORED BENT SECTION.



PILE BENT No. 280.



SECTION A-A
 TYPICAL FOR PILE BENT No. 275, 276, 277, 278, 279.



TYP. EXPANSION JOINT BENT SECTION

NOTES:

- FOR PILE SIZES, FOUNDING LEVELS & SEA BED LEVELS - REFER DRG. No. CTQ 1122/S/002.
- ROADWAY LEVEL FROM PILE BENT No. 1 TO PILE BENT No. 281 TO BE RL 8.000. FROM PILE BENT No. 282 SEAWARDS, ROADWAY LEVEL TO BE RL 7.700.
- PILE BENT No. 284 (Not drawn for clarity) IS LOCATED 18 METRES SEAWARD (EAST) OF PILE BENT No. 283.

THE CORPORATION OF THE TREASURER OF QUEENSLAND			
NO.	DATE	AMENDMENTS	
A	JAN 77	CONV. TO ROADWAY AMENDED	
B	1-2-77	TYP. CROSS SECTION HEADSTOCK LENGTH AMENDED	
C	15-3-77	Passing Bay Pile spacing amended to suit Contractor's Construction Method.	
D	18-11-77	RS. 260 & 275-278 AMENDED TO SUIT CONTRACTOR'S CONSTRUCTION METHOD.	
CSR REFERENCE NO.			
THE SUGAR BOARD BULK SUGAR TERMINALS			
B.S.T. LUCINDA - STAGE III APPROACH JETTY. GENERAL PLAN & SECTIONS.			DRAWING NO. CTQ 1122/S/001
MACDONALD WAGNER & FRIDDLE PTY. LTD.			
CSR LIMITED			

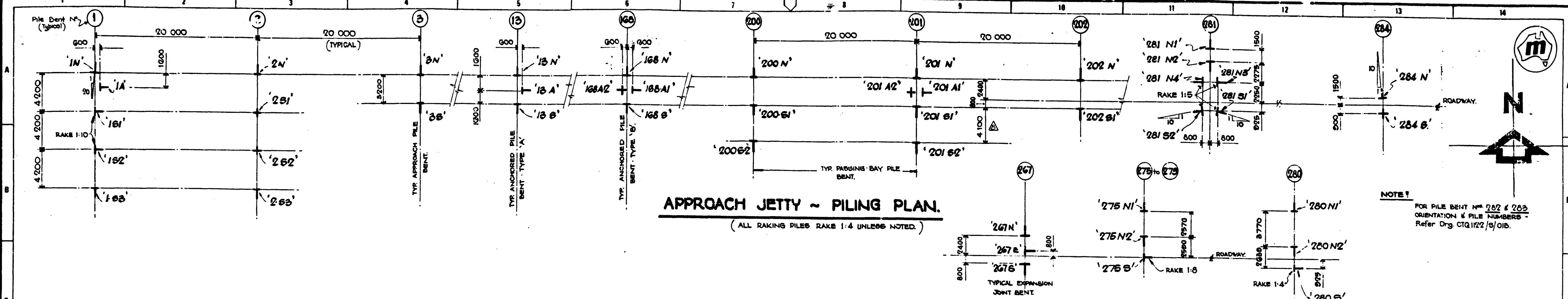


TABLE 1 - 850% D. 10mm WALL APPROACH PILES													
PILE BENT N°	1 & 2	1 & 2	3, 4, 5, 6 & 7	8 - 10	20 - 31	32 - 43	44 - 55	56 - 67	68 - 79	80 - 91	92 - 103	104 - 115	116 - 127
PILE N°	1N, 2N, 1S, 2S	1S1, 1S2, 2S1, 2S2	3N, 4N, 5N, 6N, 7N	8N, 9N, 10N, 11N, 12N, 13N	20N, 21N, 22N, 23N, 24N, 25N	32N, 33N, 34N, 35N, 36N, 37N	44N, 45N, 46N, 47N, 48N, 49N	56N, 57N, 58N, 59N, 60N, 61N	68N, 69N, 70N, 71N, 72N, 73N	80N, 81N, 82N, 83N, 84N, 85N	92N, 93N, 94N, 95N, 96N, 97N	104N, 105N, 106N, 107N, 108N, 109N	116N, 117N, 118N, 119N, 120N, 121N
CHAINAGE (Metres)	266.8 to 106.8	86.8 to 106.8	126.8 to 206.8	226.8 to 446.8	466.8 to 686.8	706.8 to 926.8	946.8 to 1166.8	1186.8 to 1406.8	1426.8 to 1646.8	1666.8 to 1886.8	1906.8 to 2126.8	2146.8 to 2366.8	2386.8 to 2606.8
PROVISIONAL TOE R.L.	-28.000	-28.000	-24.000	-24.000	-22.000	-24.500	-22.000	-22.000	-22.500	-24.000	-27.000	-24.500	-22.000
PROVISIONAL LENGTH	36.140	35.240	32.020	22.530	22.060	32.530	22.060	22.060	30.470	32.020	35.110	32.530	22.060
BED LEVEL (APPROX)	-2.000	-2.000	-2.750	-3.500	-5.500	-7.500	-8.500	-8.000	-7.750	-7.000	-6.000	-5.000	-4.000

TABLE 1 - 850% D. 10mm WALL APPROACH PILES													
PILE BENT N°	128 - 139	140 - 151	152 - 163	164 - 175	176 - 187	188 - 199	200 - 211	212 - 223	224 - 235	236 - 247	248 - 259	260 - 271	272 - 283
PILE N°	128N, 129N, 130N, 131N, 132N, 133N	140N, 141N, 142N, 143N, 144N, 145N	152N, 153N, 154N, 155N, 156N, 157N	164N, 165N, 166N, 167N, 168N, 169N	176N, 177N, 178N, 179N, 180N, 181N	188N, 189N, 190N, 191N, 192N, 193N	200N, 201N, 202N, 203N, 204N, 205N	212N, 213N, 214N, 215N, 216N, 217N	224N, 225N, 226N, 227N, 228N, 229N	236N, 237N, 238N, 239N, 240N, 241N	248N, 249N, 250N, 251N, 252N, 253N	260N, 261N, 262N, 263N, 264N, 265N	272N, 273N, 274N, 275N, 276N, 277N
CHAINAGE (Metres)	2626.8 to 2846.8	2866.8 to 3086.8	3106.8 to 3326.8	3346.8 to 3566.8	3586.8 to 3806.8	3826.8 to 4046.8	4066.8 to 4286.8	4306.8 to 4526.8	4546.8 to 4766.8	4786.8 to 5006.8	5026.8 to 5246.8	5266.8 to 5486.8	5506.8 to 5726.8
PROVISIONAL TOE R.L.	-25.000	-26.000	-24.000	-24.000	-23.000	-25.000	-25.000	-25.500	-25.000	-24.500	-24.500	-25.500	-25.000
PROVISIONAL LENGTH	33.050	34.080	32.020	32.640	31.000	33.050	33.050	33.570	37.180	32.530	37.690	33.570	32.530
BED LEVEL (APPROX)	-3.500	-2.500	-2.000	-1.500	-1.000	-0.500	-0.500	-1.500	-2.500	-3.000	-4.250	-5.500	-7.250

TABLE 2 - 920% D. 10mm WALL APPROACH PILES													
PILE BENT N°	253 - 255	256 - 267	268 - 274	275 - 280	275 - 279	281	281	282, 283	282, 283	281, 282, 283	280	284	284
PILE N°	253N, 254N, 255N	256N, 257N, 258N, 259N, 260N, 261N	268N, 269N, 270N, 271N, 272N, 273N, 274N	275N1, 276N1	275N, 276N, 277N	281N1, 281N2	281N3, 281N4	282N1, 282N2, 282N3, 282N4	282N5, 282N6	281S1, 282S1, 283S1	280S	284N	284S
CHAINAGE (Metres)	5126.8 to 5166.8	5186.8 to 5406.8	5426.8 to 5546.8	5566.8 to 5686.8	5666.8 to 5646.8	5686.8	5686.8	5706.8 to 5726.8	5706.8 to 5726.8	5686.8 to 5726.8	5666.8	5744.8	5744.8
PROVISIONAL TOE R.L.	-24.700	-26.100	-20.500	-31.000	-31.000	-33.000	-33.000	-33.000	-33.000	-33.000	-31.000	-33.000	-33.000
PROVISIONAL LENGTH	32.740	34.180	36.720	30.030	36.160	41.100	40.470	41.020	40.650	40.040	39.030	41.020	39.860
BED LEVEL (APPROX)	-2.500	-11.500	-12.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500
HEAD R.L.	7.058	7.058	7.058	6.858	6.858	6.258	6.258	6.858	6.858	6.258	6.858	6.858	6.858

TABLE 3 - ANCHOR PILES. 850% D. 10mm WALL (UNLESS NOTED)																									
PILE BENT N°	1	13	25	37	49	61	73	85	97	109	121	133	145	157	169	177	185	193	201	209	217	225	233	241	249
PILE N°	1A	13A	25A	37A	49A	61A	73A	85A	97A	109A	121A	133A	145A	157A	169A	177A	185A	193A	201A	209A	217A	225A	233A	241A	249A
CHAINAGE (Metres)	87.4	327.4	567.4	807.4	1047.4	1287.4	1527.4	1767.4	2007.4	2247.4	2487.4	2727.4	2967.4	3207.4	3447.4	3687.4	3927.4	4167.4	4407.4	4647.4	4887.4	5127.4	5367.4	5607.4	5847.4
PROVISIONAL TOE R.L.	-28.000	-24.500	-22.000	-24.500	-22.000	-22.000	-22.500	-24.000	-27.000	-24.500	-22.000	-25.000	-26.000	-24.000	-24.000	-24.000	-25.000	-25.000	-25.000	-25.000	-24.500	-24.500	-25.500	-24.500	-20.500
PROVISIONAL LENGTH	35.320	31.710	29.130	31.710	29.130	29.130	29.650	31.190	34.232	31.710	29.130	32.220	33.260	31.190	30.360	31.810	32.260	31.260	32.220	31.760	32.740	35.280	36.820	37.740	39.280
BED LEVEL (APPROX)	-2.000	-3.500	-5.500	-7.500	-8.500	-8.000	-7.750	-7.000	-6.000	-5.000	-4.000	-3.500	-2.500	-2.000	-1.500	-1.000	-0.500	-0.500	-1.500	-2.500	-3.000	-4.250	-5.500	-7.250	-11.500

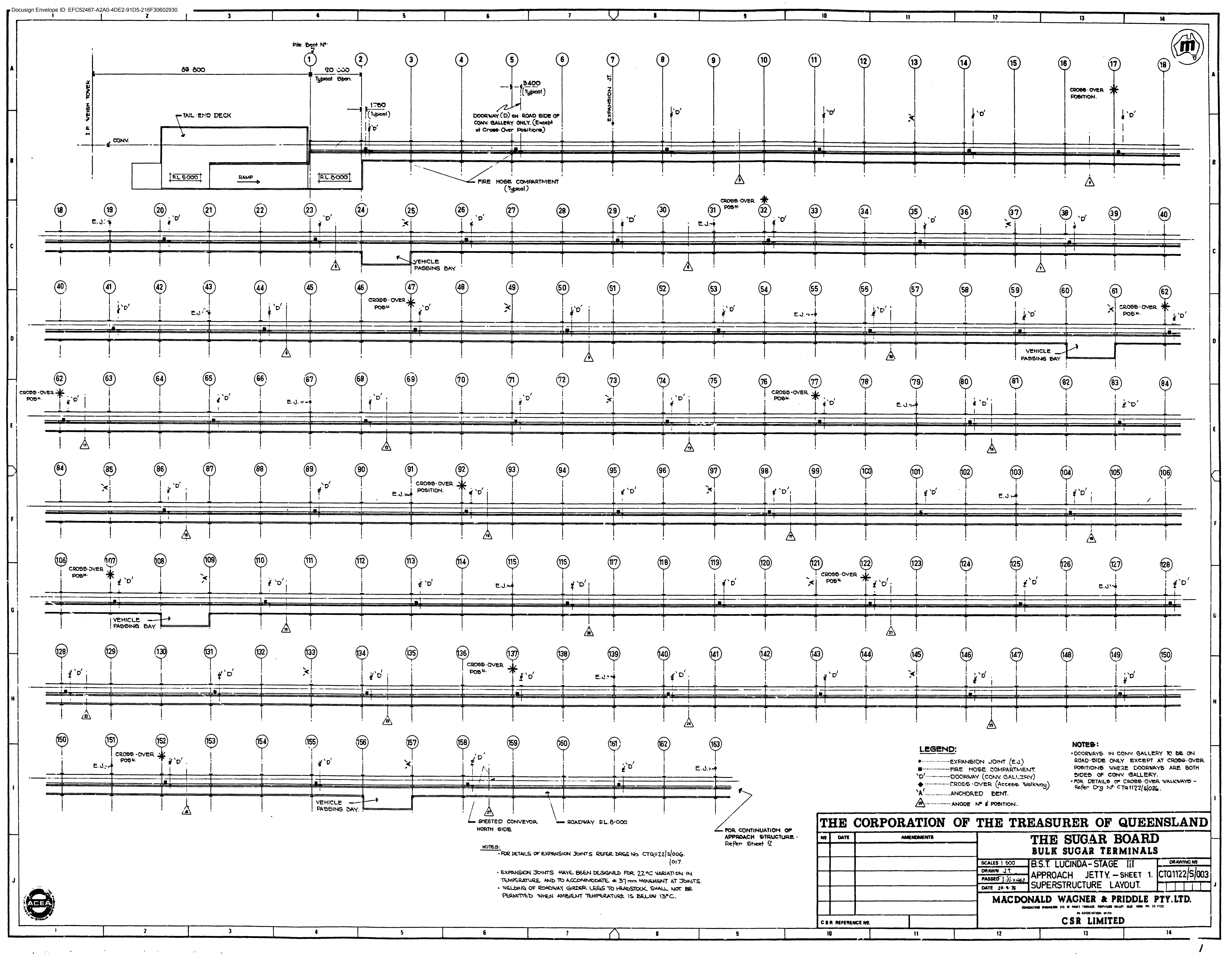
TABLE 4 - PASSING BAY PILES. 850% D. 10mm WALL.						
PILE BENT N°	24, 25	60, 61	108, 109	152, 157	200, 201	248, 249
PILE N°	24S2, 25S2	60S2, 61S2	108S2, 109S2	152S2, 157S2	200S2, 201S2	248S2, 249S2
CHAINAGE (Metres)	546.8 to 560.8	1267.8 to 1280.8	2226.8 to 2240.8	3186.8 to 3206.8	4066.8 to 4086.8	5026.8 to 5046.8
PROVISIONAL TOE R.L.	-22.000	-21.000	-24.500	-24.000	-25.500	-24.500
PROVISIONAL LENGTH	20.060	20.060	32.530	32.020	33.570	32.580
BED LEVEL (APPROX)	-5.500	-6.800	-5.000	-2.000	-1.500	-7.250

SUMMARY

GROUP	TOTAL LENGTH
TABLE 1	16 550 metres
TABLE 2	3 235 metres
TABLE 3	1 027 metres
TABLE 4	381 metres
TABLE 5	816 metres
TOTAL LENGTH 850% PILES	18 860 metres
TOTAL LENGTH 920% PILES	3 385 metres

NOTES:

1. Pile Head R.L. for all piles to be R.L. 7.058 unless noted.
2. Anchor Pile Head R.L. to be R.L. 6.258.
-





THE CORPORATION OF THE TREASURER OF QUEENSLAND					
NO.	DATE	AMENDMENT:			
A.	29-11-76	FEE HOSE COMPARTMENT ADDED AT BENT 105.			
C.S.R. REFERENCE NO.					

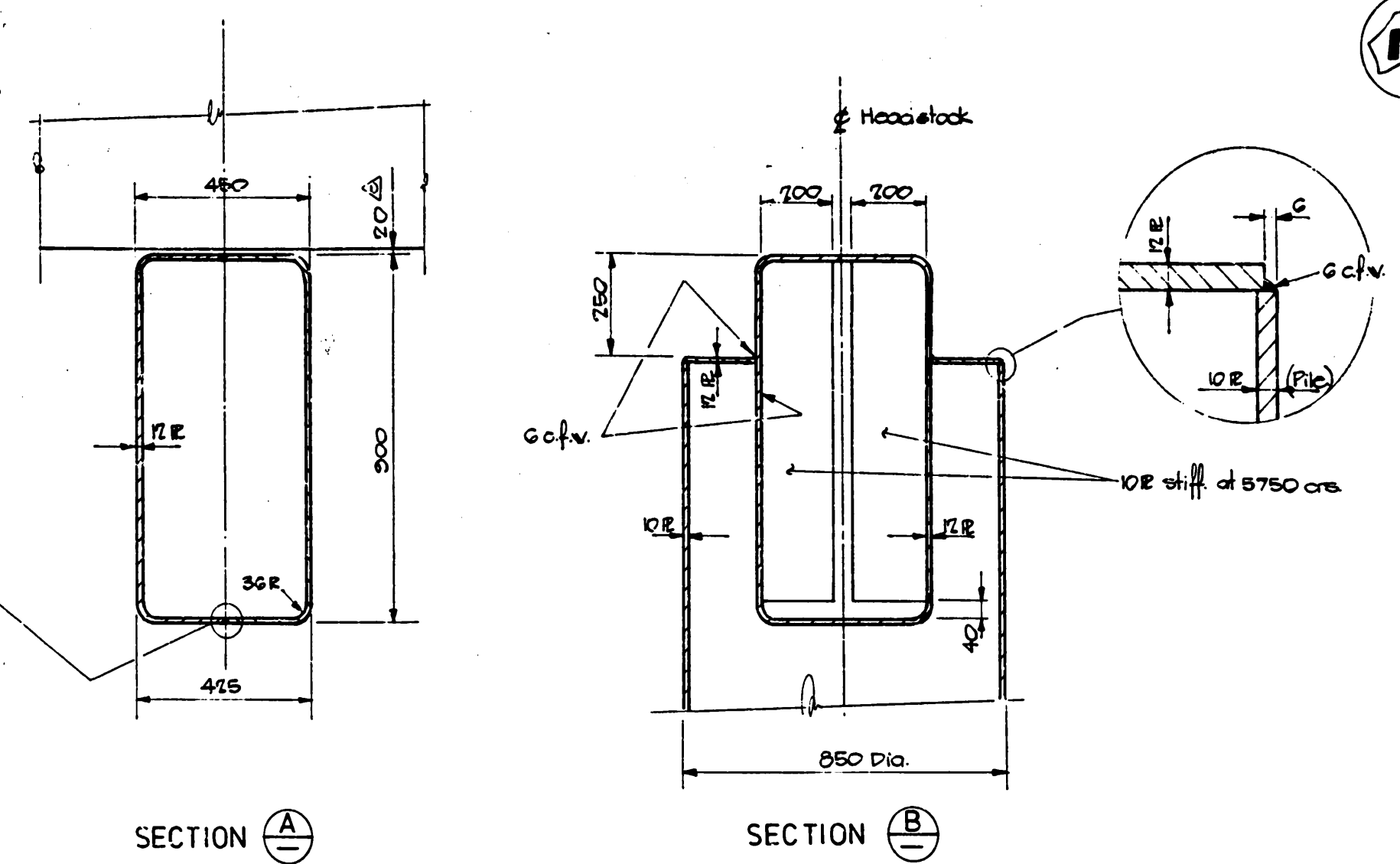
THE SUGAR BOARD BULK SUGAR TERMINALS	
SCALES 1:500 DRAWN J.T. PASSED <u>J.W. DRY</u> DATE 28-8-76	B.S.T. LUCINDA STAGE III APPROACH JETTY.—SHEET 2. SUPERSTRUCTURE LAYOUT.
DRAWING NO	CTQ1122/S1004
A	[] [] [] [] [] [] [] []

MACDONALD WAGNER & PRIDDLE PTY.LTD.

CONSULTING ENGINEERS 212 ST PAULS TERRACE PORTLAND WHARF OLD GOLD PH 32 7132

IN ASSOCIATION WITH

CSR LIMITED



Refer Proj. No. CP-11172/S-015
for details of allowable
tolerances for headstocks
& piles.

⌀ Pile Bent

SECTION (C)

7 min.

12 RE

7-5-8 RE

45 min.

10 RE

Incomplete penetration
single-bevel butt weld.

89 ϕ $\times$ 4.0 wall conveyor pipe stringer.

Conveyor idler support bracket (By others)

Suspended conveyor idlers (By others)

THE CORPORATION OF THE TREASURER OF QUEENSLAND		
NO	DATE	AMENDMENTS
A	10-12-76	Roadway construction altered.
B	4-1-77	Dimension changed.
C	18-1-77	Roadway & to Conveyor & altered. Neoprene bearing thk. amended. Dimensions added.
D	18-1-77	Dimensions changed.
E	1-2-77	Headstock length altered to 7900
C S R REFERENCE NO.		

THE SUGAR BOARD BULK SUGAR TERMINALS	
SCALES 1:20:110 DRAWN R. Holmes PASSED DATE DEC 76.	B.S.T. LUCINDA STAGE III APPROACH JETTY TYPICAL CROSS-SECTION.
	DRAWING NO CT01122/S/005 E

MACDONALD WAGNER & PRIDDLE PTY.LTD.
 CONSULTING ENGINEERS 212 ST MARKS TERRACE, NORTHQUEEN HILL QLD 4058 PH 31 7112
 IN ASSOCIATION WITH
CSR LIMITED

φ Pile Bent & Headstock.

φ Conveyor

φ Roadway

φ Headstock.

G cont. fillet weld.

12" Beel 12" (cut to suit)

RL. G-420 Δ

Incomplete penetration single bevel butt weld.

G cont. fillet weld.

SEAWARD. →

φ Raked Pile 850% Hovoll.

ELEVATION

4- 200. 12 E STIFFENERS
C.F.V. TO INSIDE H'STOCK.
Δ Refer Drg CTQ1122/S/005
for SHIT* details.

φ Headstock.

φ Pile Bent

φ Pile.

PLAN

ANCHOR PILE AT EXPANSION JOINT.

ELEVATION

SECTION A

315 φ's of 10R stiff in Headstock.

340 (Side R)

327 (Stiff)

315 (Cover)

10 Cover R

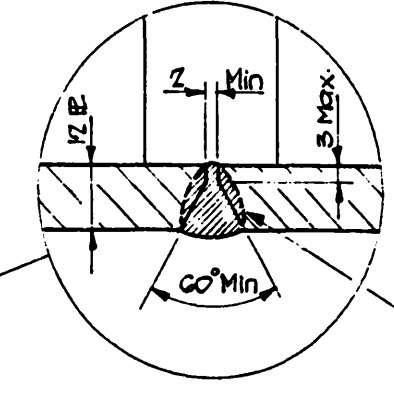
12 R stiff

12 R stiff

12 Side R

12 Base R

10 R stiffeners
G cont. fillet weld
all round both sides



Qualified incomplete penetration single V butt weld.

SECTION B

850 %

NOTE

For Type B' Anchored Pile Bent - Refer Drg No CTQ1122/S/007
For Typical Cross-Section of Approach Jetty - refer Drg No CTQ1122/S/005.

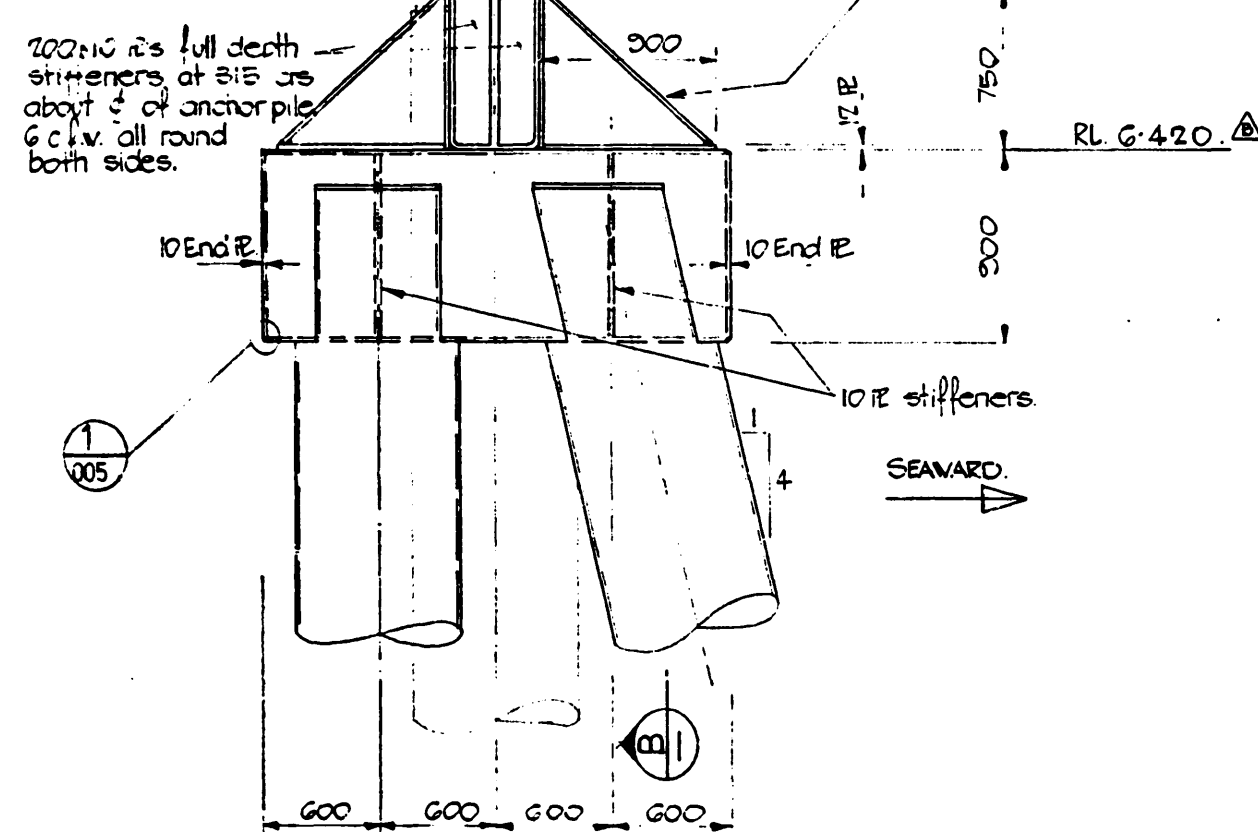
THE CORPORATION OF THE TREASURER OF QUEENSLAND				
NO	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS	
A	20-12-76	Roadway construction altered.	SCALES 1/25 1/10	B.S.T. LUCINDA STAGE III APPROACH JETTY TYPE 'A' TYPICAL ONE-PILE ANCHOR BENT.
B	1-2-77	DIM. TO PILE & FROM ROAD & AMEND. RL. HEADSTOCK 1/25 AMENDED.	DRAWN R. Holmes	
C	23-8-77	PLATE SIZES AMENDED.	PASSED	
			DATE DEC-76	
			DRAWING NO. CTQ1122/S/006	
			C	
CSR REFERENCE NO.			MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 213 ST PAULS TERRACE PORTFOLIO VALLEY - OLD 4006 PH 52 7123 IN ASSOCIATION WITH	
			CSR LIMITED	



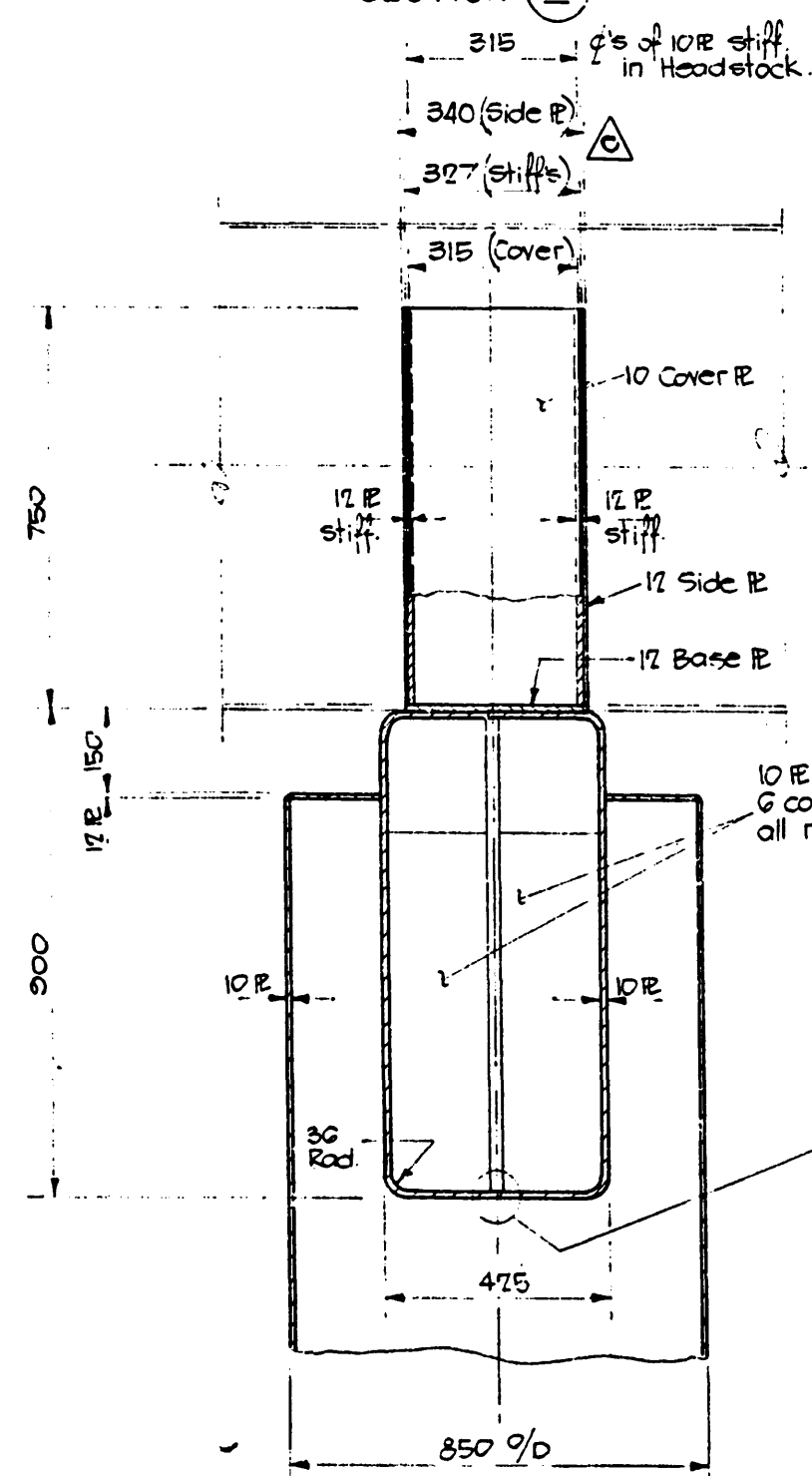
File Bent & Headstock

Conveyor

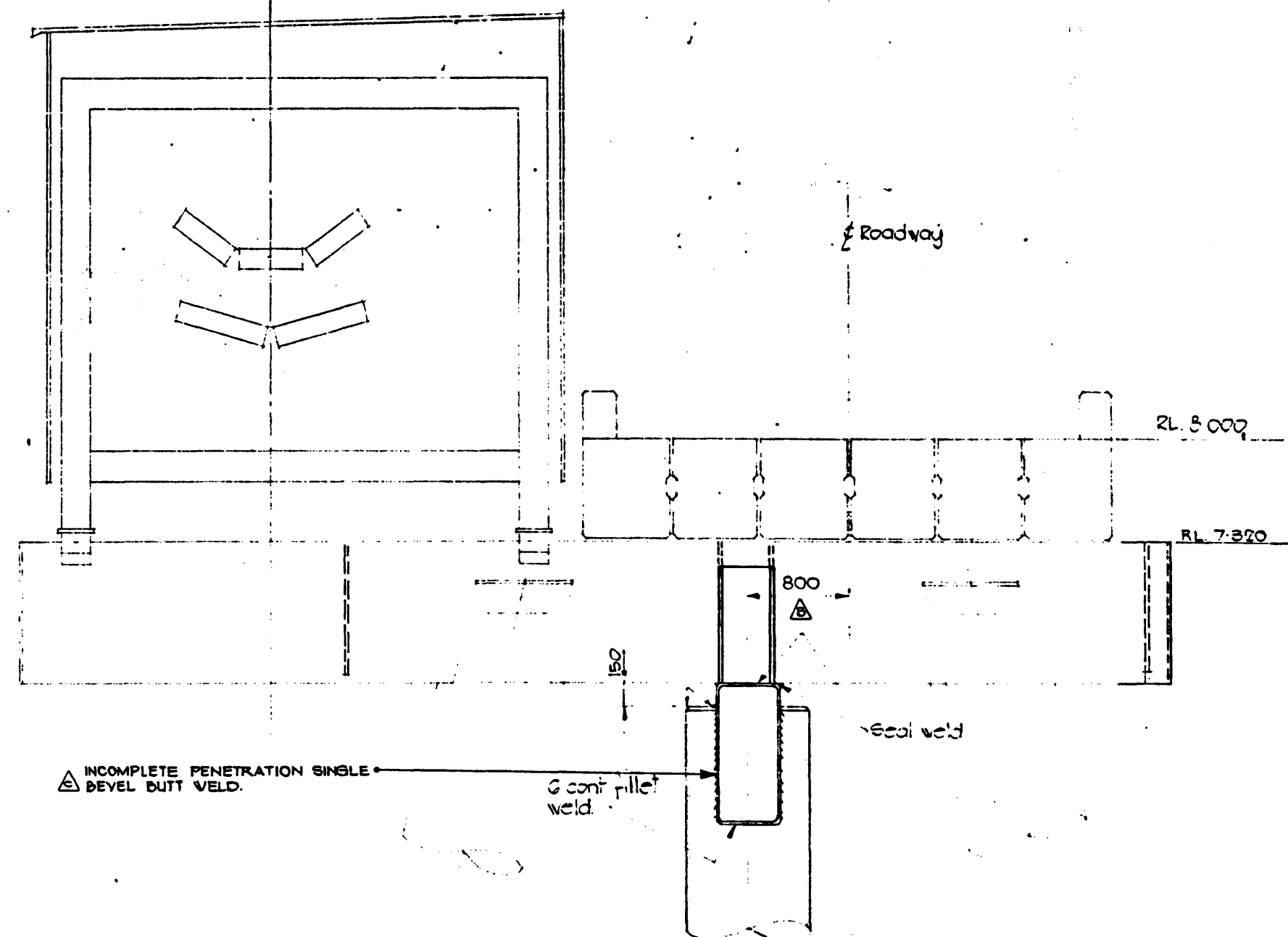
Roadway



SECTION A

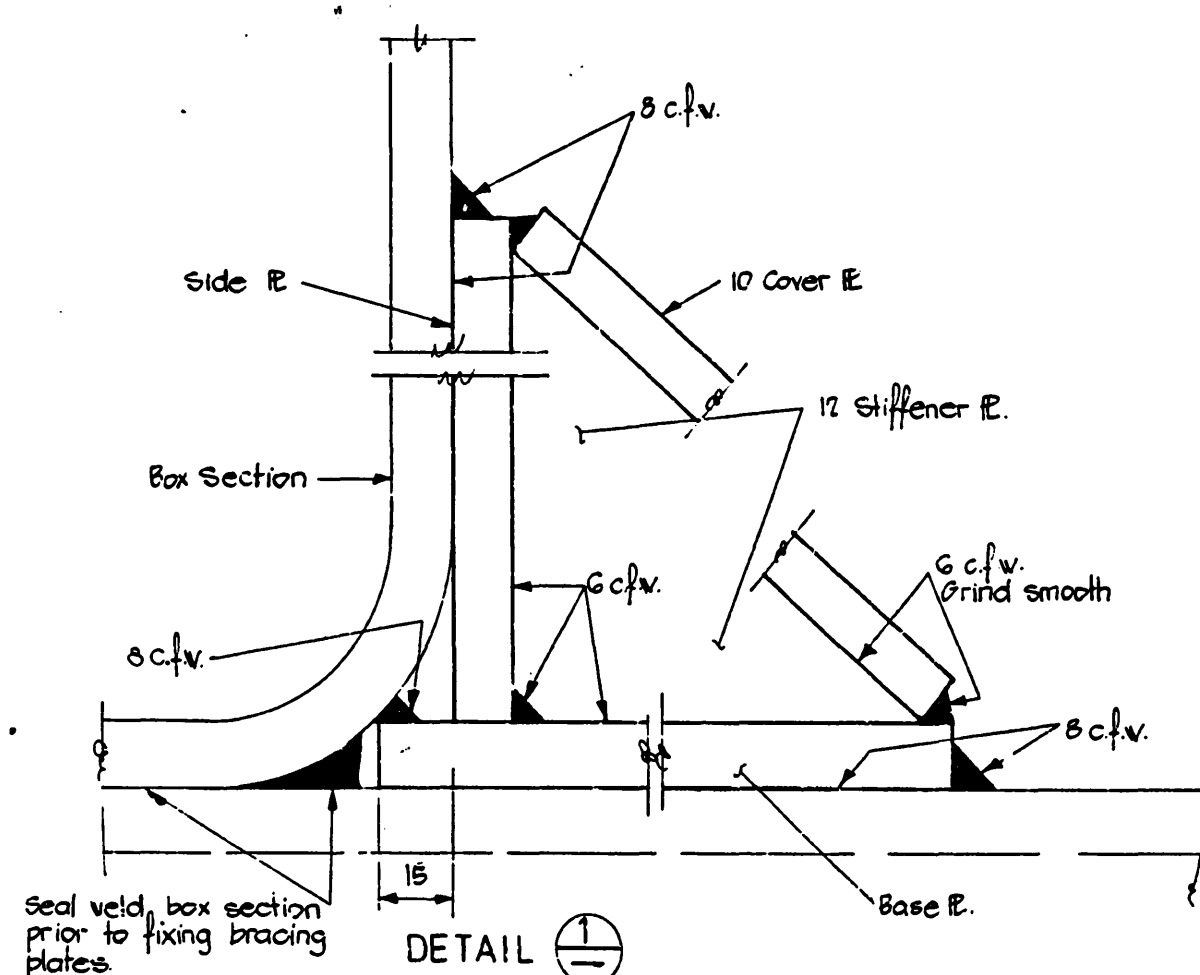


SECTION B



ELEVATION

of Raked Pile
650% 10 wall.
NB 370% 10 wall for pile bent
no 761 & 774

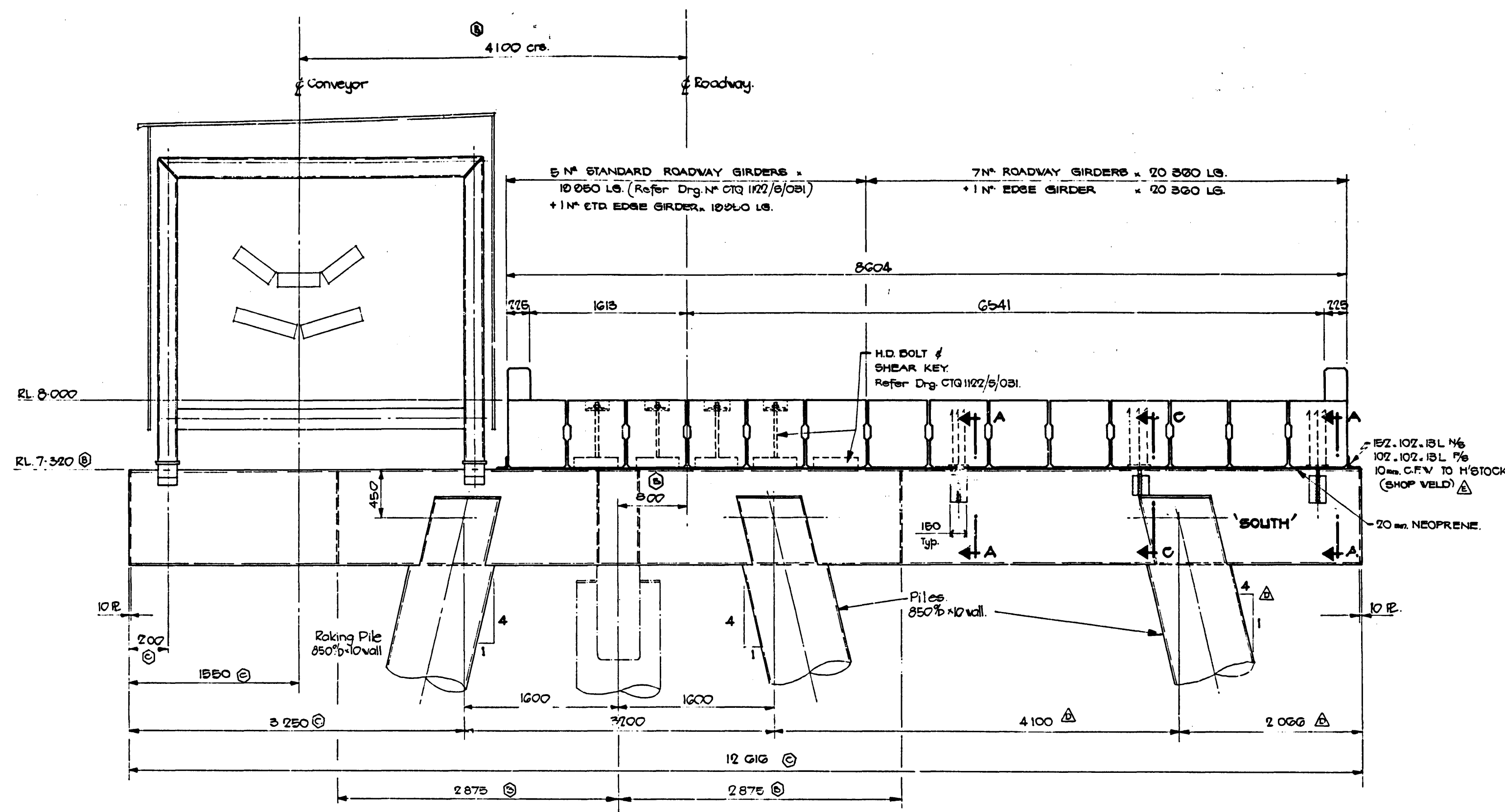


DETAIL

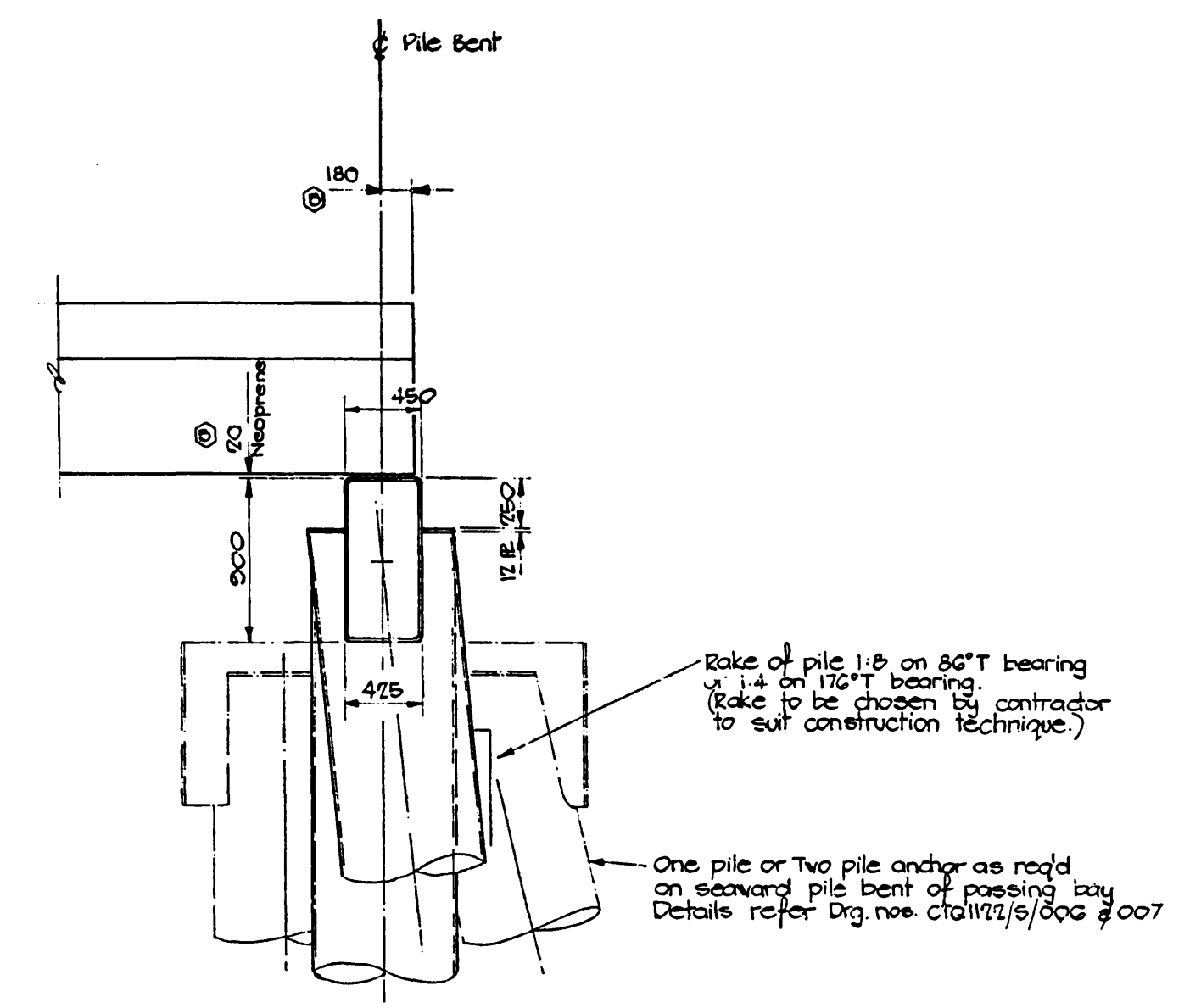
NOTE

For 'Type A' Anchor Pile Bent - Refer Drg. No. CTQ1111/s/006
For Typical Cross Section of Approach - (850% pile) Refer Drg. No. CTQ1111/s/005
(850% pile)

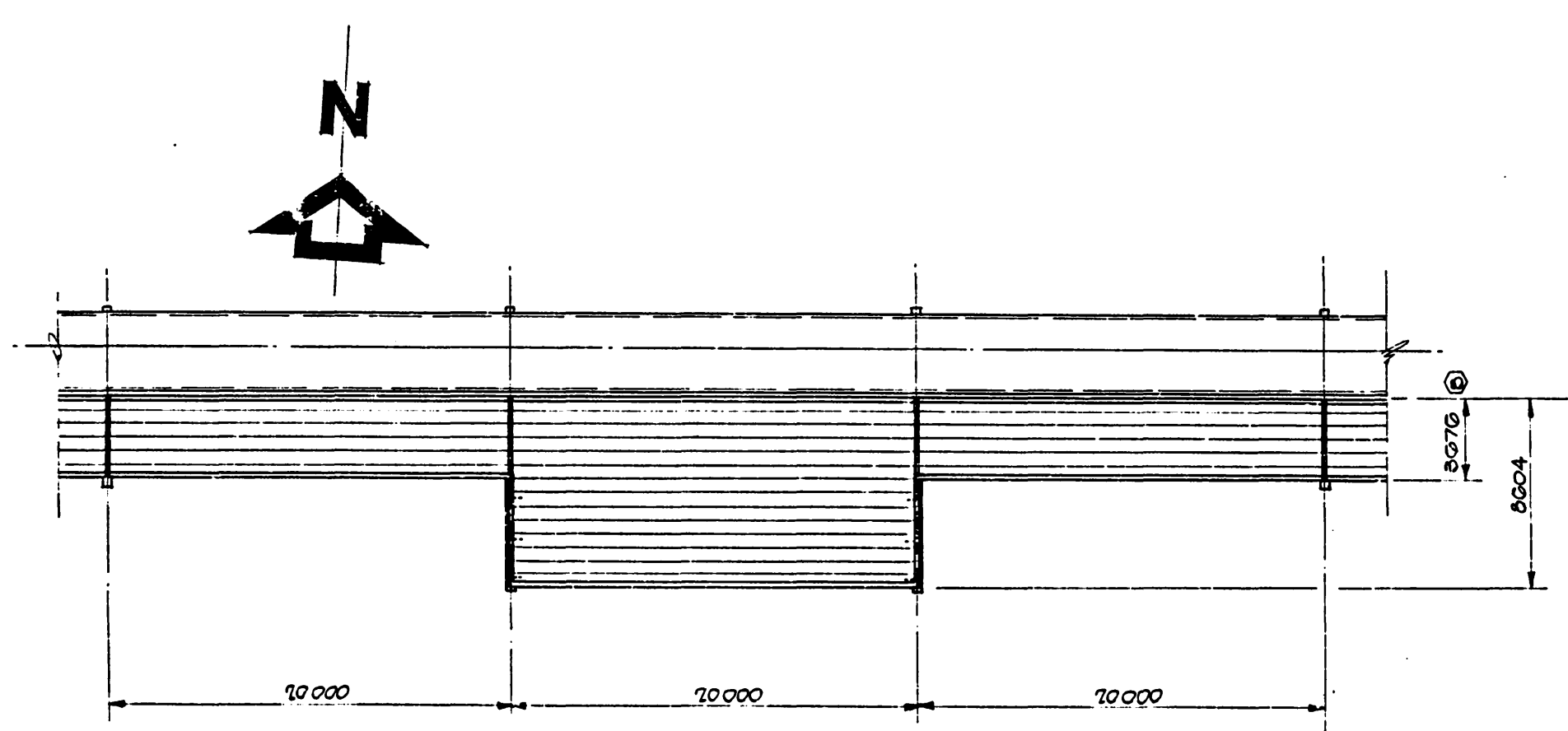
THE CORPORATION OF THE TREASURER OF QUEENSLAND					
NO	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS		
A	20.12.76	Roadway construction altered.	SCALES 1:25 1:10 DRAWN R. Holmes PASSED DATE DEC. 76	B.S.T. LUCINDA STAGE III APPROACH JETTY TYPE 'B' TYPICAL TWO-PILE ANCHOR BENT.	DRAWING NO CT01122/9/007
B	1.2.77	DIM. TO PILE & FROM ROAD & AMENDED. R.L. AMEN.			
C	20.8.77	PLATE SIZES AMENDED.			
			MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 215 & 216 HALL STREET, NORTHBRIDGE, SYDNEY, N.S.W. 1511 IN ASSOCIATION WITH CSR LIMITED		
CSR REFERENCE NO.					



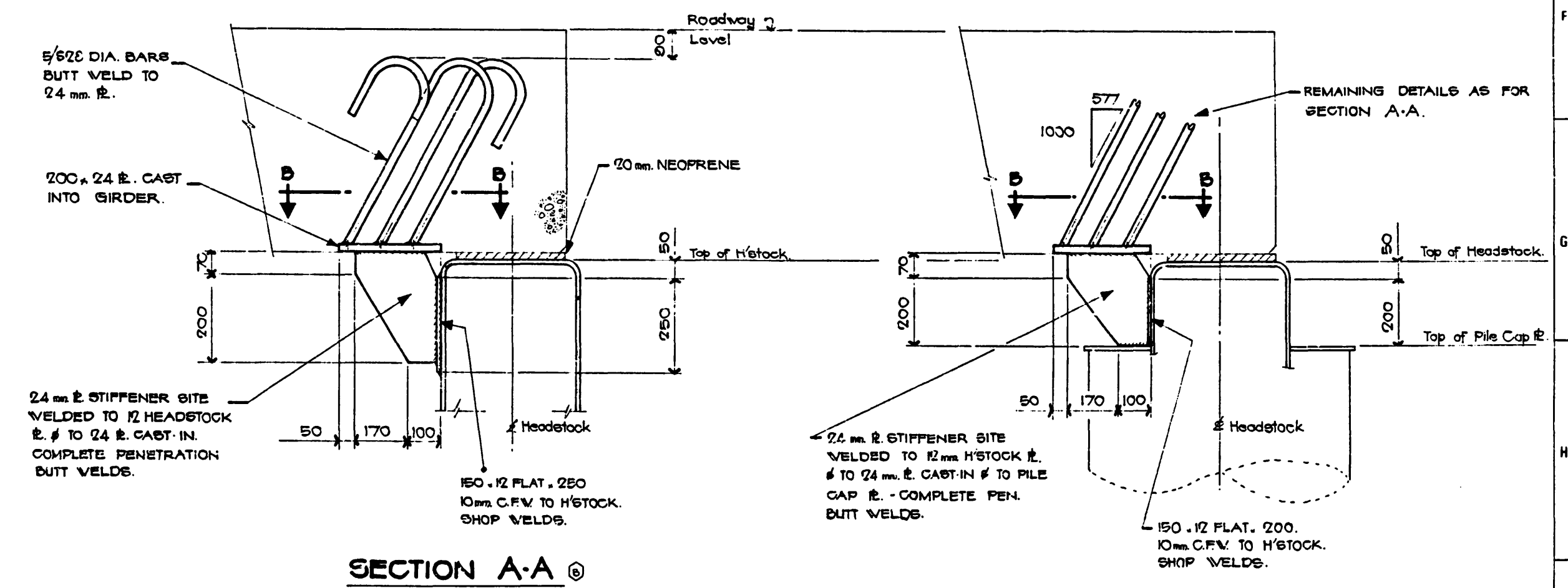
TYP. SECTION THRU. PASSING BAY



END VIEW

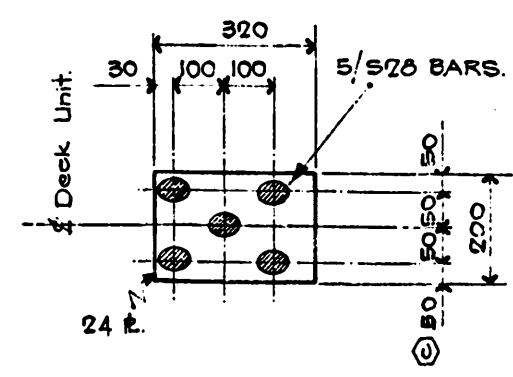


TYPICAL PASSING BAY LAYOUT.
For locations of Passing Bays refer Drg. N° CTQ1122/S/001.



SECTION A-A

SECTION C-C



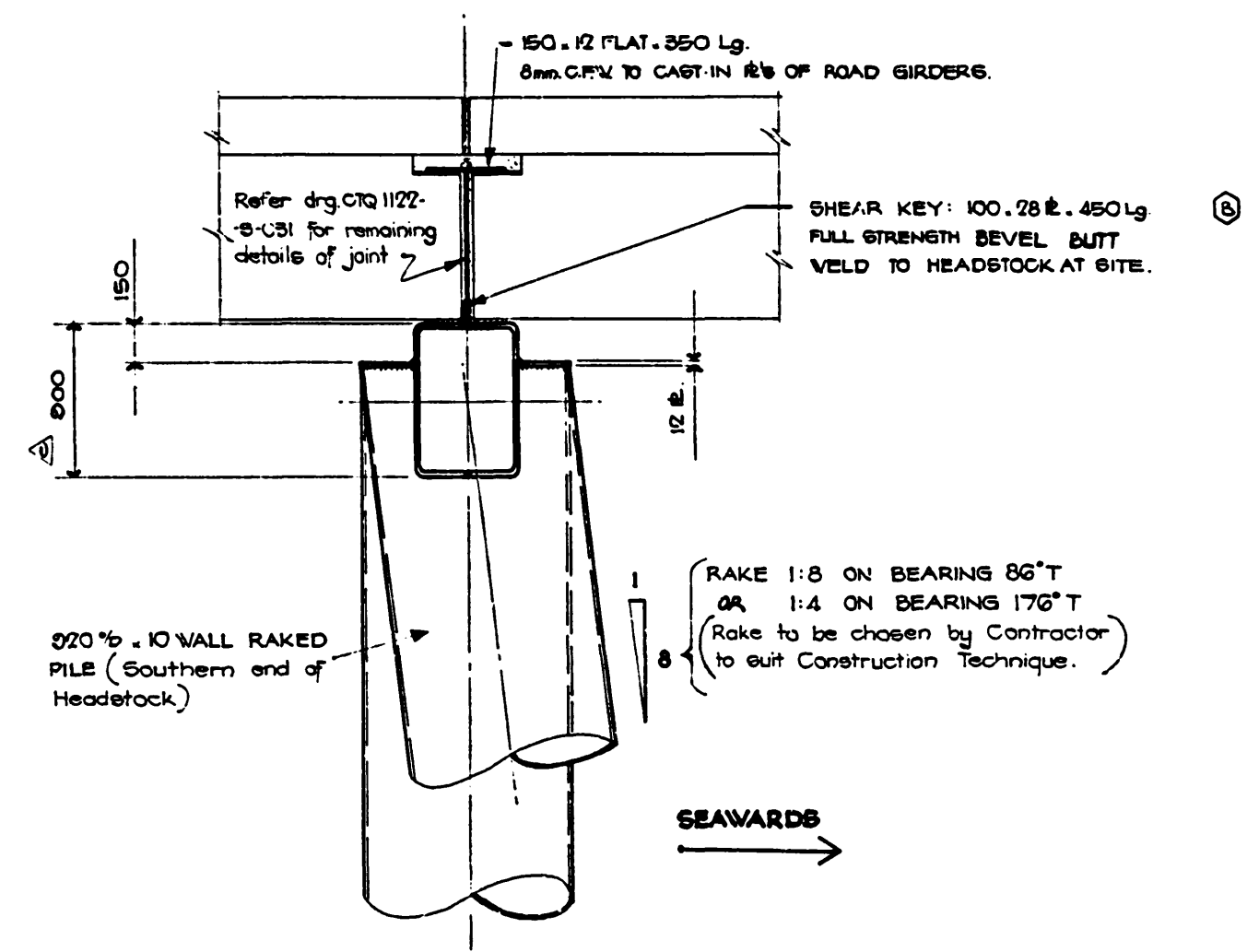
SECTION B-B

NOTES:

- THE CONTRACTOR MAY MODIFY BOX GIRDER DETAILS TO SUIT POST-TENSIONING TECHNIQUE, SUBJECT TO ENGINEERS APPROVAL.
- FOR STANDARD CONC. ROADWAY GIRDERS & STANDARD JOINT DETAILS - REFER DRG. CTQ1122/S/001.
- FOR REMAINING HEADSTOCK & PILE DETAILS - REFER DRG. N° CTQ1122/S/005.

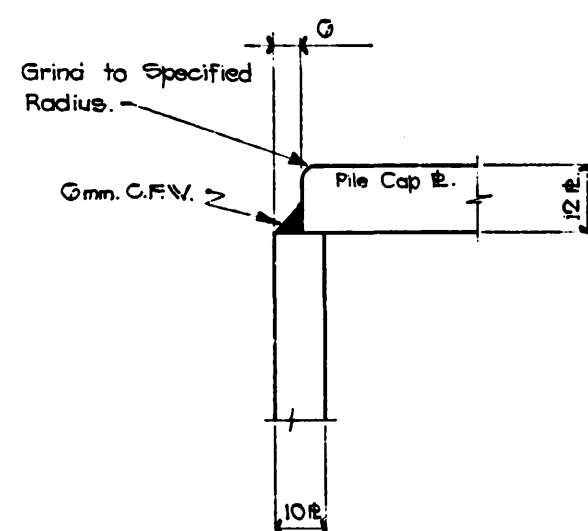
THE CORPORATION OF THE TREASURER OF QUEENSLAND					THE SUGAR BOARD BULK SUGAR TERMINALS		DRAWING NO.	
NO.	DATE	AMENDMENTS			SCALES 1:75 1:100	B.S.T. LUCINDA STAGE III APPROACH JETTY PASSING BAY DETAILS.	CTQ1122/S/008	E
A	20.12.76	Roadway construction altered.			DRAWN R. Holmes			
B	JAN 77	Conv. of dim. to Road & amend. Deck unit hold down details added. Girder Length alt.			PASSED			
C	1.2.77	HEADSTOCK LENGTH ALTERED.			DATE DEC.76			
D	16.5.77	Pile Spacing amended to suit Contractor Construction Methods.						
E	12.10.77	Shop weld edge angles to suit Contractor						
C.S.R. REFERENCE NO.					MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST PAUL'S TERRACE PORTLAND VALLEY QLD 4006 PH 32 7122 IN ASSOCIATION WITH CSR LIMITED			



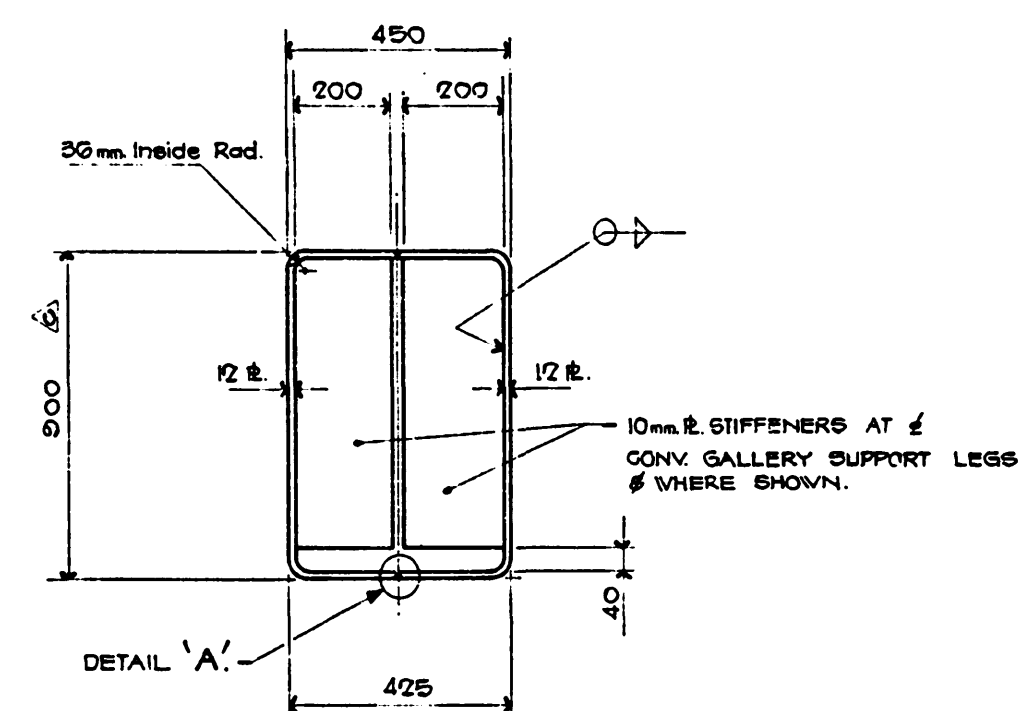


1:20.

Scale: 1:20

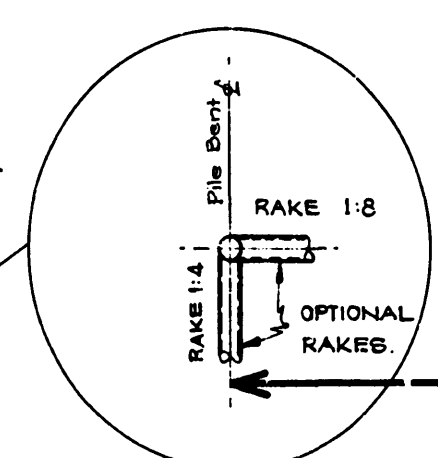


DETAIL 'C'.



NOTES:

1. FOR LOCATION OF PILE BENTS - REFER GEN. PLAN DRG. N° CTQ 1122/S/001
2. FOR A TYPICAL PILE BENT CROSS-SECTION REFER DRG. N° CTQ 1122/S/005
3. ALL WELDS - Gmm. C.F.W. UNLESS NOTED.
4. FOR STANDARD ROADWAY JOINT DETAILS - REFER DRG. N° CTQ 1122/S/031.

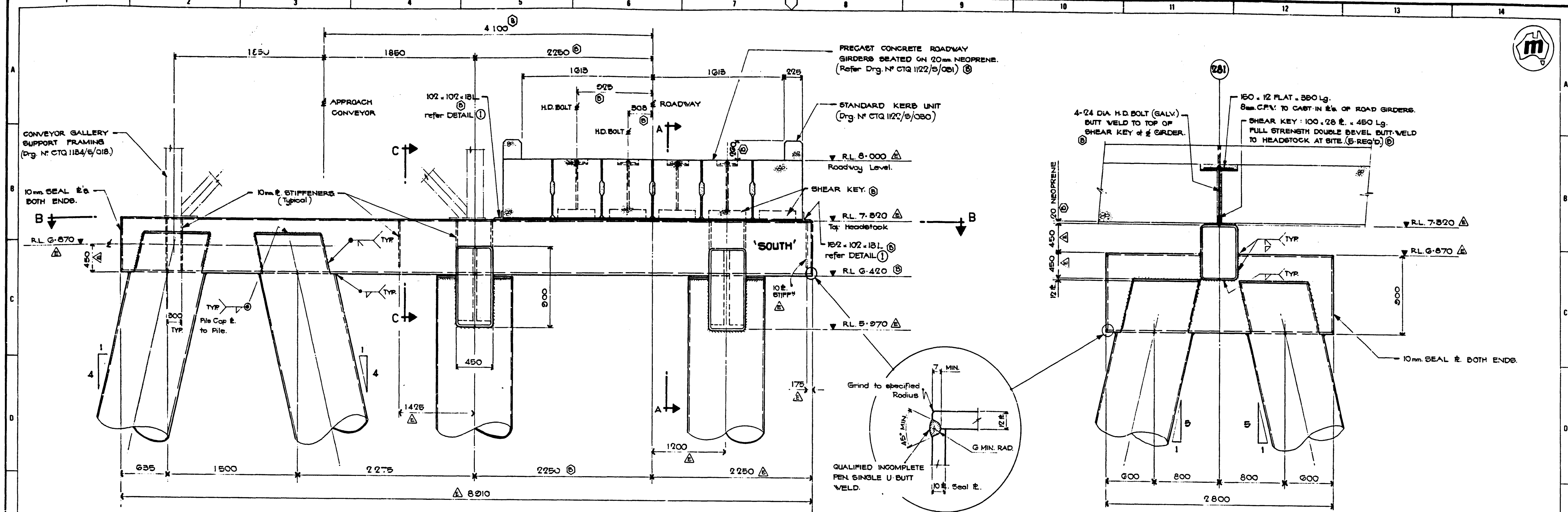


NOTE: All Files take 1:4 unless noted otherwise.

THE CORPORATION OF THE TREASURER OF QUEENSLAND		
NO.	DATE	AMENDMENTS
A	20-12-78	Roadway construction altered.
B	JAN-79	↳ ROAD TO 4 CONV. RL'S AMENDED
C	15-11-77	PILE LOCATIONS & HEADSTOCK LENGTH AMENDED TO SUIT CONTRACTOR'S CONSTRUCTION
C.S.R. REFERENCE NO		

THE SUGAR BOARD BULK SUGAR TERMINALS	
SCALES 1:10 U.N. DRAWN J.T. PASSED DATE DEC. '78	B.S.T. LUCINDA STAGE III APPROACH JETTY. PILE BENTS NO 275-279 inc.
	DRAWING NO CTN225/5/010 C

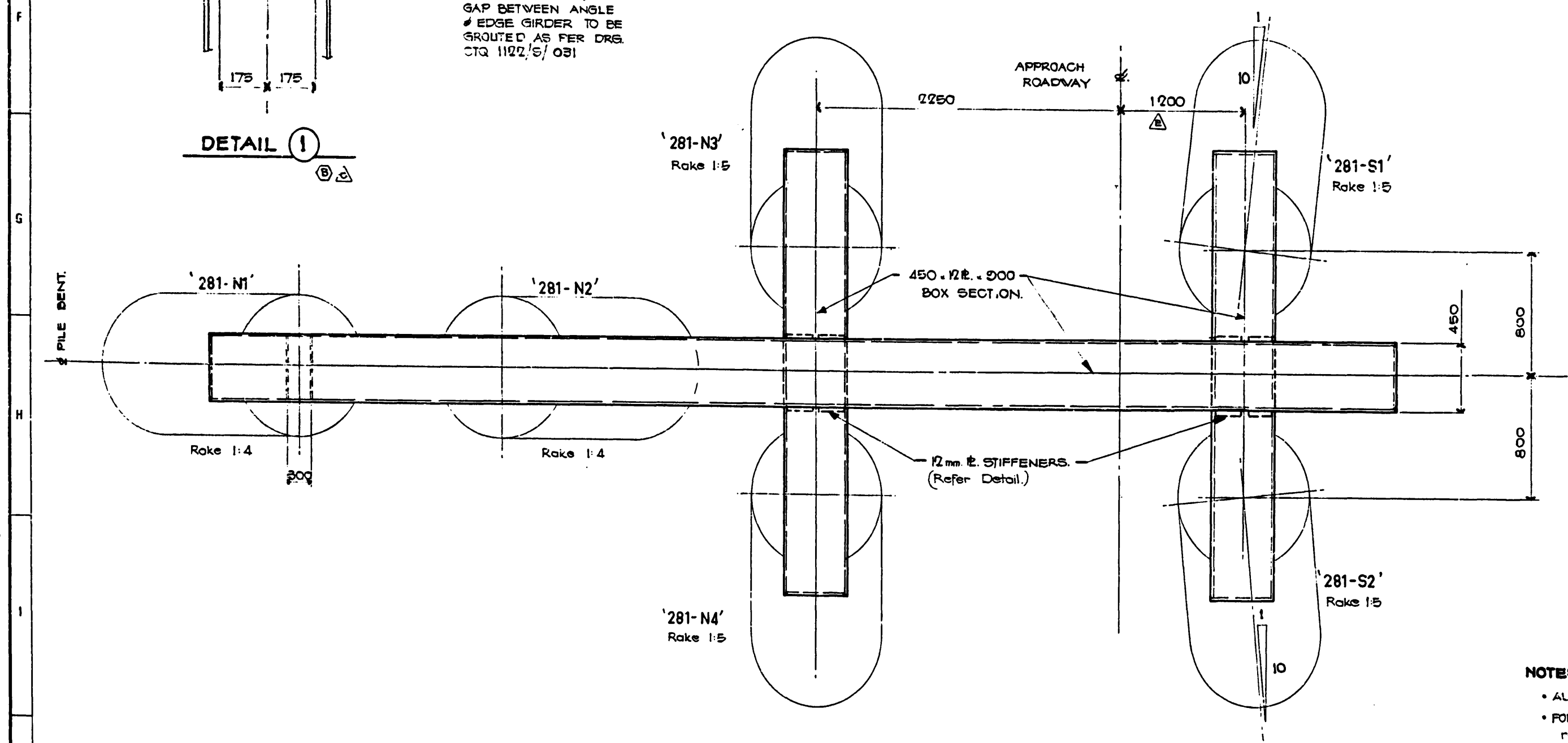
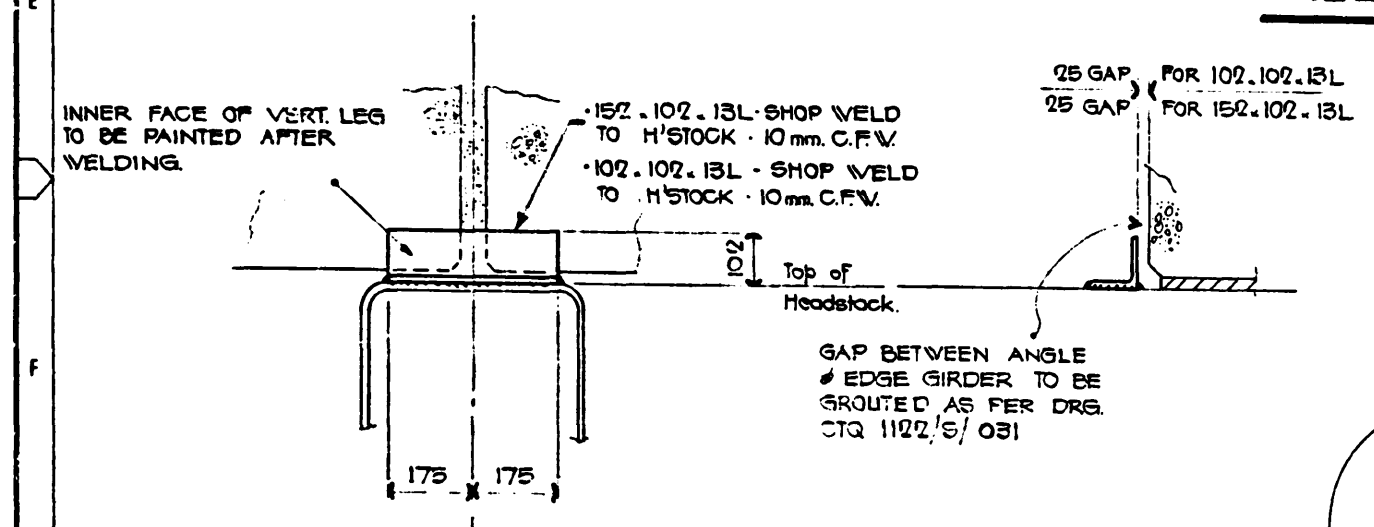
MACDONALD WAGNER & PRIDDLE PTY.LTD.
 CONSULTING ENGINEERS 213 ST PAULS TERRACE, PORTPHU VALLEY, QLD. 4080 PH 32 7123
 IN ASSOCIATION WITH
CSR LIMITED



PILE BENT 281

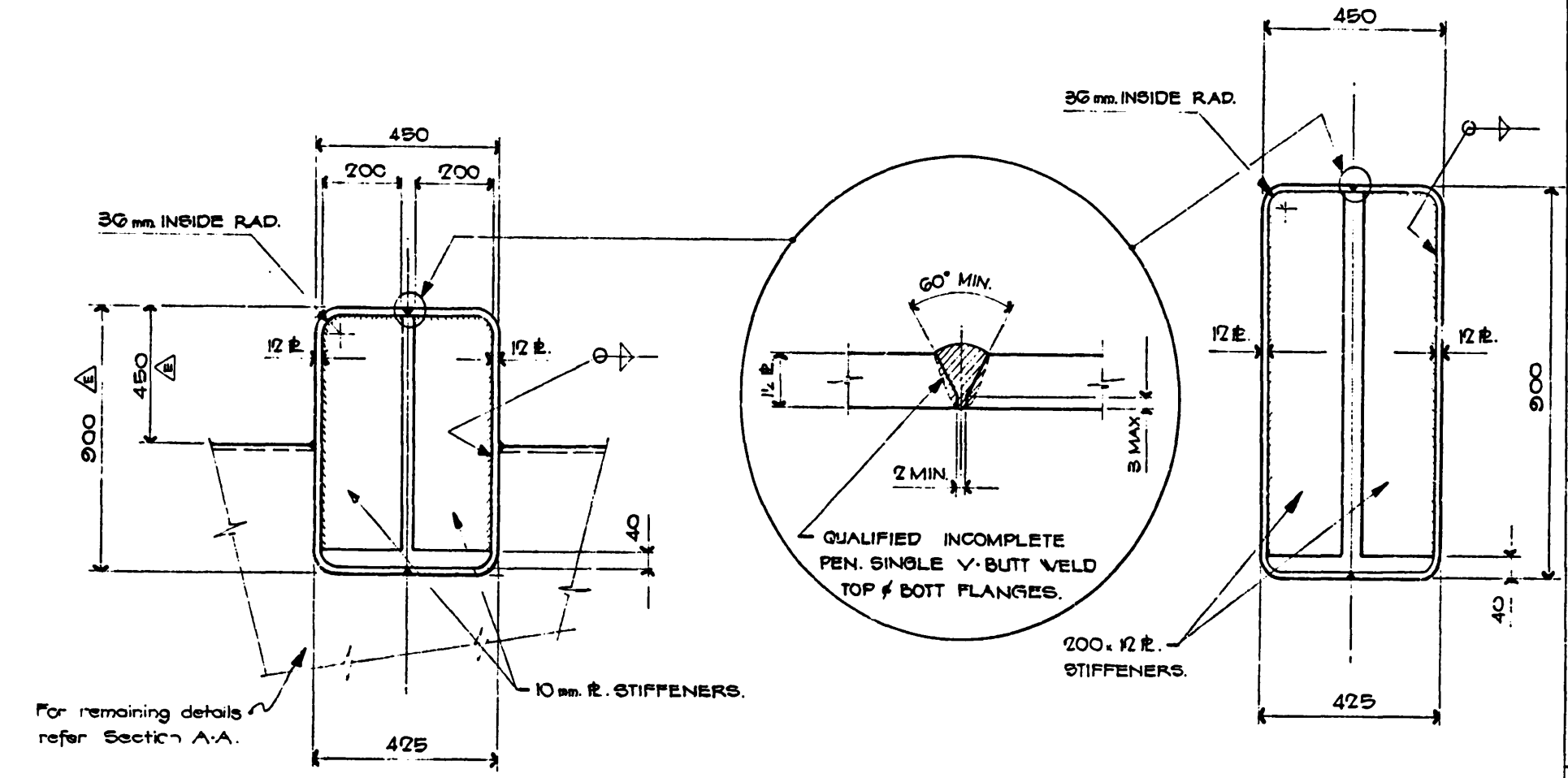
SEAL @ DETAIL

SECTION A-A



SECTION B-B

NOTE: ALL PILES 920 o.d. x 10mm WALL



450 x 12 L. x 900 BOX SECT. DETAIL

Scale 1:10

- NOTES:
- ALL WELDS 6mm C.F.V. UNLESS NOTED
 - FOR DETAILS OF STD. DECK UNITS - refer drg. n° CTG 1122/S/031.

NOTED

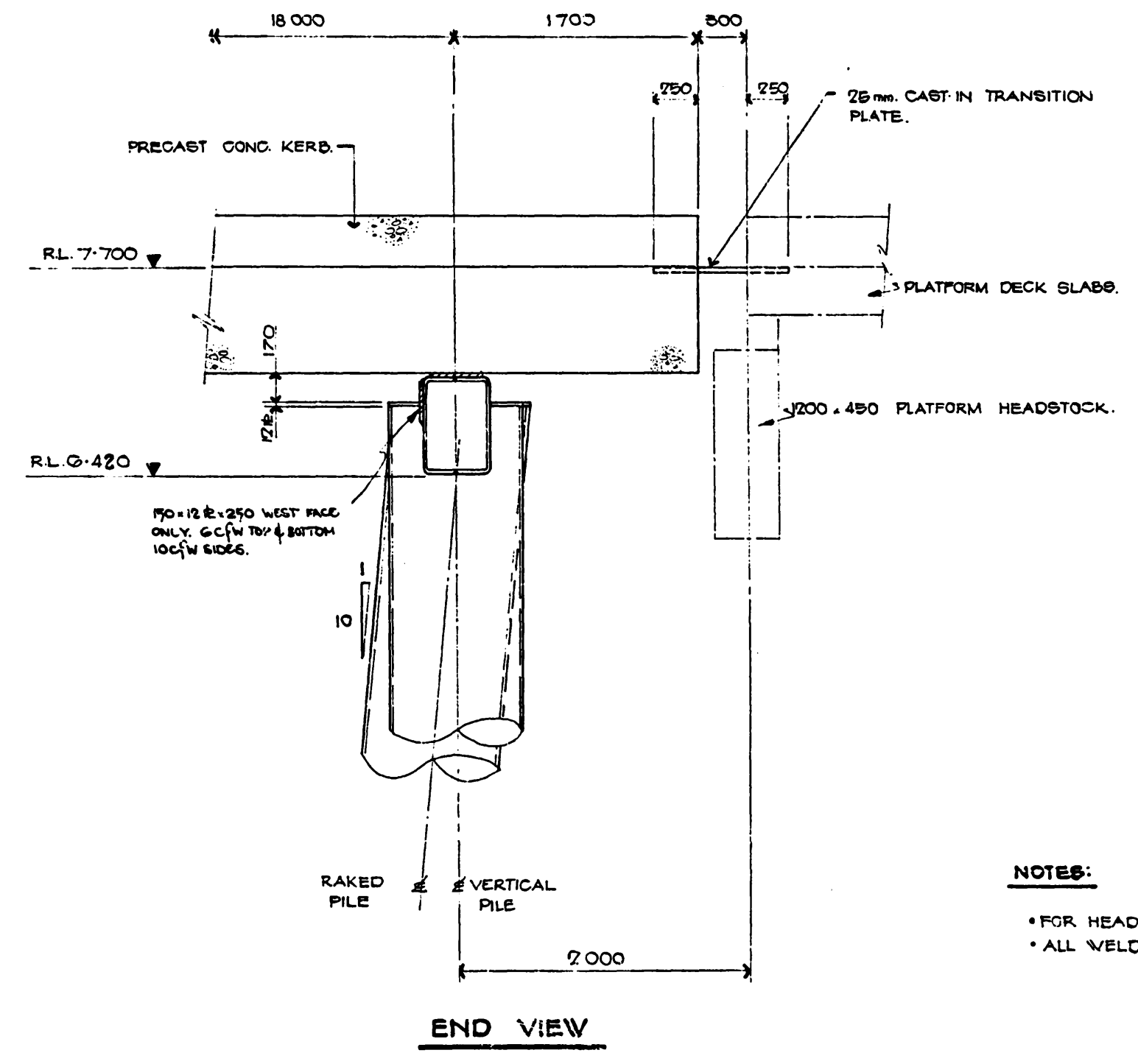
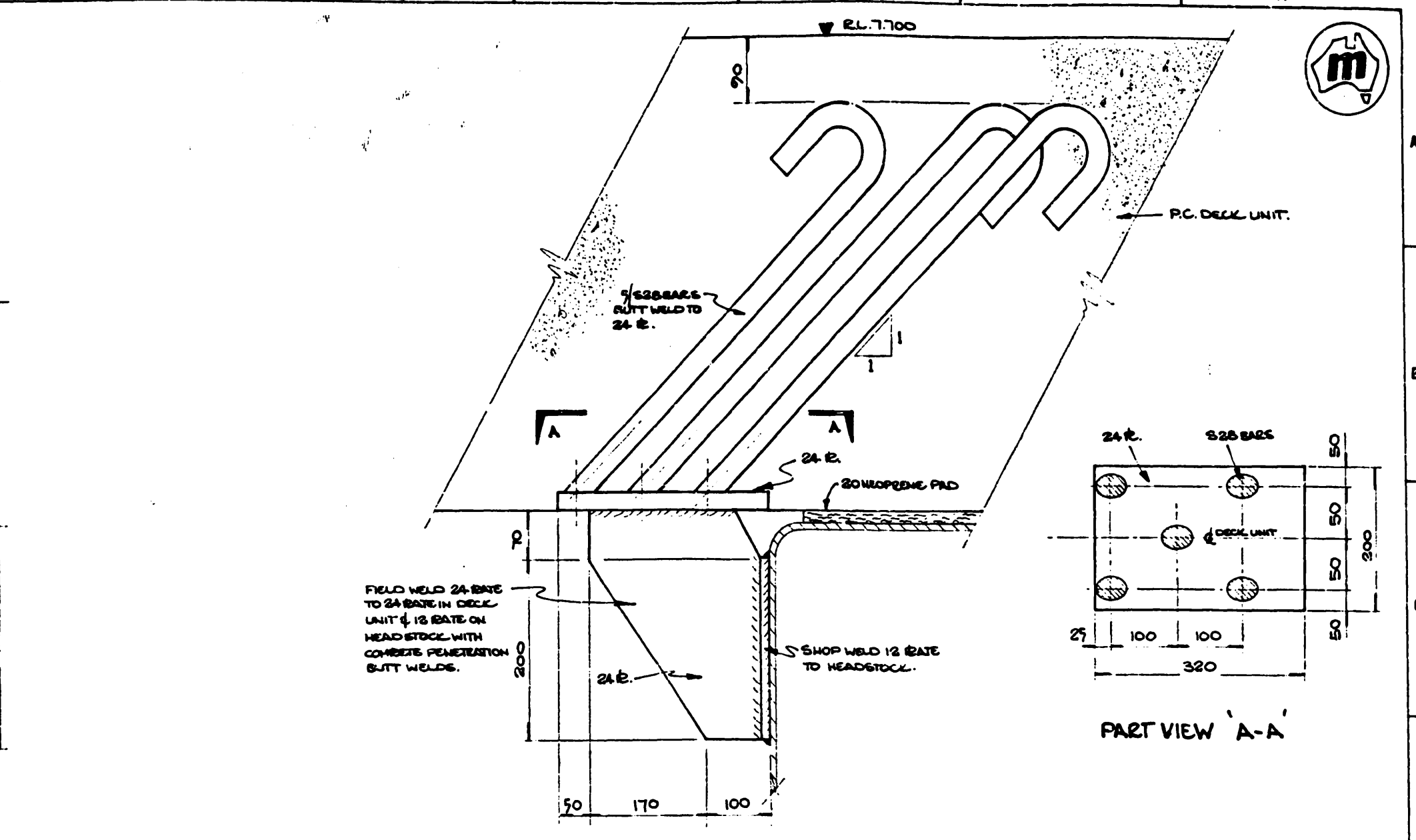
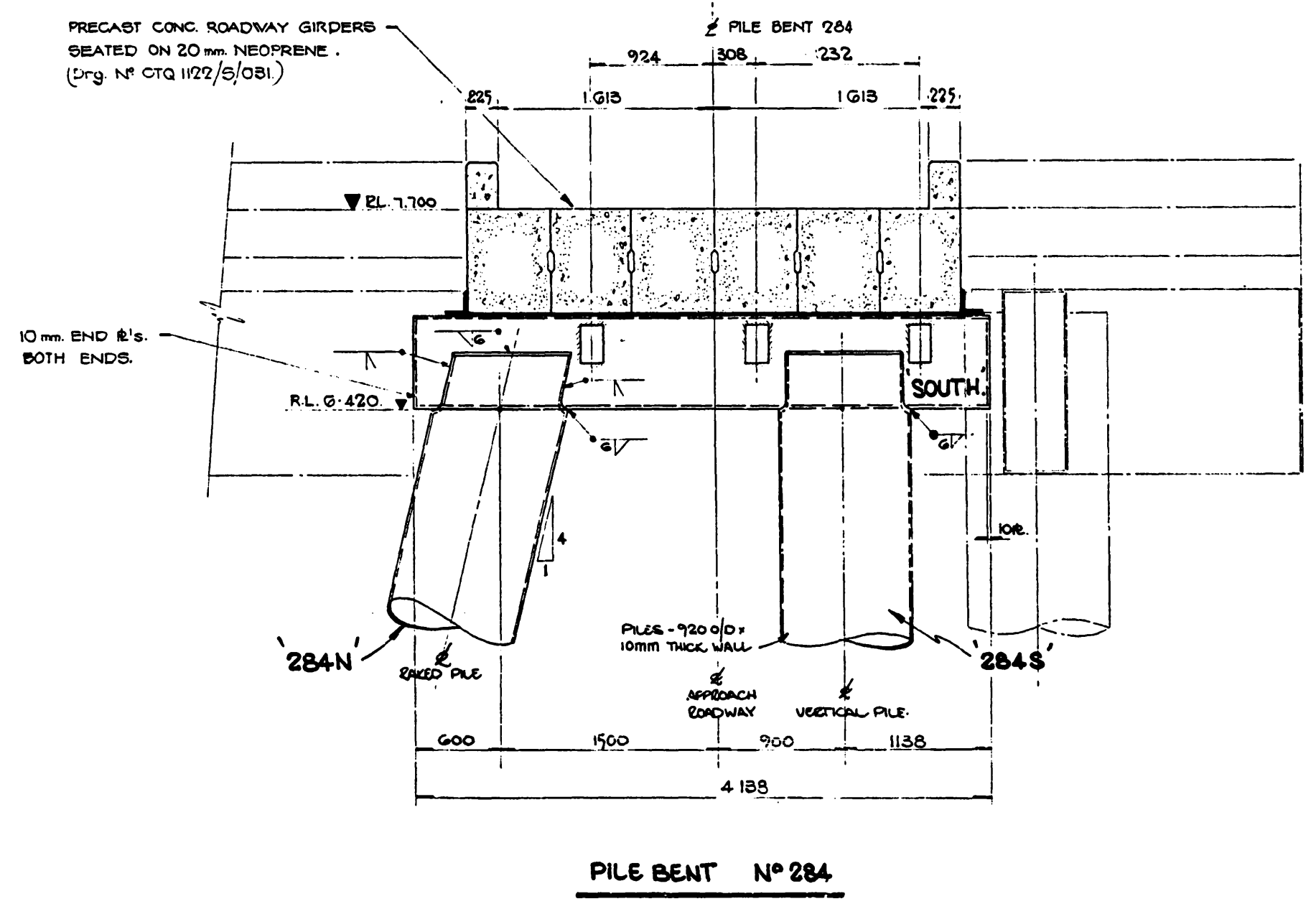
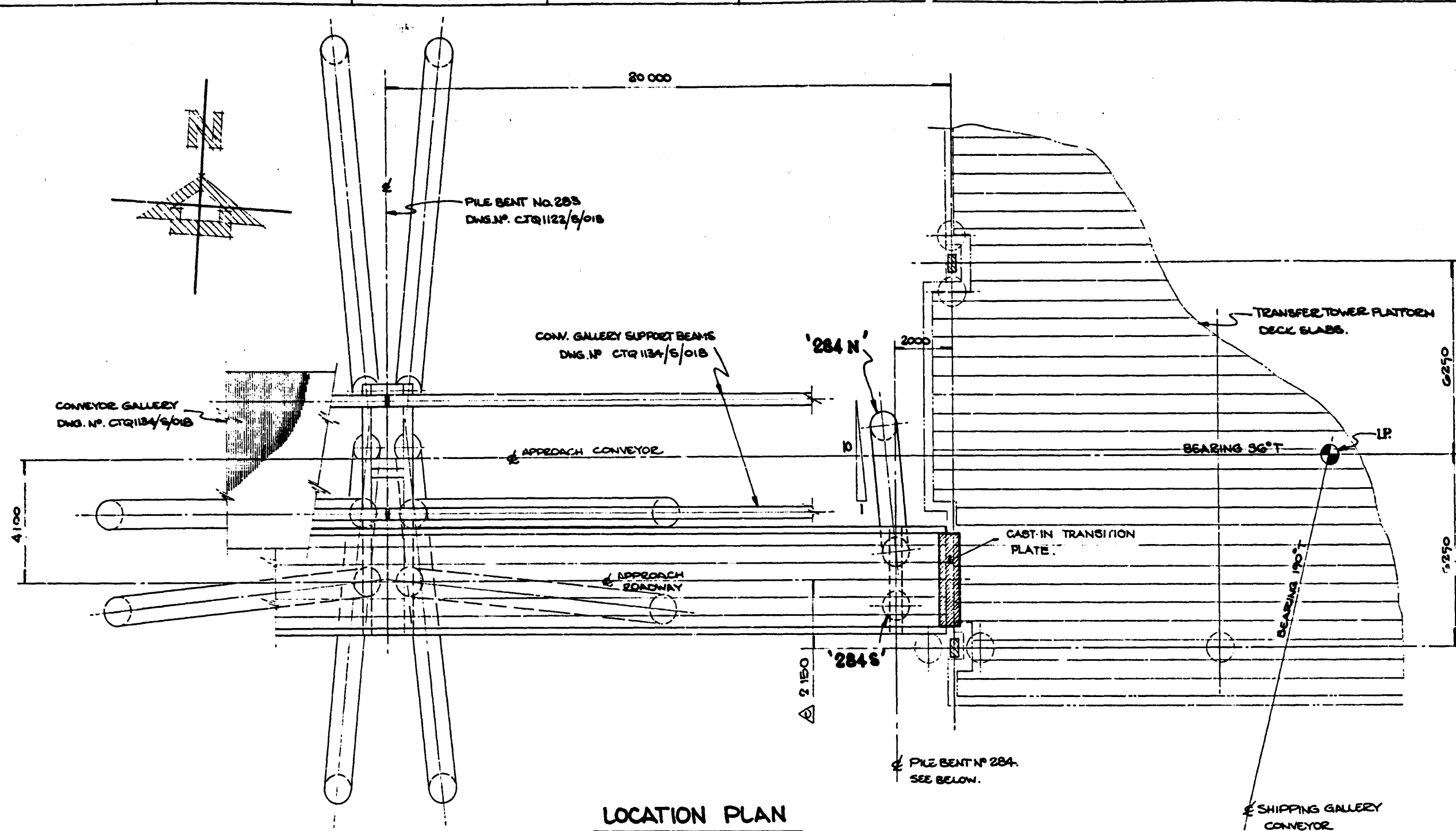
THE CORPORATION OF THE TREASURER OF QUEENSLAND			
NO.	DATE	AMENDMENTS	
A	20.12.78	Roadway construction altered.	THE SUGAR BOARD BULK SUGAR TERMINALS SCALES 1:20 DRAWN J.T. PASSED DATE DEC. '78 B.S.T. LUCINDA STAGE III APPROACH JETTY. PILE BENT NO. 281. DRAWING NO. CTG 1122/S/012 E
B	JAN. 77	DIMENSION 4100 WAS 4000. NEOPRENE AMEND	
C	10.5.77	DETAIL ① AMENDED	
D	10.5.77	SITE WELDING OF 152 L AMENDED TO SHOP WELDING TO SUIT CONTRACTOR.	
E	18.11.77	PILE BENT AMENDED TO SUIT CONTRACTORS' CONSTRUCTION METHOD.	
C.S.R. REFERENCE NO.			

MACDONALD WAGNER & PRIDDLE PTY. LTD.

CONSULTING ENGINEERS 113 ST. HALLS TERRACE, PORTHURST VALLEY, QLD 4008 PH 51 7123

IN ASSOCIATION WITH

CSR LIMITED

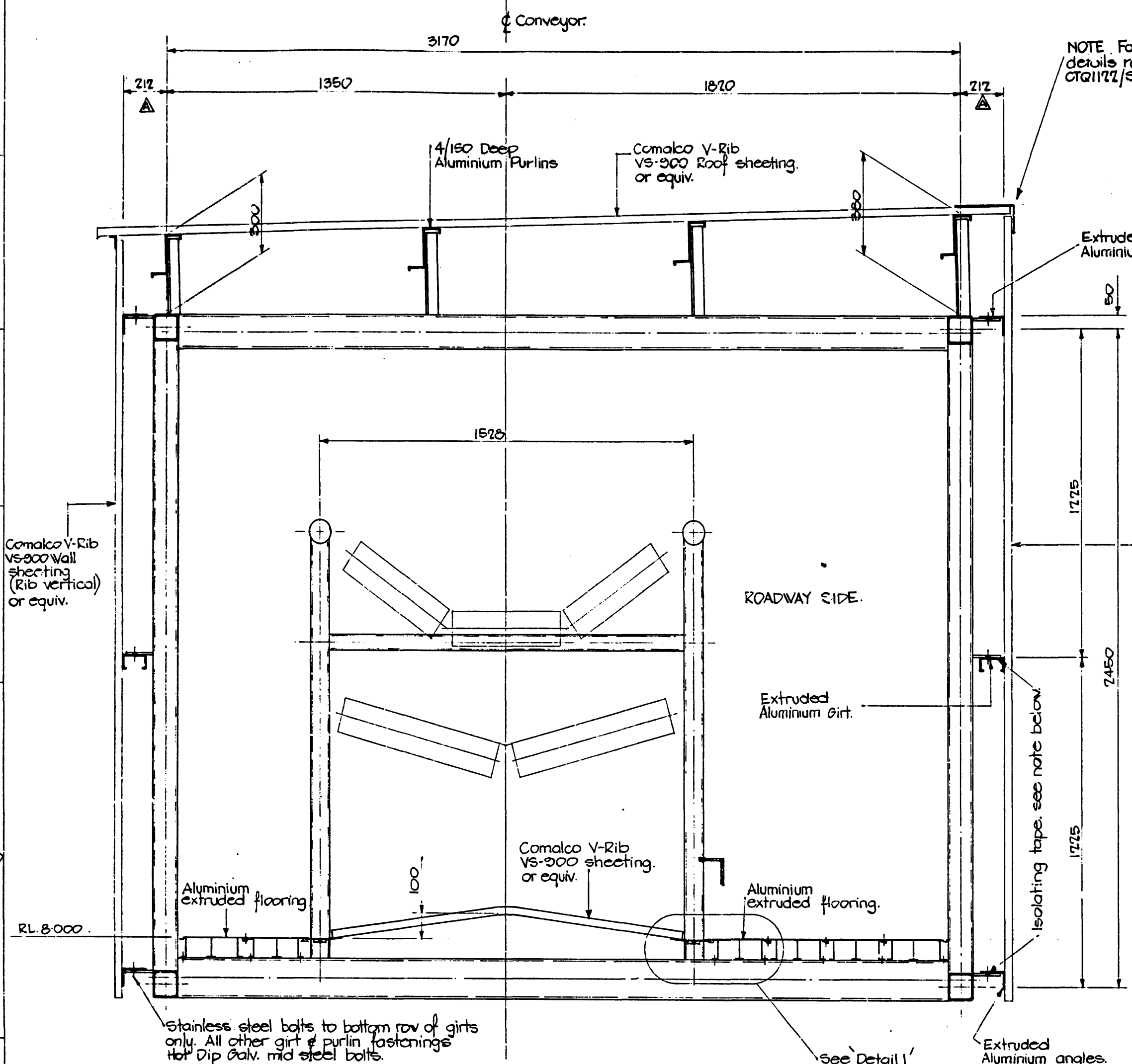


NOTES:

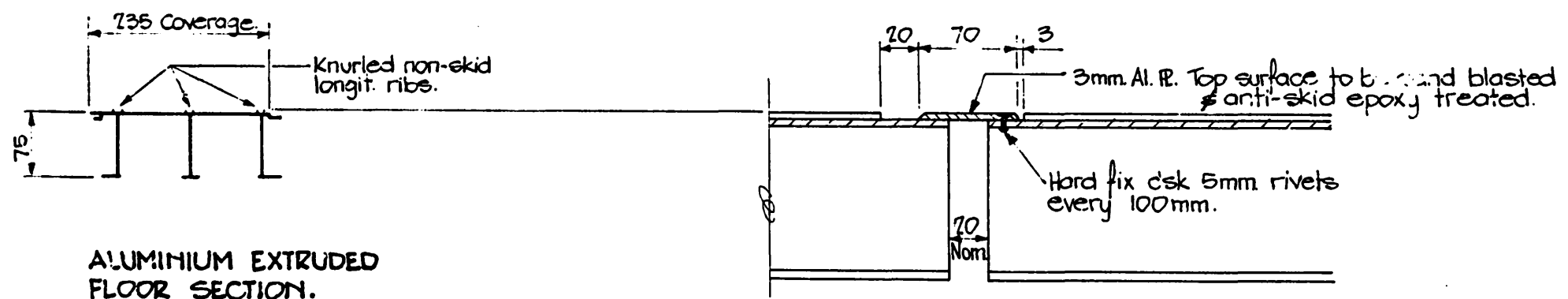
- FOR HEADSTOCK DETAILS - REFER DRG. CTQ 1122/5/010
- ALL WELDS 6 mm C.F.W. UNLESS NOTED.

THE CORPORATION OF THE TREASURER OF QUEENSLAND									
THE SUGAR BOARD BULK SUGAR TERMINALS					DRAWING NO.				
B.S.T. LUCINDA STAGE III APPROACH JETTY PILE BENT No. 284					CTQ 1122/5/014				
MACDONALD WAGNER & PRIDDLE PTY. LTD.					C				
CONSULTING ENGINEERS 215 ST. PAULS TERRACE, PORTHURST VICT. 3207. AUST. INC. 22 7123					IN ASSOCIATION WITH				
CSR LIMITED					CSR REFERENCE NO.				
NO. DATE AMENDMENTS					SCALE 1:100 & 1:25				
A 20-12-76 Roadway construction altered					DRAWN R.K.L.				
B JAN 77 FILE CENTRES AMENDED DECK UNIT HO BRACKET DETAILS ADDED					PASSED				
C SEPT 77 Dimension altered for Bent Position.					DATE DEC 79				

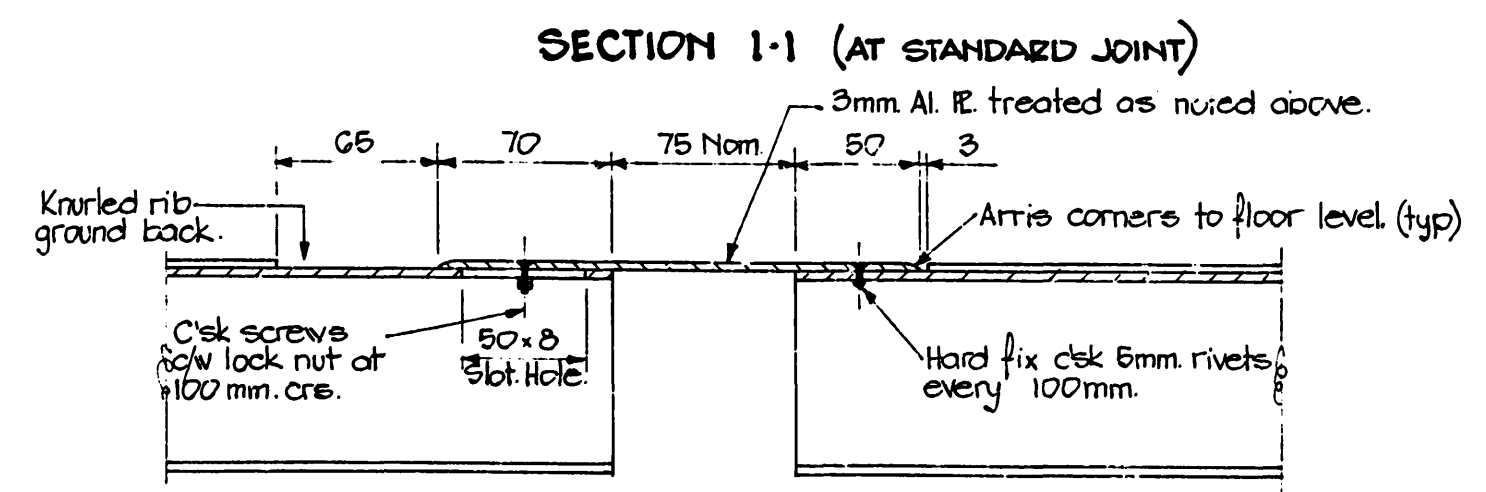




TYPICAL SECTION.
Scale: 1:10.

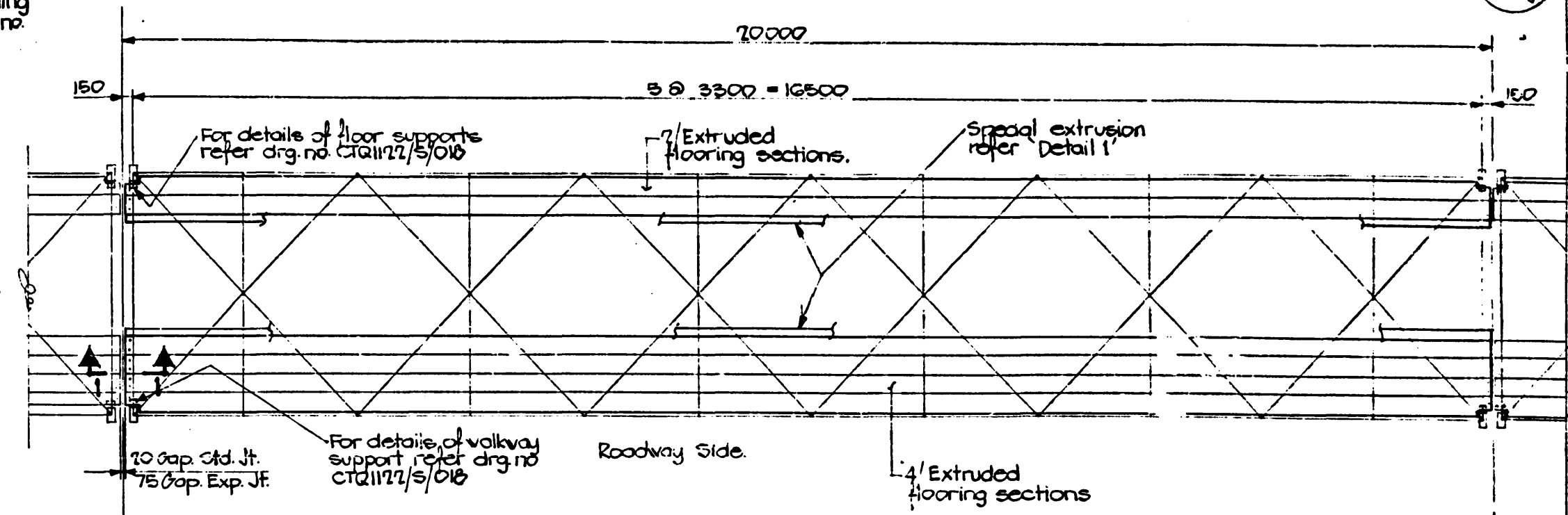


ALUMINIUM EXTRUDED FLOOR SECTION.

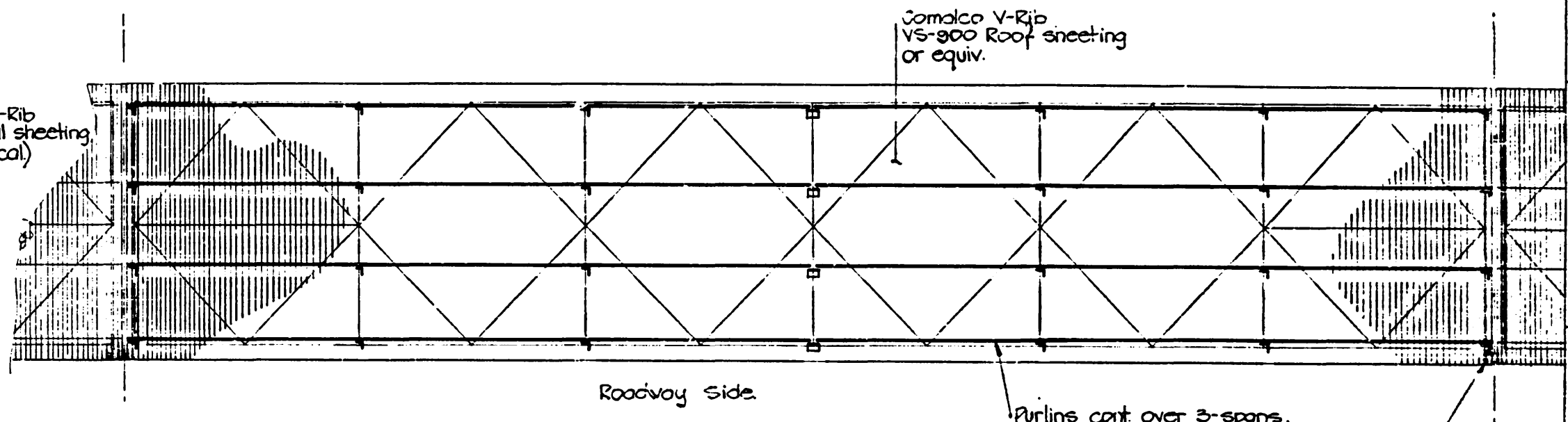


SECTION 1-1 (AT EXPANSION JOINT)

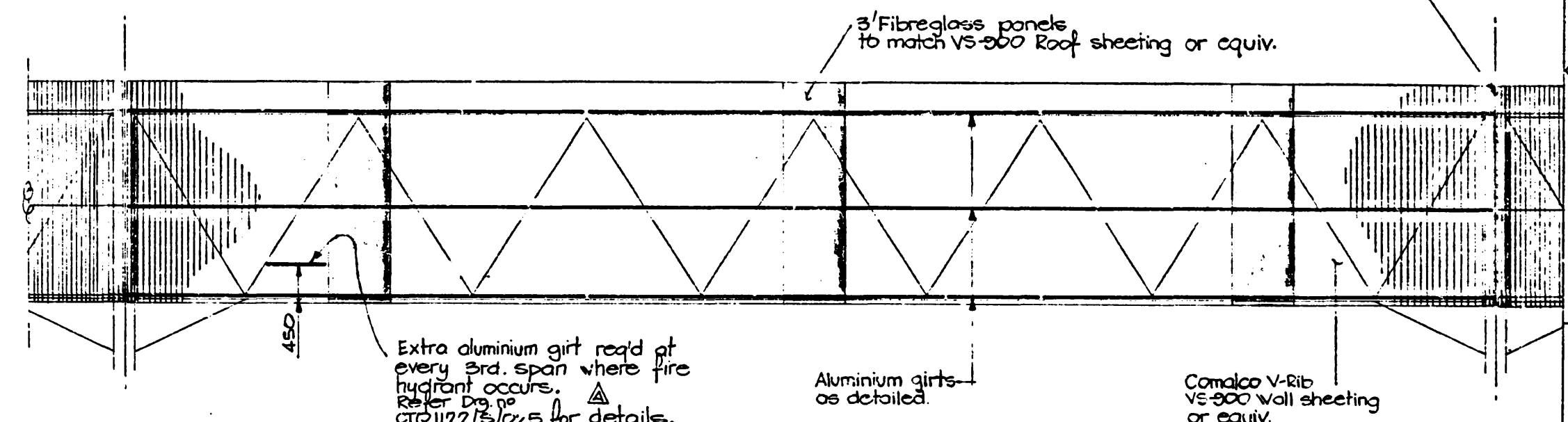
NOTE: For all flashing details refer drg no CTO1122/5/021



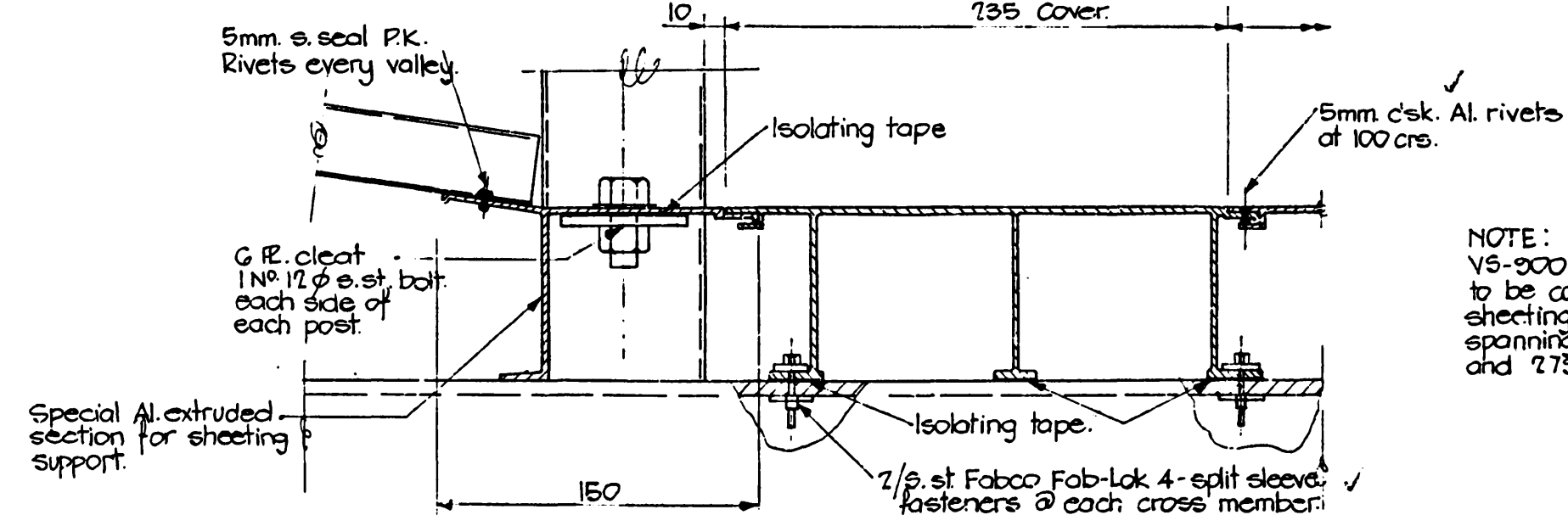
FLOOR PLAN.



ROOF PLAN.



SIDE ELEVATION. (ROADWAY SIDE)



DETAIL 1 Scale: 1:2.

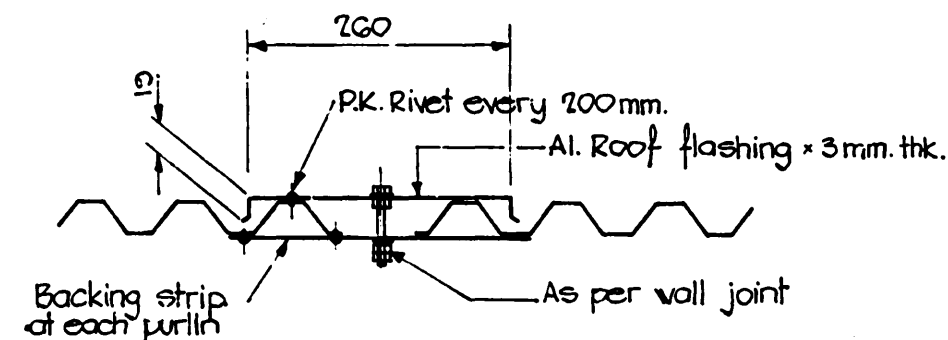
NOTES.
All contact surfaces between aluminium & steel eg. floor sections, purlins & girts, to be isolated with the following tape or its equiv. PLICOFLEX 340-20 Butyl/PVC laminated tape to be applied with PLICOFLEX NP 310 Primer in accordance with manufacturer's instructions.
All Aluminium components, cladding, floors, purlins & girts to be assembled & fixed to complete steel galleries on shore prior to its erection in place on headstock. Cover flashings roof, walls & floors to be completed in place after galleries have been welded to headstock.

NOTE:
VS-300 sheeting under conveyor belt is to be covered with 1mm thk. flat aluminium sheeting riveted to V-Rib for galleries spanning between pile bents no 1 to 3 and 273-281 only.

THE CORPORATION OF THE TREASURER OF QUEENSLAND									
THE SUGAR BOARD BULK SUGAR TERMINALS									
NO		DATE		AMENDMENTS					
A		24-2-77		DIMENSION ALTERED. PURLIN CLEAT ALTERED. GIRT ADDED.					
DRAWN		2-2-77		BY: [Signature]					
PASSED		17/2/77		BY: [Signature]					
DATE		17/2/77		BY: [Signature]					
CSR REFERENCE NO.									
SCALES 1:10		1:50							
DRAWN 2-2-77									
PASSED 17/2/77									
DATE 17/2/77									
B.S.T. LUCINDA STAGE III APPROACH JETTY. STANDARD SPAN CLADDING									
MACDONALD WAGNER & PRIDDLE PTY. LTD.									
CONSULTING ENGINEERS 312 ST PAULS TERRACE NORTHQUEEN BEACH QLD 4058 PH 52 1712									
IN ASSOCIATION WITH									
CSR LIMITED									

WALL SHEET. FLASHING JOINT

Cornalco V-Rib VS-900
wall sheeting or equiv.



ROOF SHEET. DETAIL AT TRUSS JOINT

NOTE: Provide similar cover flashing over VS-200 sheeting beneath conveyor belt. (3 backing strips reqd.)

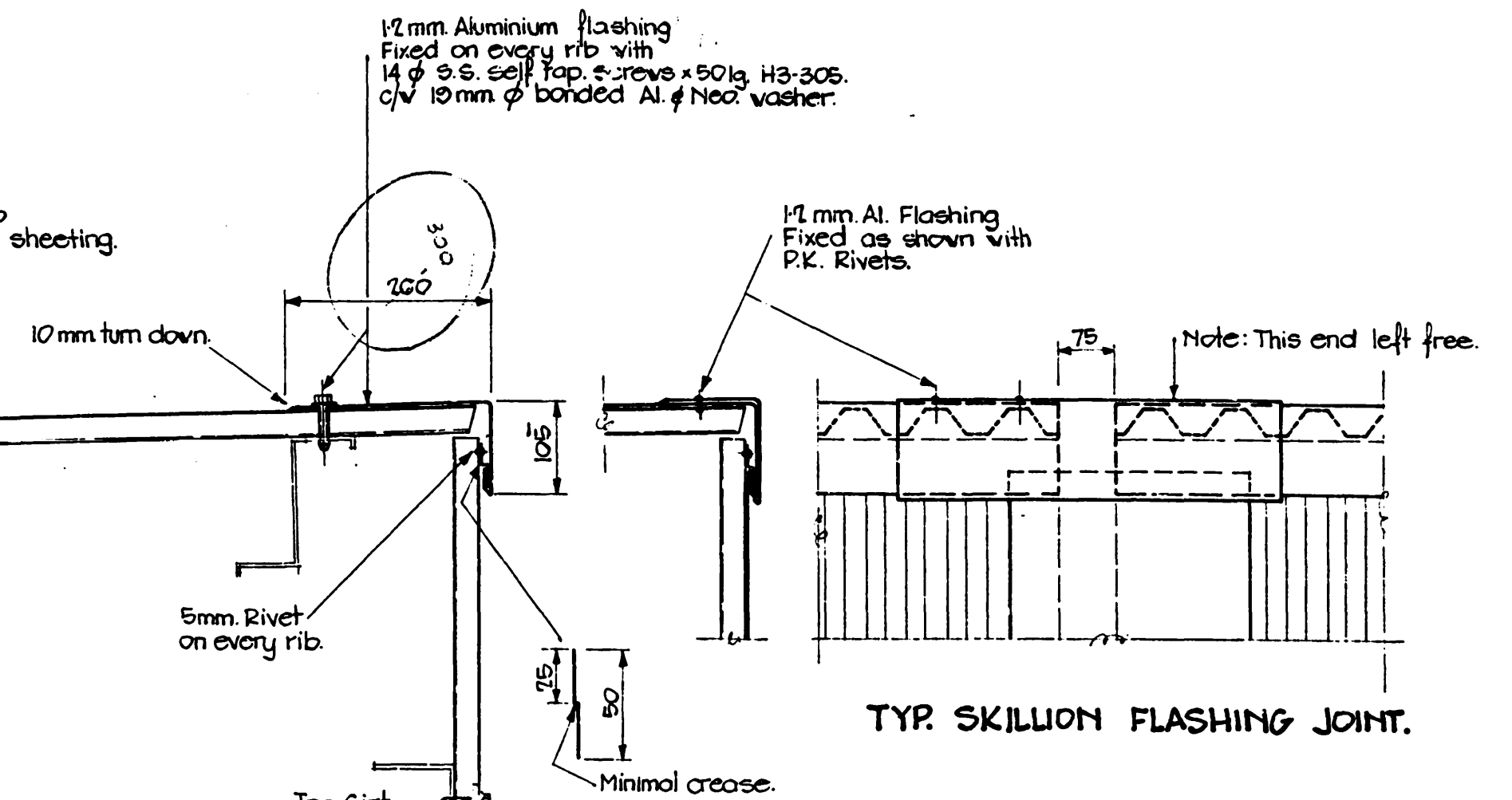
FIXING NOTES:

STITCHING:
Fix all Aluminium roof & wall sheeting at
200 mm. c/s. using 5mm. dia. P.K. Al. self sealing rivets.

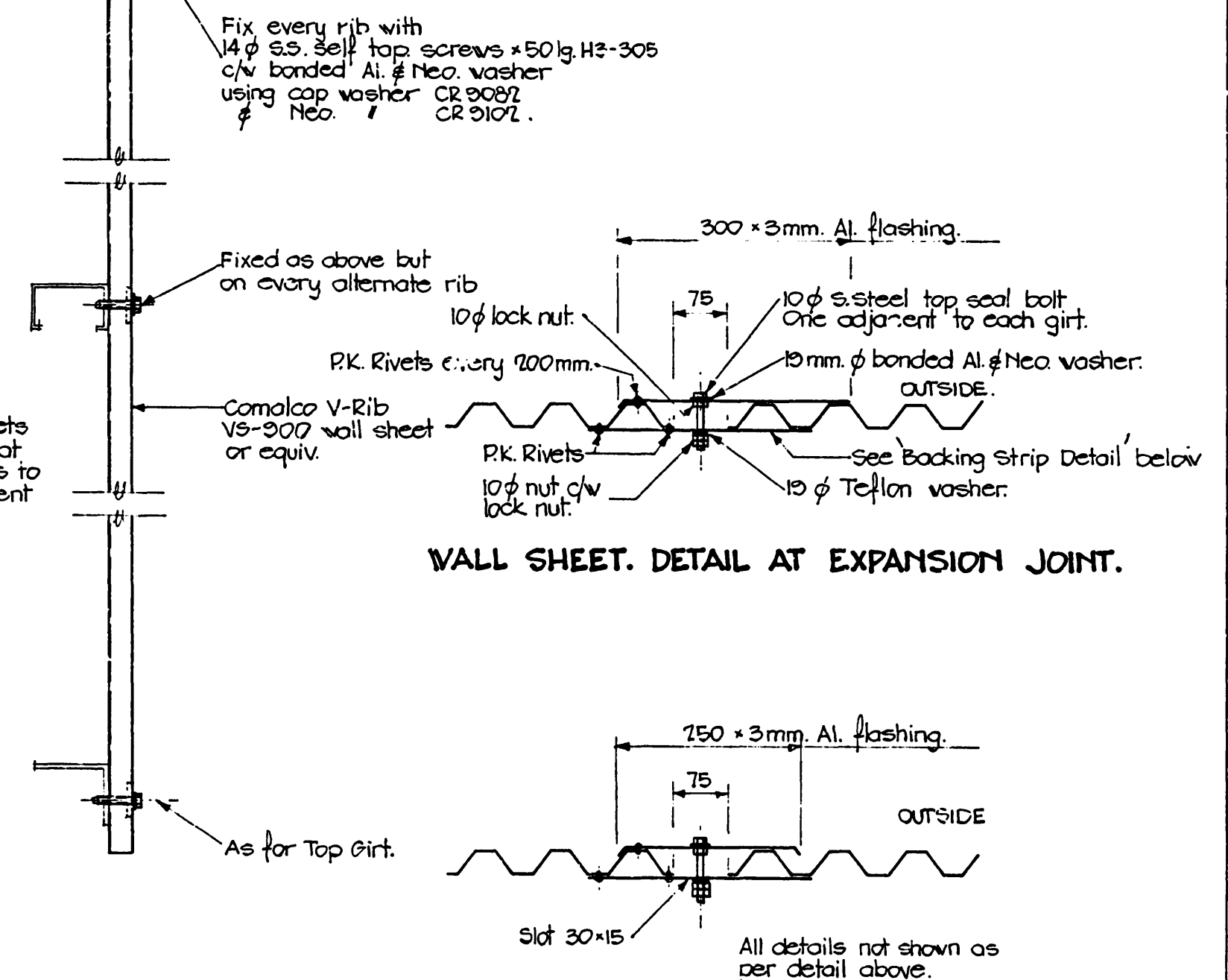
FIBRE GLASS STITCHING:
Fix at 100mm. crs. using 5mm. ϕ P.K. rivets with washers front & back of sheeting.

CORNERS:
Fix of every rib where noted above & adjacent
to head & tail end structures.

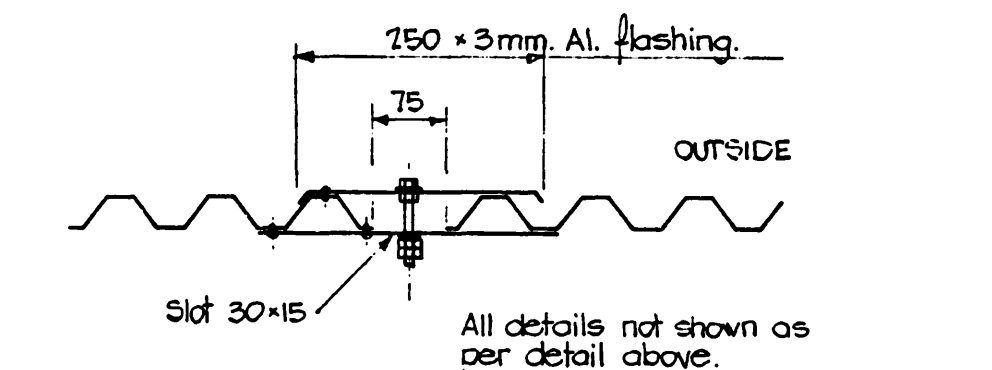
P.K. RIVETS :
Denotes:-Aluminium self sealing 5mm. ϕ rivets
used in conjunction with 12mm. ϕ Al. & Neo. bonded
washers. Sizes -ADG4, G6, & G8 to suit.



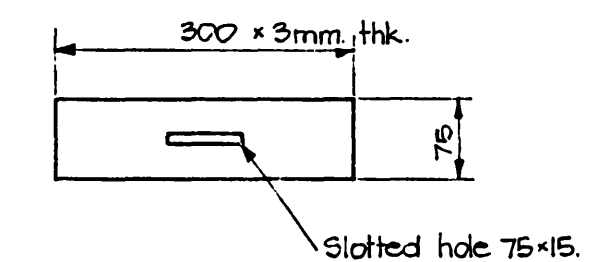
TYP. SKILLION FLASHING JOINT.



WALL SHEET. DETAIL AT EXPANSION JOINT.



WALL SHEET. DETAIL AT STANDARD JOINT.



BACKING STRIP DETAIL.
Provide Backing Strip adjacent to each girt.

THE CORPORATION OF THE TREASURER OF QUEENSLAND							
No.	DATE	AMENDMENTS					
C.S.R. REFERENCE No.							

**THE SUGAR BOARD
BULK SUGAR TERMINALS**

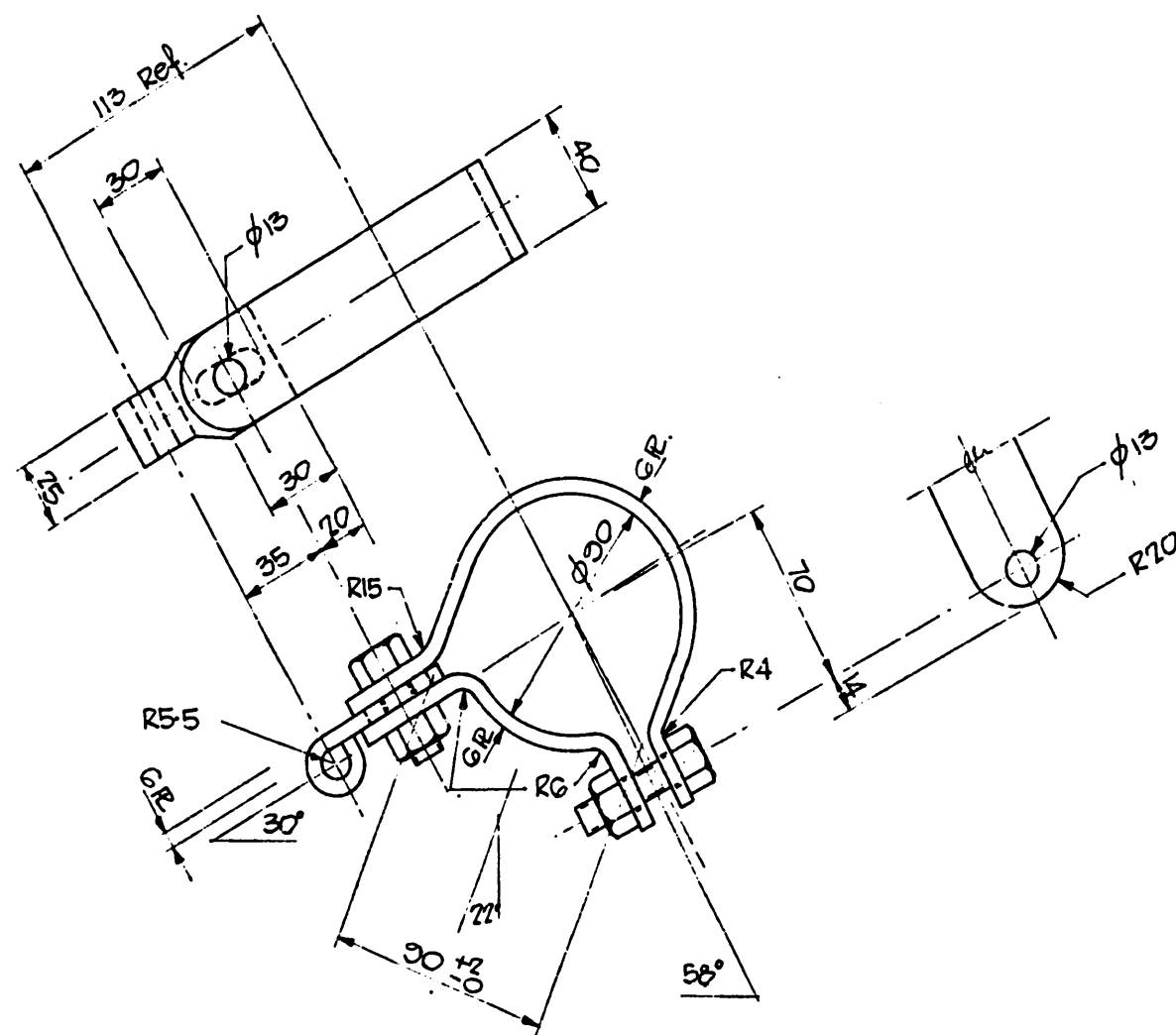
SCALES 1:5	B.S.T. LUCINDA STAGE III	DRAWING NO.
DRAWN K.G. HOLMES	APPROACH JETTY	CTQ1122/S/021
PASSED [signature]	GALLERY FLASHING DETAILS	
DATE AUG 76		

MACDONALD WAGNER & PRIDDLE PTY.LTD.

CONSULTING ENGINEERS 313 ST. PAULS TERRACE PORTLAND VALLEY, QLD 4086 PH 53 7123

IN ASSOCIATION WITH

C.S.R. LIMITED



Return Idler Brackets to be supplied & welded to stringer support leg under sub-contract No 4C.

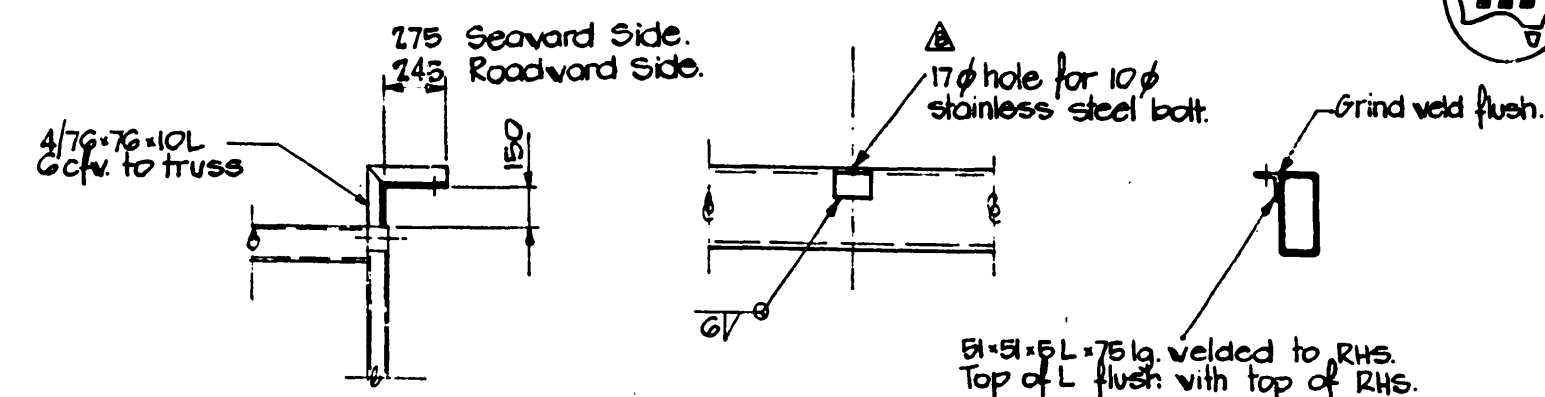
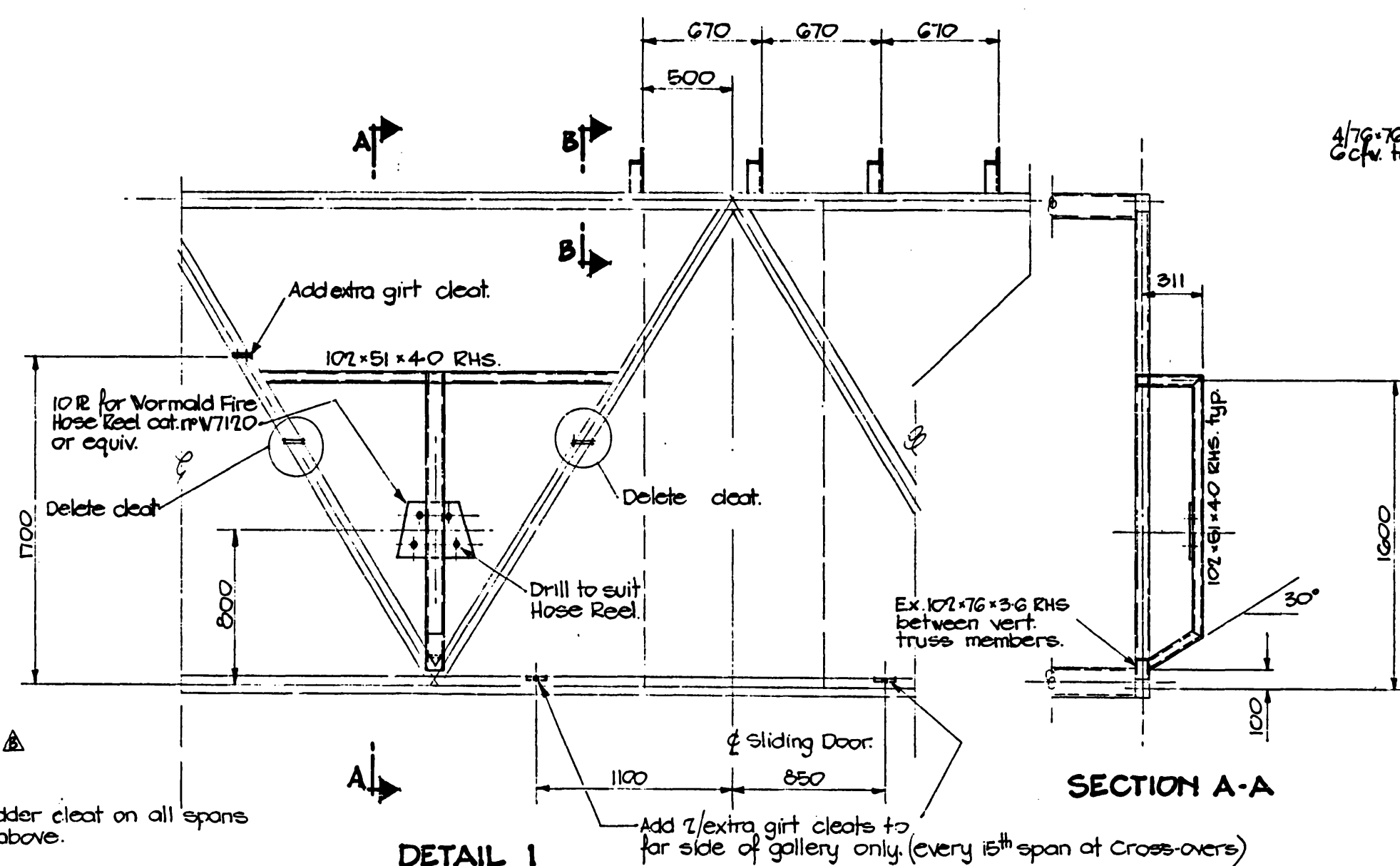


HOLD Y

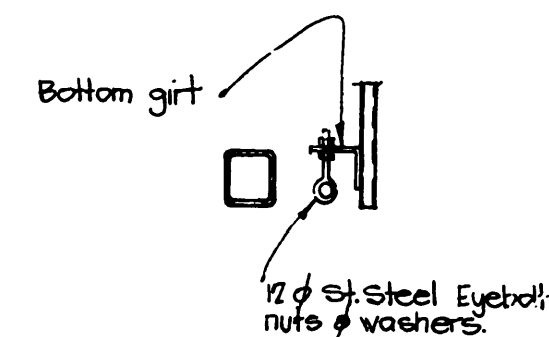


IN ASSOCIATION WITH
CSR LIMITED





CABLE LADDER FIXING CLEAT.



SECTION C-C. 
ANODE SUPPORT.

DETAIL 1

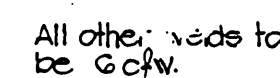
Expansion Joint

Anchored Bent

Expansion Joint

• Fixed End
○ Free End

For location of Anchored bents & expansion joints - Refer dwg. no CTQ1171/S/003 & 004.



TOP CHORD

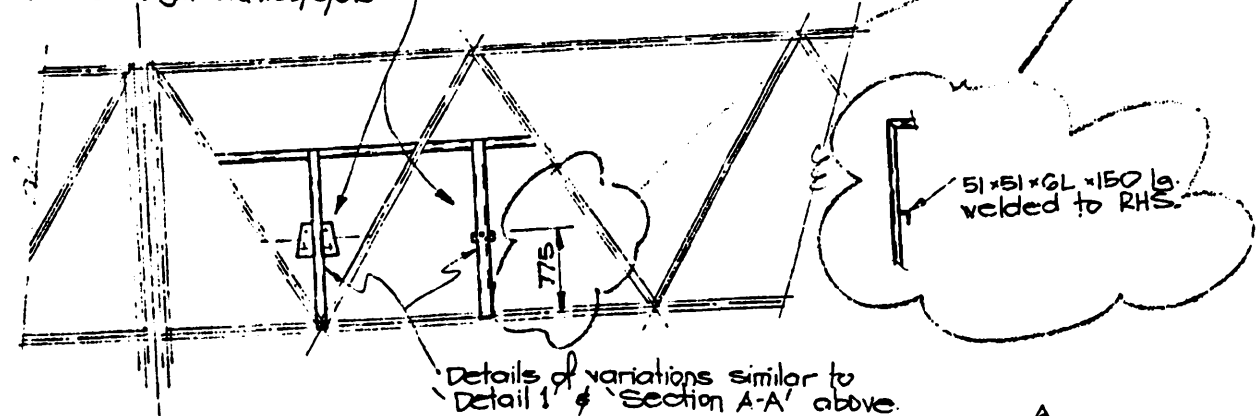
SIDE ELEVATION

BOTTOM CHORD.

WELD DETAILS.

NOTE: All other details as per
standard truss Drg. no C121122/5/018

NA For positions of Hose Reels &
Fire Hydrants see Drg No CTQ 1122/S/012



VARIATIONS TO INCLINED GALLERIES

GENERAL NOTES.

1. All exposed corners to be ground smooth to a min. radius of 1mm.
2. Welding, grinding & alignment of the entire gallery section & conv. stringers to be completed prior to commencement of surface preparations & painting.
3. Final painting of structure to be completed prior to the installation of aluminium girts, purlins, floors, cladding, cable trays etc.

FABRICATION NOTES.

CHORDS.

Butt welds in all 102x102 RHS to be qualified complete penetration butt welds made over an internal backing ring. Butt welds to be located not more than 500mm from the intersection of 102x76 web diagonals & will be permitted only outside the middle third of the truss spans.

SIDE WEB MEMBERS

Fillet welds to face of chords - Gmm. cont.
Welds to edge of chords - Detail 1 drg. no. CTA1127/5/018
Welds at intersection of web members - Detail 1 drg. no. CTA1127/5/018
Grind flush where necessary to permit assembly of top & bottom chord cross members.

TOP & BOTTOM CHORD CROSS MEMBERS.

Welds to faces of side trusses-Gmm. cont. fillet.
Weld to edges of top & bottom chord - Detail 1 drg. no CTQ1177/5/018

TOP & BOTTOM CHORD BRACING

Cont. 5mm. fillet to all connections except where welds are made to the exposed thickness of members, weld size shall be limited to the member thickness.

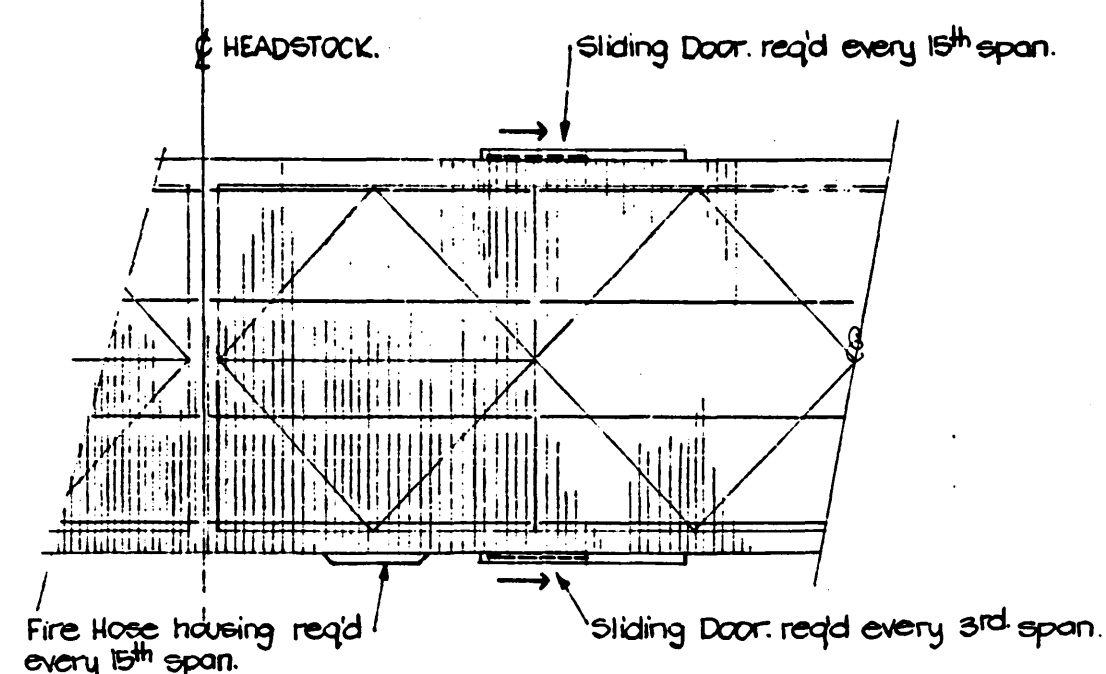
END FRAMES.

For welding of corners refer Detail 2 drg. no CTQ1122/S/018

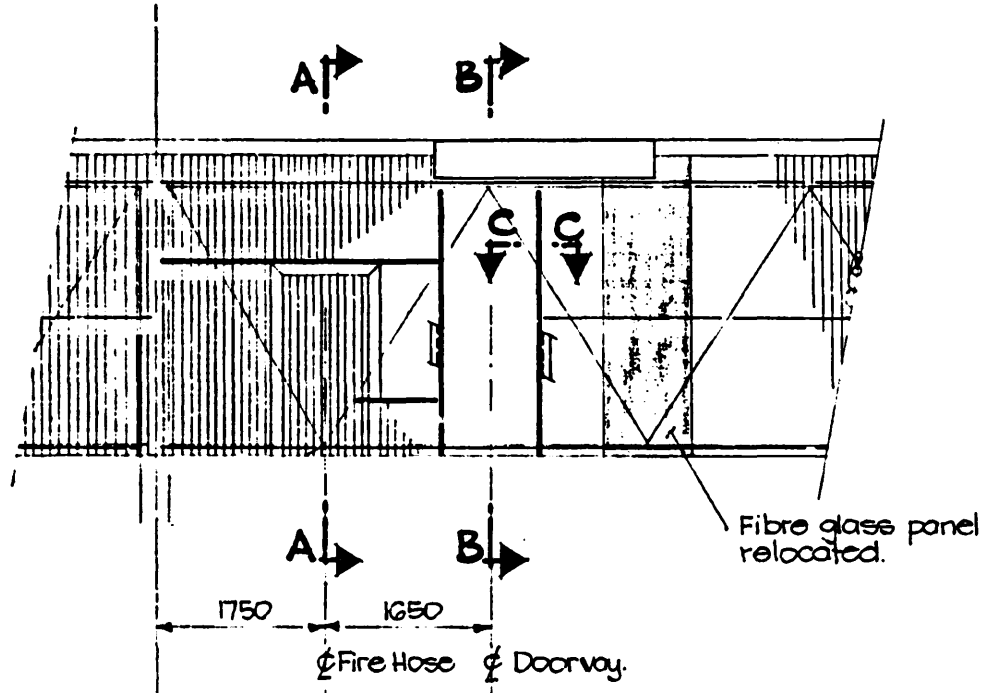
GENERAL

All RHS shall conform to AS1163. Gd. 250 shall be free from all oil or paint during fabrication. Ends of all members shall be saw cut. Incorrect fit-up of joints will subject the member to rejection by the Engineer. Alternatively the Engineer may require fillet weld sizes to be increased to compensate for incorrect fill-up of joints. Refer Specification.

THE CORPORATION OF THE TREASURER OF QUEENSLAND									
NO	DATE	AMENDMENTS		THE SUGAR BOARD BULK SUGAR TERMINALS SCALES 1:100 1:20 DRAWN E.G. HOLMES PASSED <i>mm</i> DATE <i>14/6/72</i> B.S.T. LUCINDA STAGE III APPROACH JETTY. VARIATIONS TO STD. GALLERY DRAWING NO CTQ1122/5/024					
A	4-1-77	Diagonal brace omitted.							
B	24-2-77	HOLE DIA. ALTERED & TRA BRACKETS ADDED. VARIATIONS TO INCLUDE GALLERIES APPROD.		MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST. PAULS TERRACE PORTSMOUTH VALLEY, QLD. 4006 PH. 317122 IN ASSOCIATION WITH C.S.R. LIMITED					
				C.S.R. REFERENCE NO. 1 2 3 4 5 6 7 8 9 10					

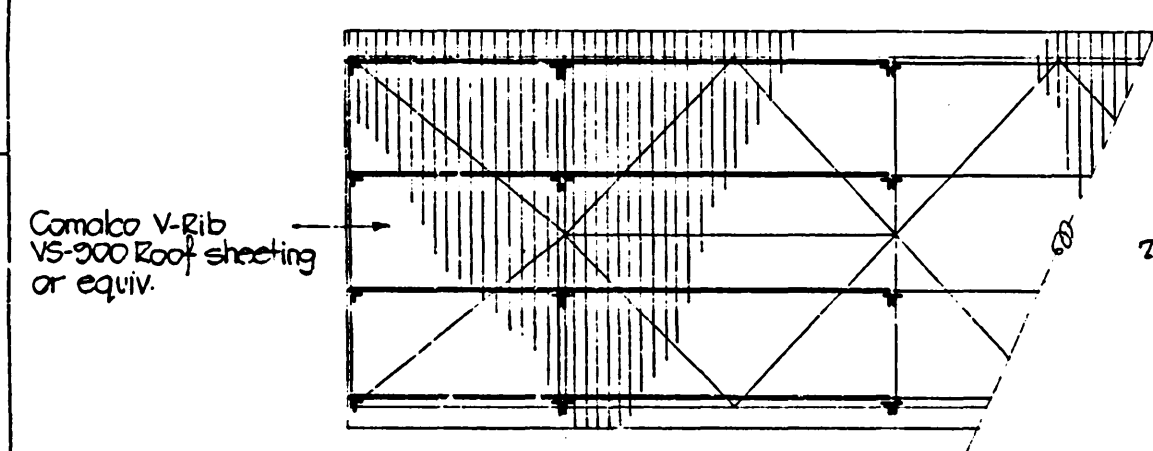


ROOF PLAN

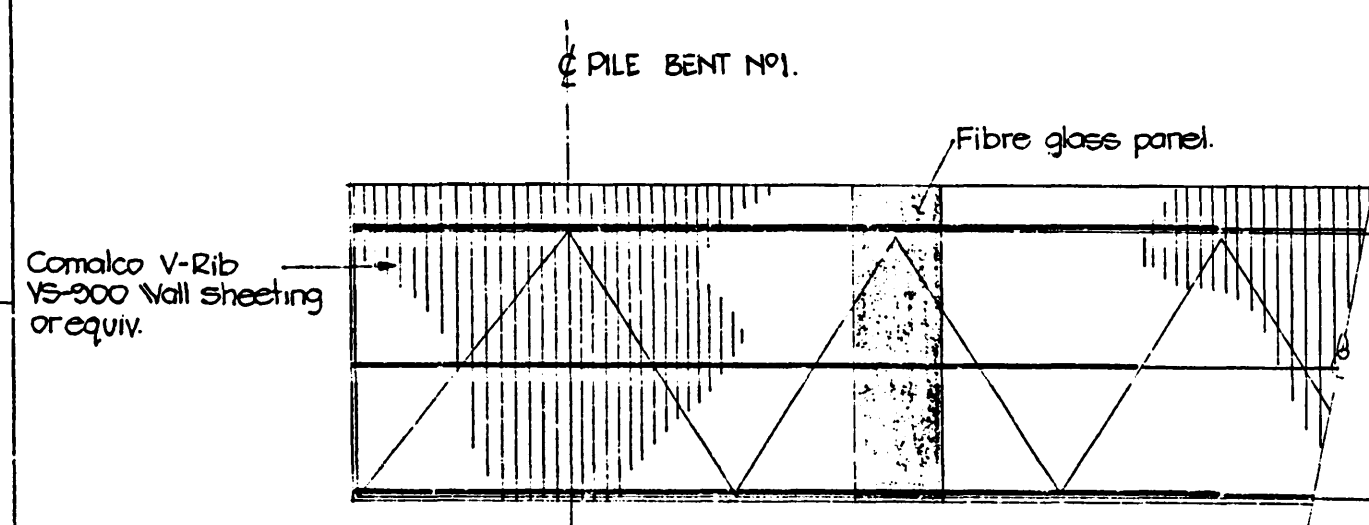


SIDE ELEVATION. (ROADWAY SIDE)

VARIATIONS TO STANDARD TRUSS.

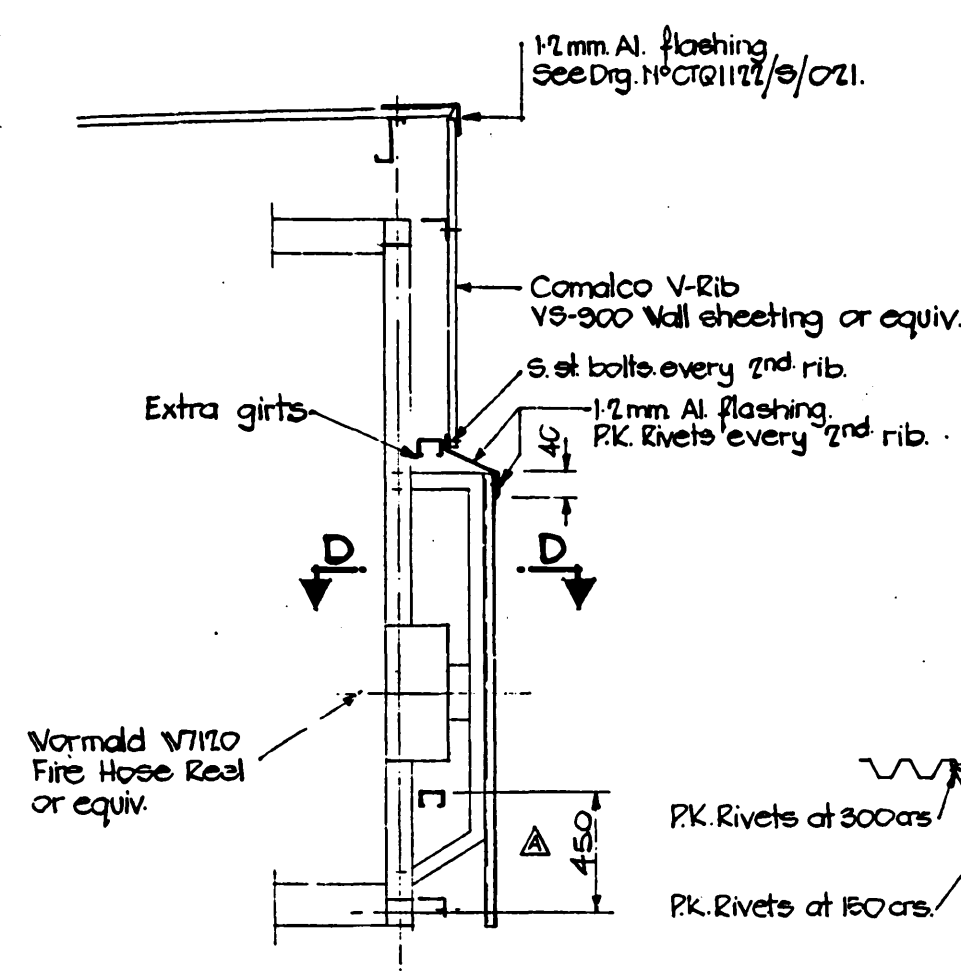


ROOF PLAN.

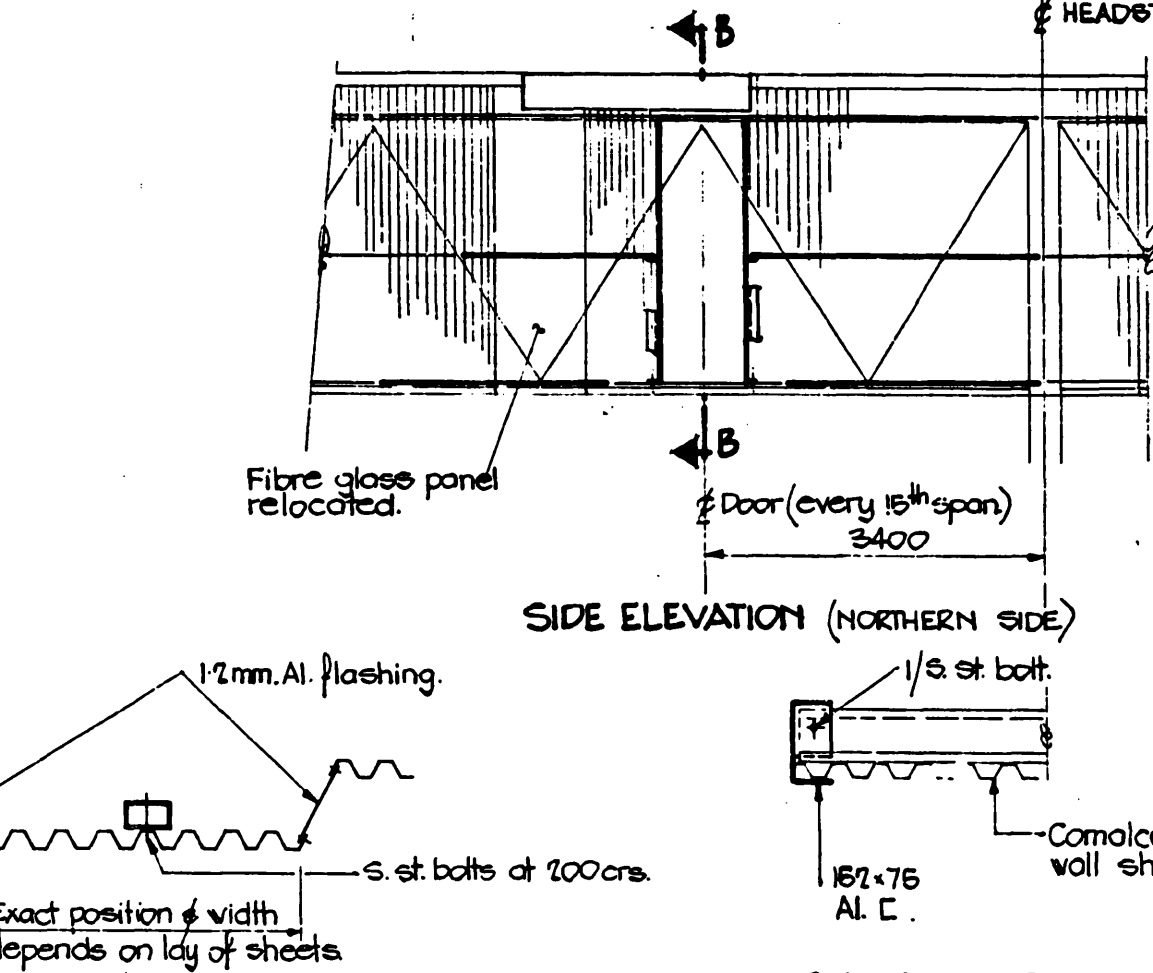


SIDE ELEVATION.

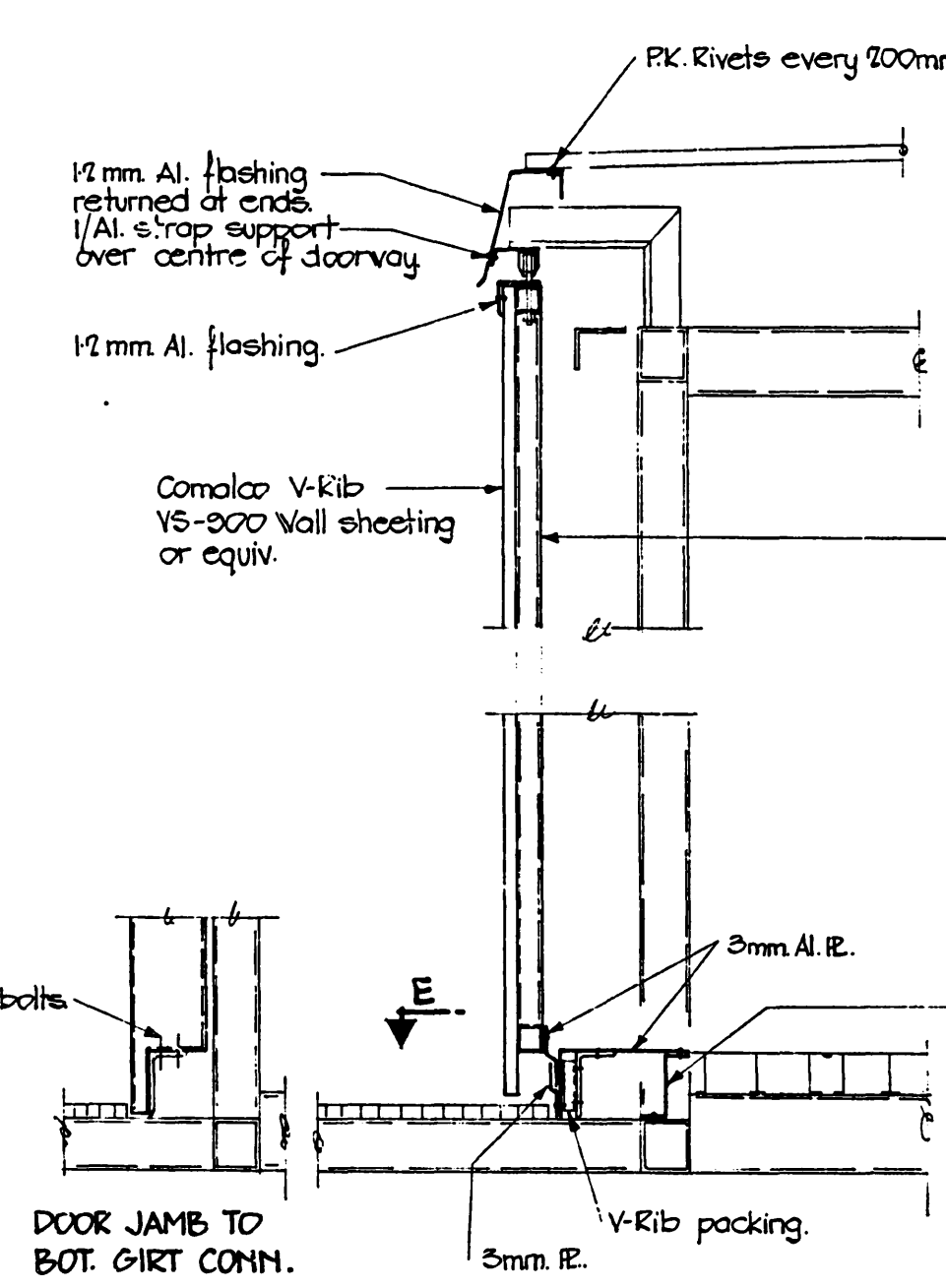
VARIATIONS TO TAIL END TRUSS.



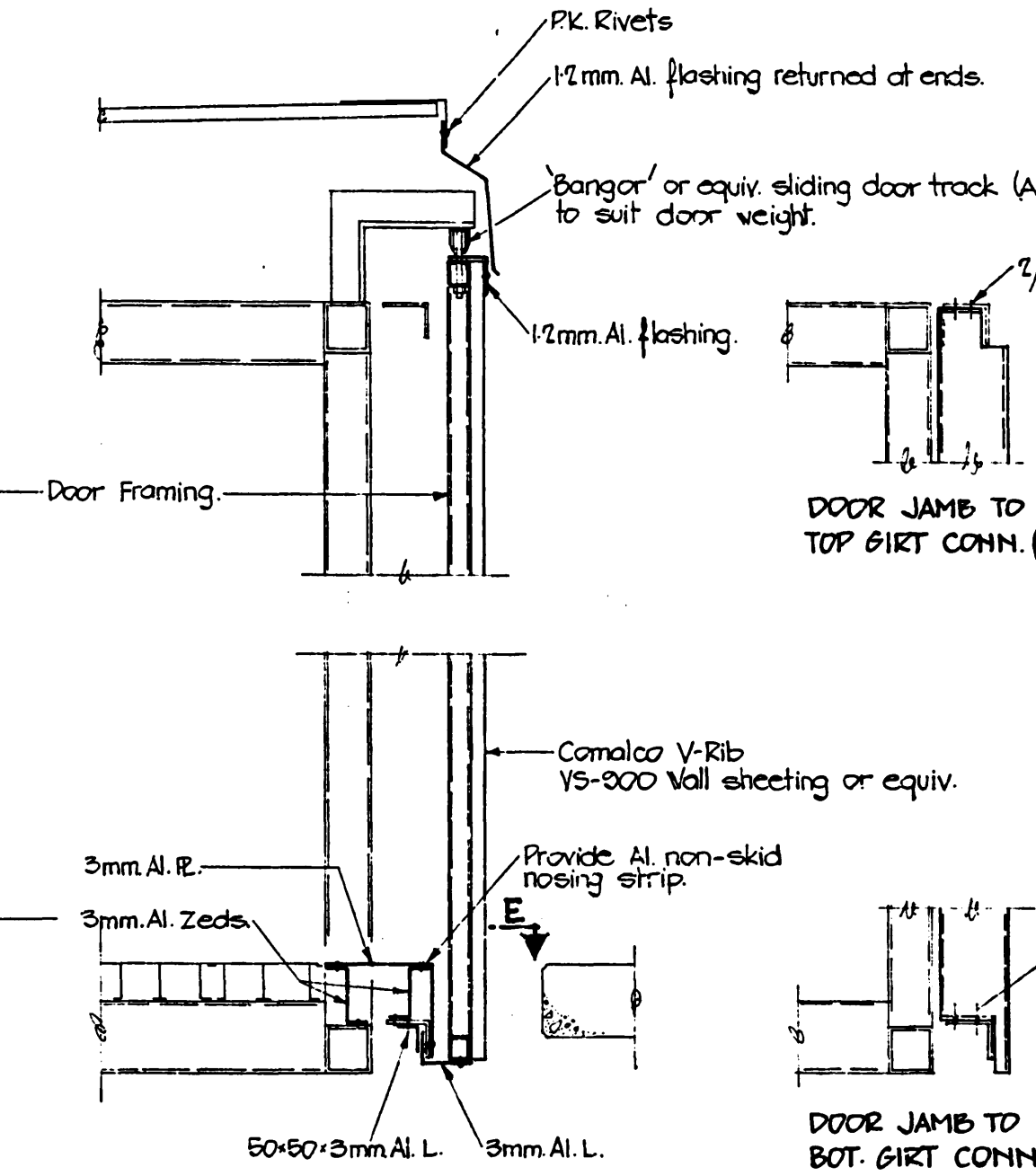
SECTION A-A



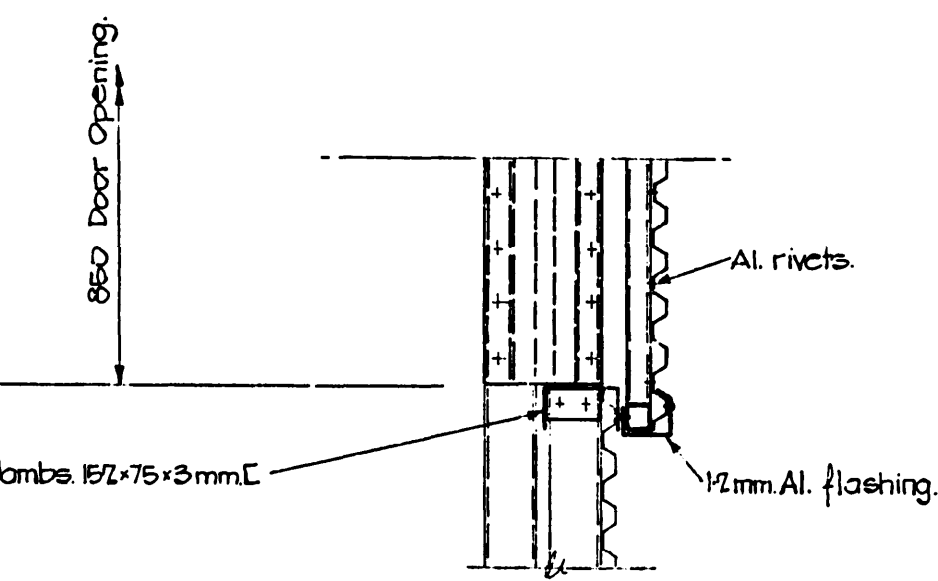
SECTION D-D.



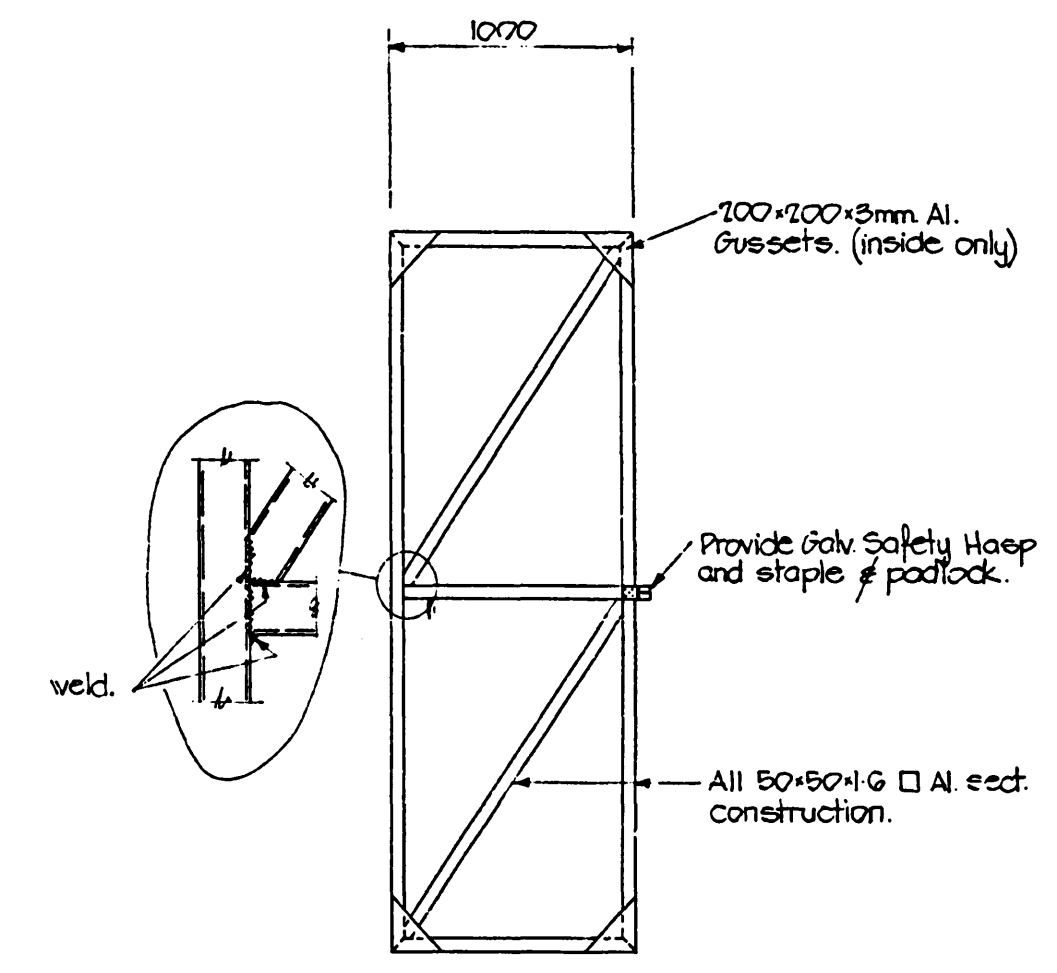
SECTION B-B



SECTION C-C.

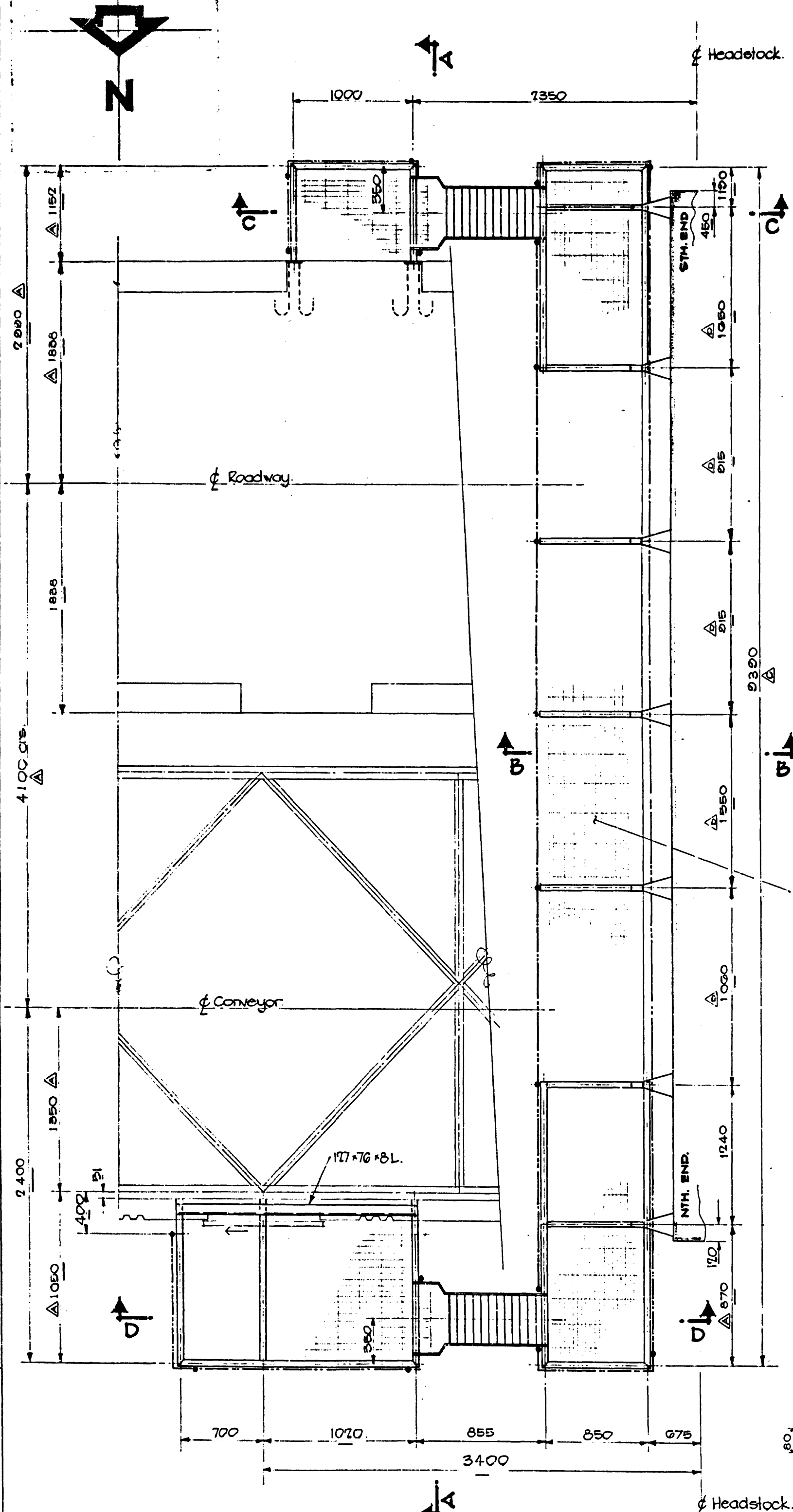


PLAN SECT. E-E.



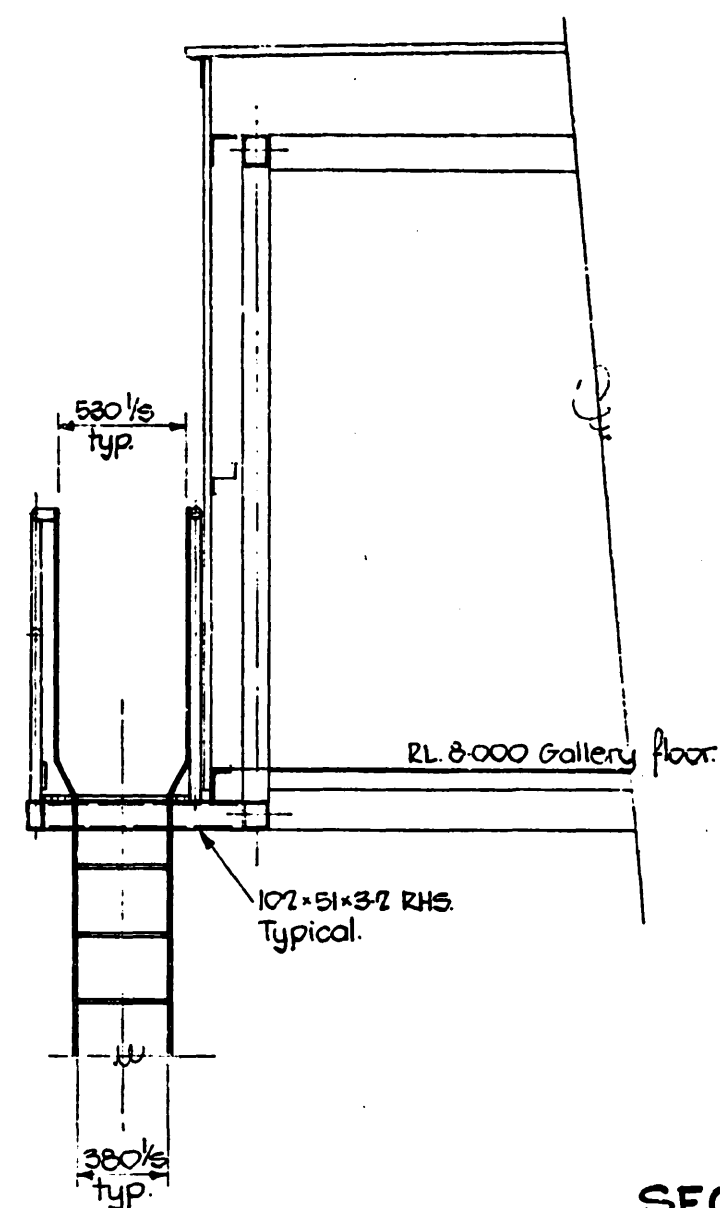
AL. SLIDING DOOR (INSIDE VIEW)

THE CORPORATION OF THE TREASURER OF QUEENSLAND									
NO.	DATE	AMENDMENTS							
A	28.1.77	EXTRA GIRT ADDED.							

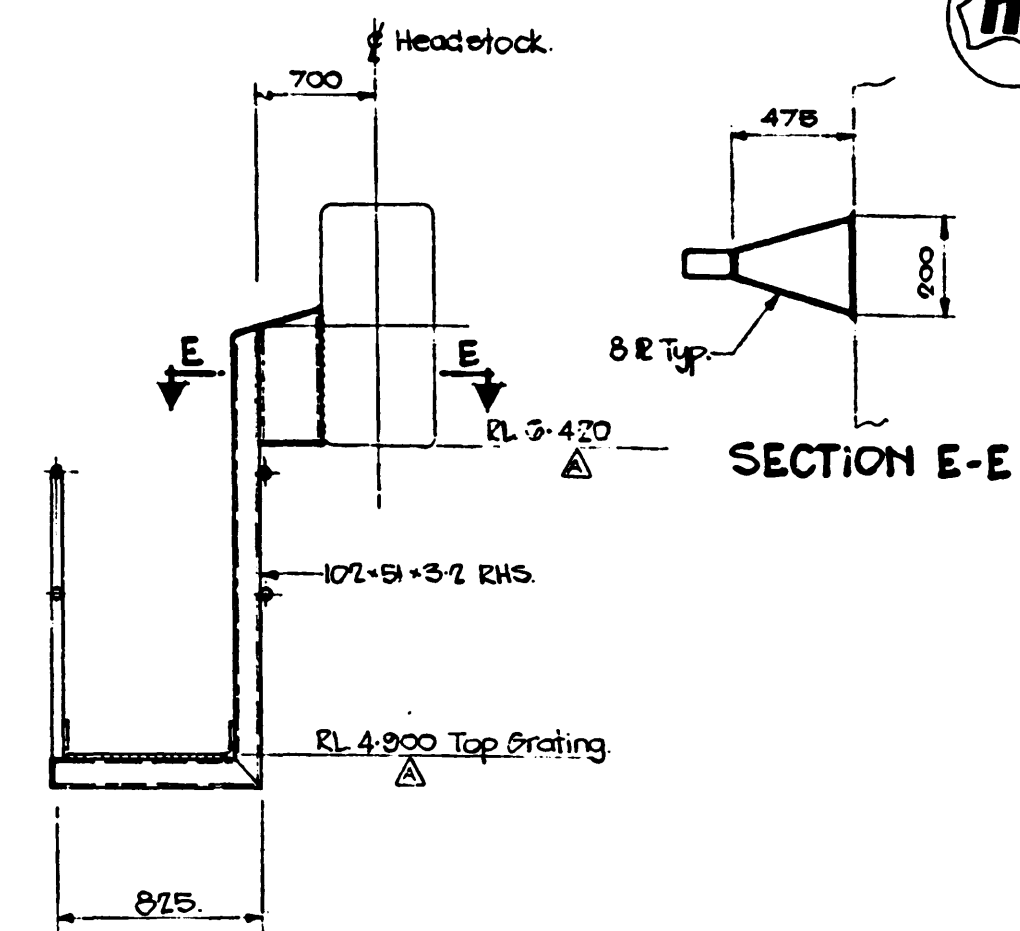
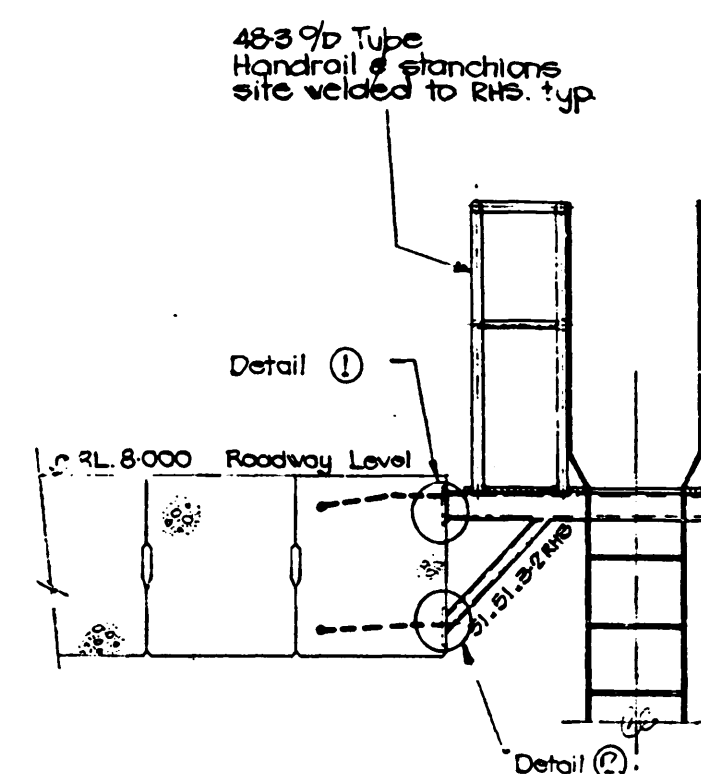


PLAN ON CROSS-OVER.

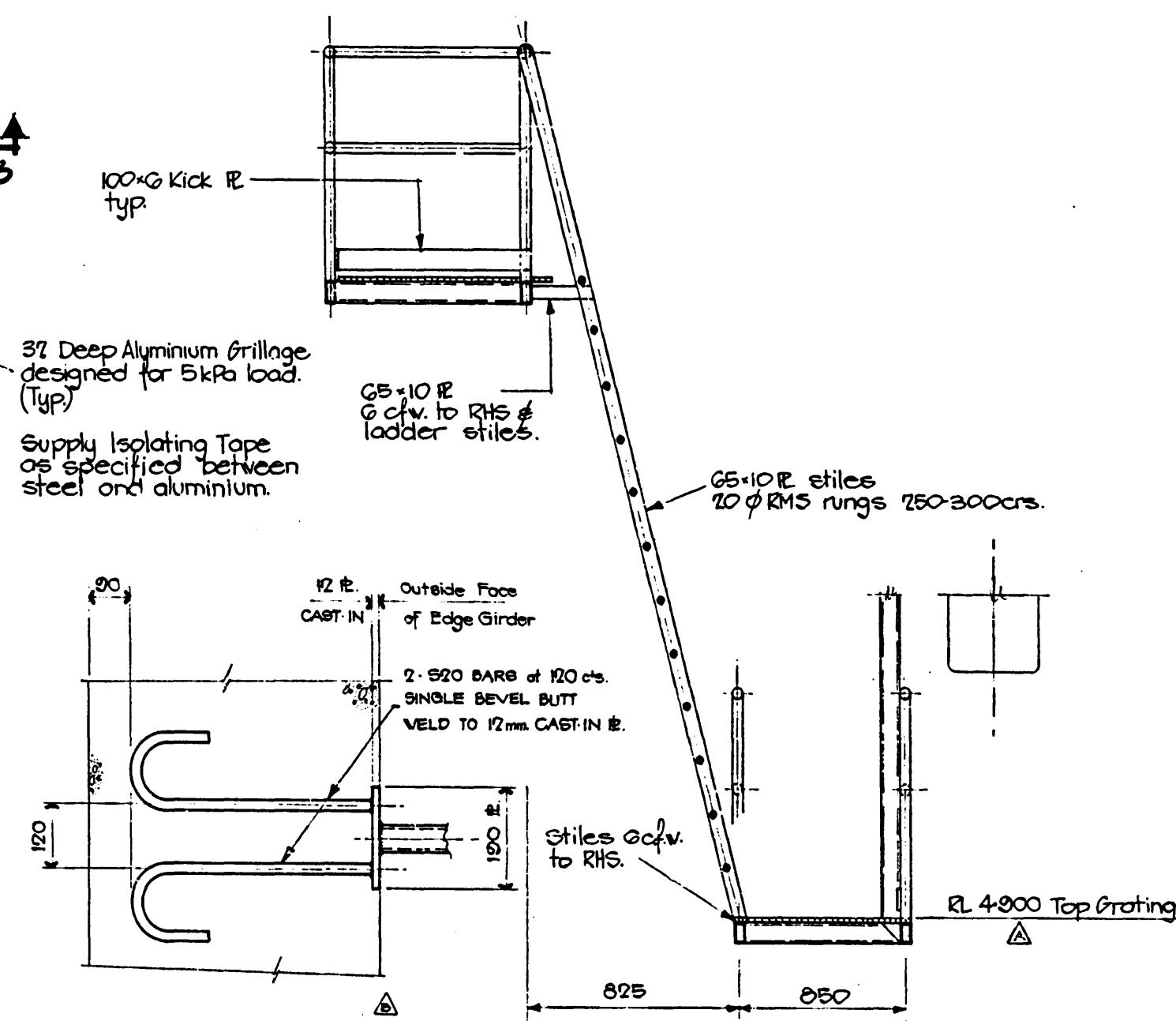
For locations refer 'Super Structure Layout Sh. i #2'.



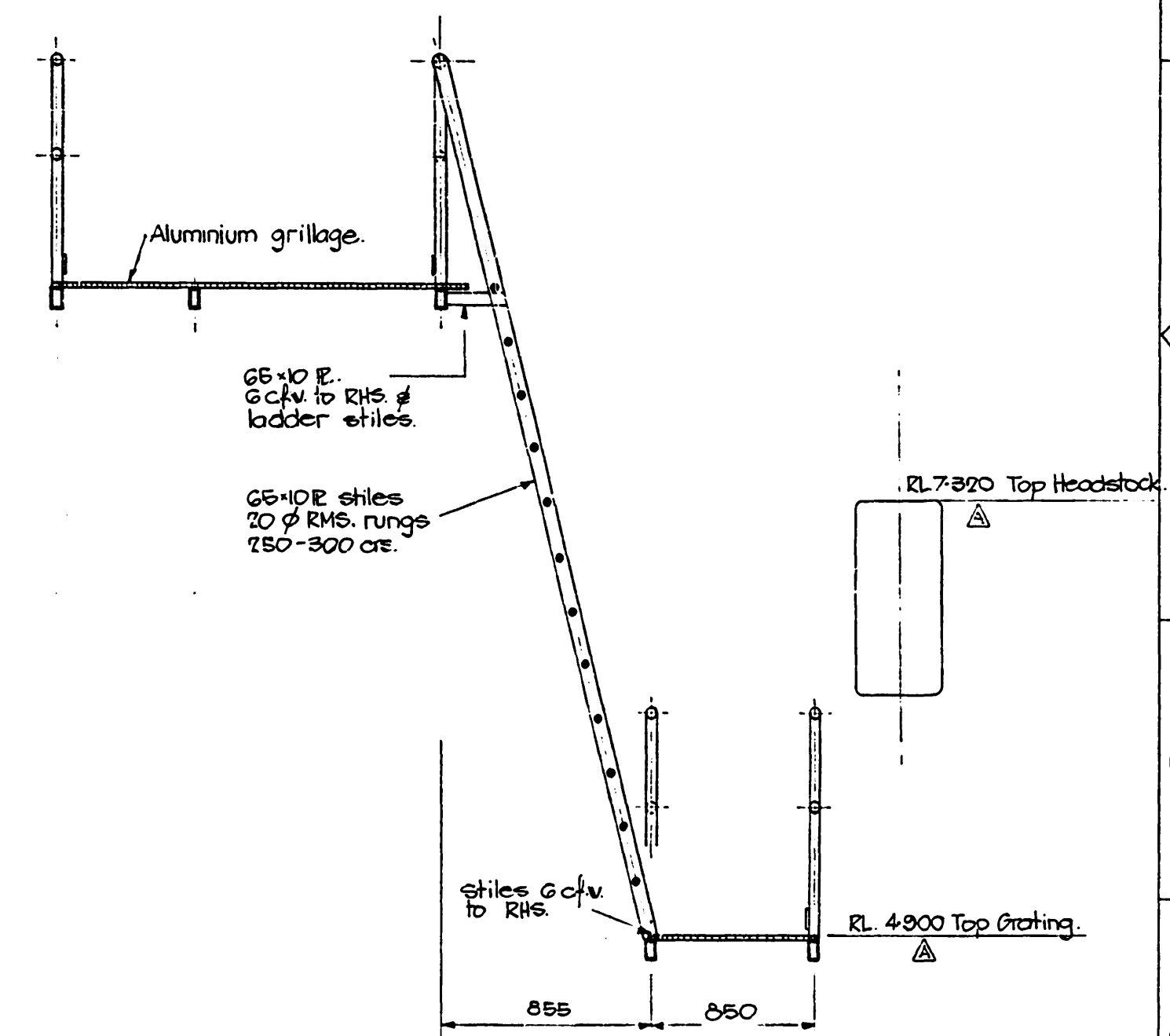
SECTION A-A



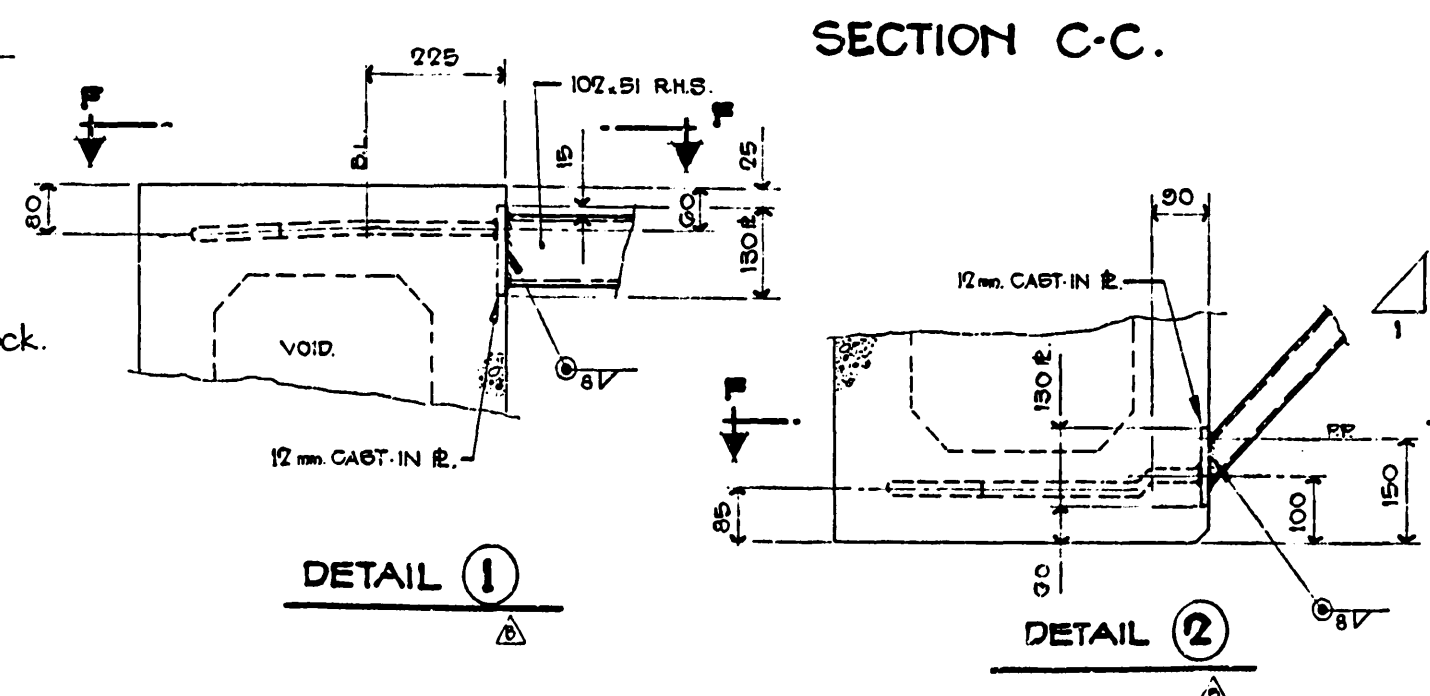
SECTION B-B



SECTION F.F

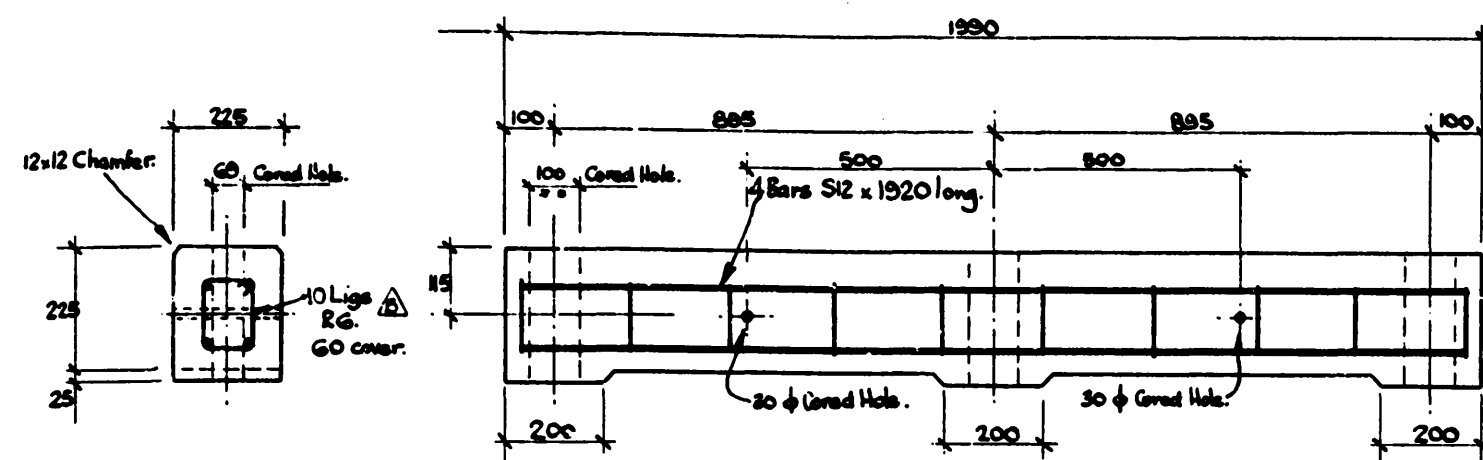


SECTION D-D



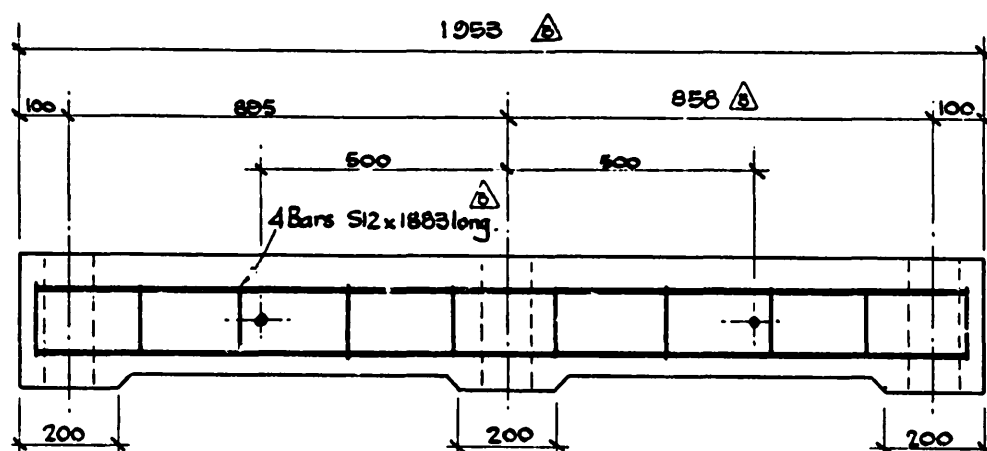
SECTION C-C.

THE CORPORATION OF THE TREASURER OF QUEENSLAND			
№	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS
A	20-1-77	Drw. revised for conc. roadway girder design	SCALES 1:20 DRAWN R & HOLMES PASSED <i>may</i> DATE <i>Aug 76</i> B.S.T. LUCINDA STAGE III APPROACH JETTY CROSS-OVER DETAILS. DRAWING NO. CTQ127/5026 E
B	2-2-77	Details 1 & 2 amended for Cost-in & with Tendons. Section FF added.	
C	26-5-77	9/A DIMENSION ALTERED.	
D	25-10-77	Walkway Hangers Relocated.	
E	27-10-77	Walkway Hangers Extended to clear Piles at extreme tolerance position	
C.S.R. REFERENCE NO.			MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 212 ST PAULS TERRACE, PORTHURST VALLEY, QLD. 4050. PH. 32 7121 IN ASSOCIATION WITH C.S.R. LIMITED



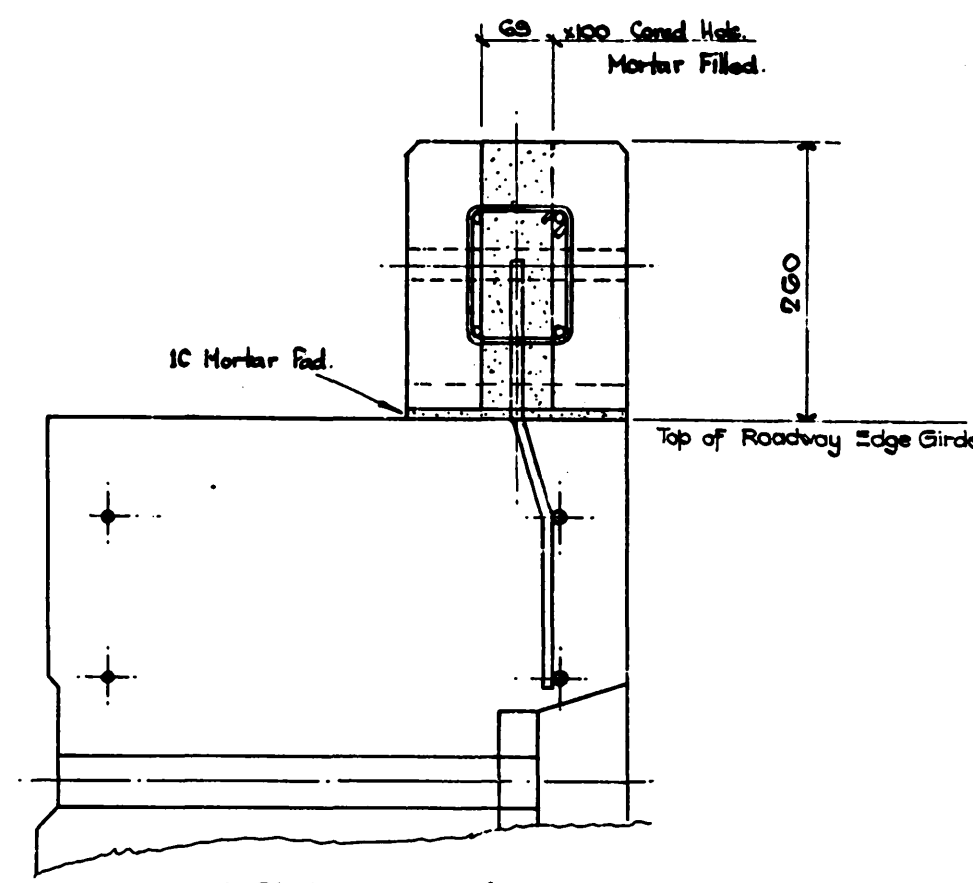
STANDARD KERB UNIT.

1:10.



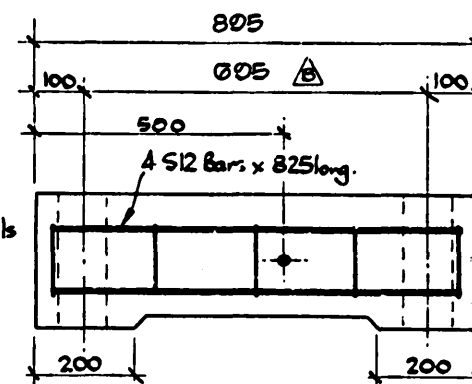
KERB UNIT AT EXPANSION JOINT.

1:10.



ASSEMBLY OF KERB TO DECK UNIT.

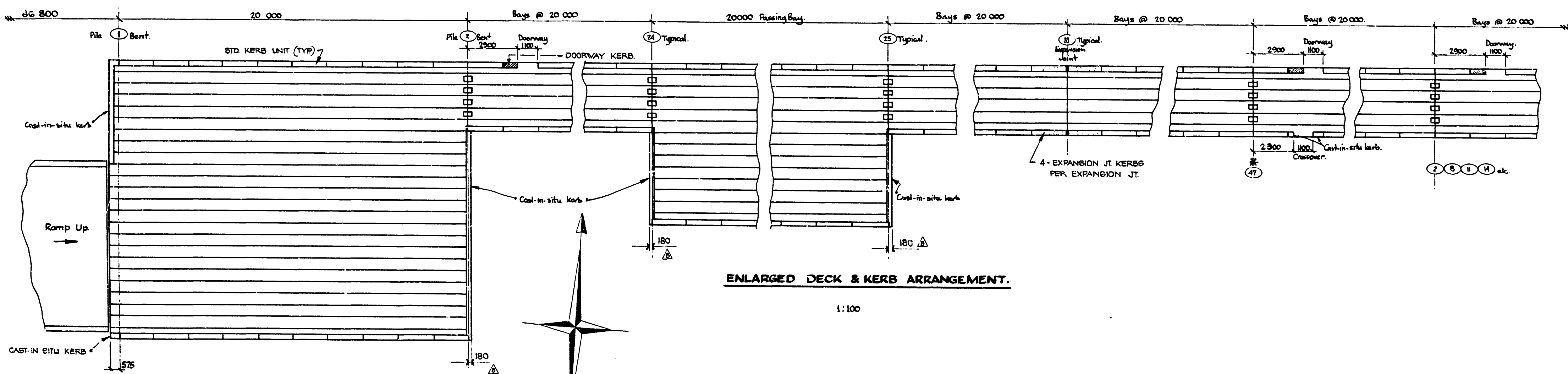
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DOORWAY KERB.

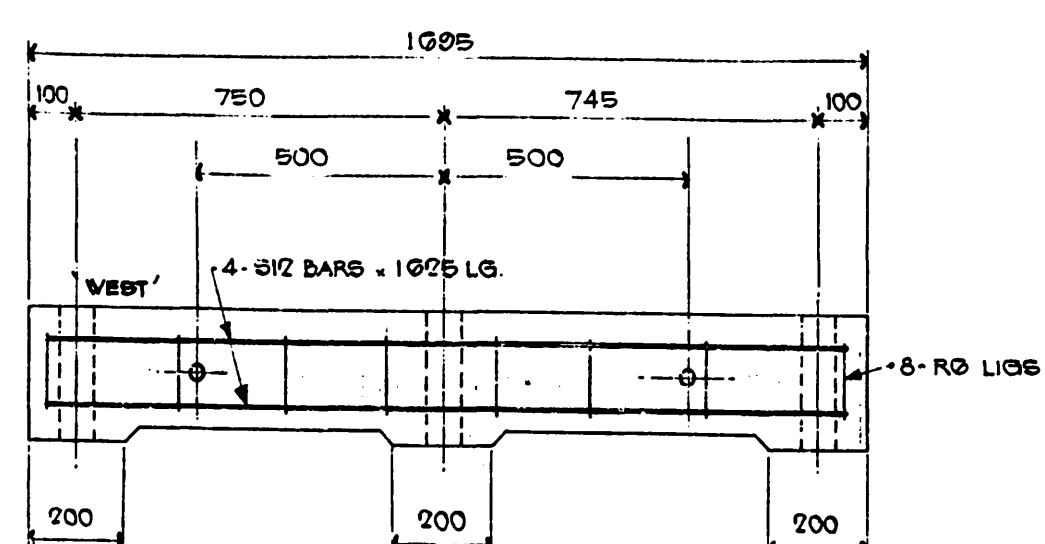
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Bay No.	Type	Bay No.	Type	Bay No.	Type	Bay No.	Type	Bay No.	Type	Bay No.	Type	Bay No.	Type	Bay No.	Type
1	Expansion	51	Standard	101	Doorway	151	Expansion	201	Standard	251	Doorway	301	Expansion	351	Standard
2	Doorway	52	Standard	102	Expansion	152	Doorway	202	Expansion	252	Standard	302	Doorway	352	Expansion
3	Standard	53	Doorway	103	Standard	153	Expansion	203	Doorway	253	Expansion	303	Standard	353	Doorway
4	Expansion	54	Expansion	104	Doorway	154	Standard	204	Expansion	254	Doorway	304	Expansion	354	Standard
5	Doorway	55	Standard	105	Expansion	155	Doorway	205	Standard	255	Expansion	305	Doorway	355	Expansion
6	Expansion	56	Doorway	106	Standard	156	Expansion	206	Doorway	256	Standard	306	Expansion	356	Doorway
7	Standard	57	Expansion	107	Doorway	157	Standard	207	Expansion	257	Doorway	307	Expansion	357	Standard
8	Doorway	58	Standard	108	Expansion	158	Doorway	208	Standard	258	Expansion	308	Doorway	358	Expansion
9	Expansion	59	Doorway	109	Standard	159	Expansion	209	Doorway	259	Standard	309	Expansion	359	Doorway
10	Standard	60	Expansion	110	Doorway	160	Standard	210	Expansion	260	Doorway	310	Expansion	360	Standard
11	Doorway	61	Standard	111	Expansion	161	Doorway	211	Standard	261	Expansion	311	Doorway	361	Expansion
12	Expansion	62	Doorway	112	Standard	162	Expansion	212	Doorway	262	Standard	312	Expansion	362	Doorway
13	Standard	63	Expansion	113	Doorway	163	Standard	213	Expansion	263	Doorway	313	Standard	363	Expansion
14	Doorway	64	Standard	114	Expansion	164	Doorway	214	Standard	264	Expansion	314	Doorway	364	Standard
15	Expansion	65	Doorway	115	Standard	165	Expansion	215	Doorway	265	Standard	315	Expansion	365	Doorway
16	Standard	66	Expansion	116	Doorway	166	Standard	216	Expansion	266	Doorway	316	Standard	366	Expansion
17	Doorway	67	Standard	117	Expansion	167	Doorway	217	Standard	267	Expansion	317	Doorway	367	Standard
18	Expansion	68	Doorway	118	Standard	168	Expansion	218	Doorway	268	Standard	318	Expansion	368	Doorway
19	Standard	69	Expansion	119	Doorway	169	Standard	219	Expansion	269	Doorway	319	Standard	369	Expansion
20	Doorway	70	Standard	120	Expansion	170	Doorway	220	Standard	270	Expansion	320	Doorway	370	Standard
21	Expansion	71	Doorway	121	Standard	171	Expansion	221	Doorway	271	Standard	321	Expansion	371	Doorway
22	Standard	72	Expansion	122	Doorway	172	Standard	222	Expansion	272	Doorway	322	Standard	372	Expansion
23	Doorway	73	Standard	123	Expansion	173	Doorway	223	Standard	273	Expansion	323	Doorway	373	Standard
24	Expansion	74	Doorway	124	Standard	174	Expansion	224	Doorway	274	Standard	324	Expansion	374	Doorway
25	Standard	75	Expansion	125	Doorway	175	Standard	225	Expansion	275	Doorway	325	Standard	375	Expansion
26	Doorway	76	Standard	126	Expansion	176	Doorway	226	Standard	276	Expansion	326	Doorway	376	Standard
27	Expansion	77	Doorway	127	Standard	177	Expansion	227	Doorway	277	Standard	327	Expansion	377	Doorway
28	Standard	78	Expansion	128	Doorway	178	Standard	228	Expansion	278	Doorway	328	Standard	378	Expansion
29	Doorway	79	Standard	129	Expansion	179	Doorway	229	Standard	279	Expansion	329	Doorway	379	Standard
30	Expansion	80	Doorway	130	Standard	180	Expansion	230	Doorway	280	Standard	330	Expansion	380	Doorway
31	Standard	81	Expansion	131	Doorway	181	Standard	231	Expansion	281	Doorway	331	Standard	381	Expansion
32	Doorway	82	Standard	132	Expansion	182	Doorway	232	Standard	282	Expansion	332	Doorway	382	Standard
33	Expansion	83	Doorway	133	Standard	183	Expansion	233	Doorway	283	Standard	333	Expansion	383	Doorway
34	Standard	84	Expansion	134	Doorway	184	Standard	234	Expansion	284	Doorway	334	Standard	384	Expansion
35	Doorway	85	Standard	135	Expansion	185	Doorway	235	Standard	285	Expansion	335	Doorway	385	Standard
36	Expansion	86	Doorway	136	Standard	186	Expansion	236	Doorway	286	Standard	336	Expansion	386	Doorway
37	Standard	87	Expansion	137	Doorway	187	Standard	237	Expansion	287	Doorway	337	Standard	387	Expansion
38	Doorway	88	Standard	138	Expansion	188	Doorway	238	Standard	288	Expansion	338	Doorway	388	Standard
39	Expansion	89	Doorway	139	Standard	189	Expansion	239	Doorway	289	Standard	339	Expansion	389	Doorway
40	Standard	90	Expansion	140	Doorway	190	Standard	240	Expansion	290	Doorway	340	Standard	390	Expansion
41	Doorway	91	Standard	141	Expansion	191	Doorway	241	Standard	291	Expansion	341	Doorway	391	Standard
42	Expansion	92	Doorway	142	Standard	192	Expansion	242	Doorway	292	Standard	342	Expansion	392	Doorway
43	Standard	93	Expansion	143	Doorway	193	Standard	243	Expansion	293	Doorway	343	Standard	393	Expansion
44	Doorway	94	Standard	144	Expansion	194	Doorway	244	Standard	294	Expansion	344	Doorway	394	Standard
45	Expansion	95	Doorway	145	Standard	195	Expansion	245	Doorway	295	Standard	345	Expansion	395	Doorway
46	Standard	96	Expansion	146	Doorway	196	Standard	246	Expansion	296	Doorway	346	Standard	396	Expansion
47	Doorway	97	Standard	147	Expansion	197	Doorway	247	Standard	297	Expansion	347	Doorway	397	Standard
48	Expansion	98	Doorway	148	Standard	198	Expansion	248	Doorway	298	Standard	348	Expansion	398	Doorway
49	Standard	99	Expansion	149	Doorway	199	Standard	249	Expansion	299	Doorway	349	Standard	399	Expansion
50	Doorway	100	Standard	150	Expansion	200	Doorway	250	Standard	300	Expansion	350	Doorway	400	Standard



ENLARGED DECK & KERB ARRANGEMENT.

1:100



SPECIAL KERB UNIT.

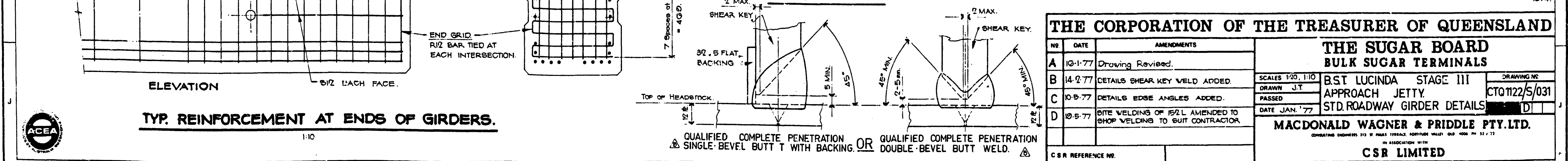
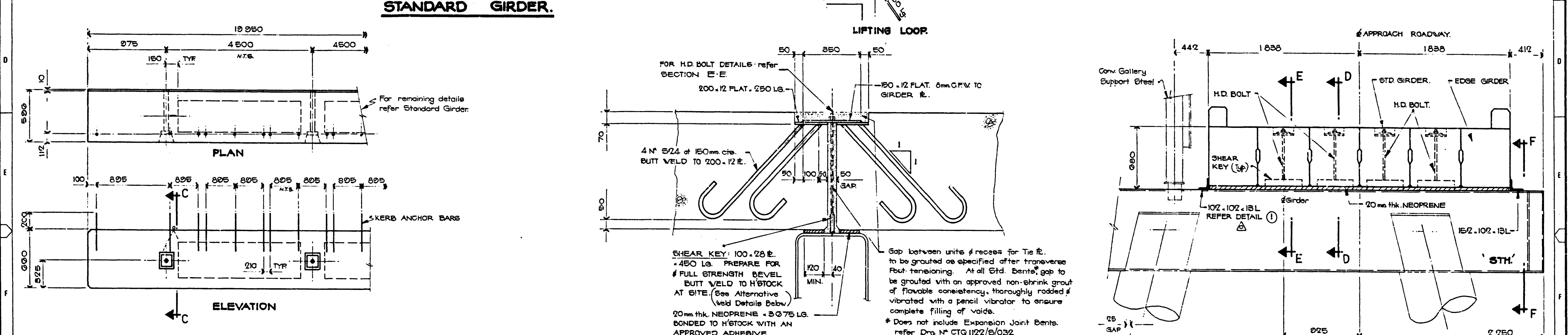
FOR PILE BENT NO 284 ONLY.

1:10

NOTES:

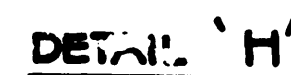
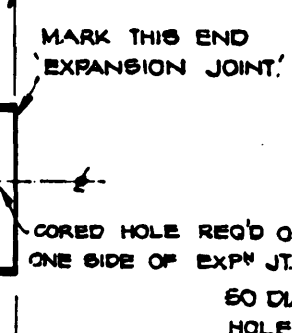
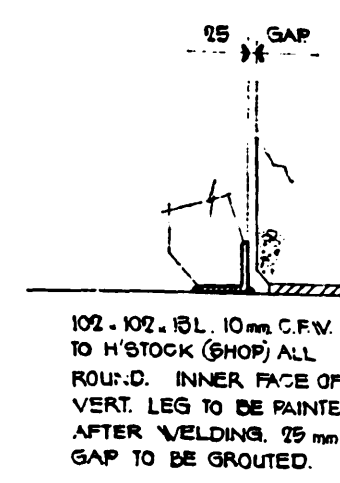
1. CONCRETE: Reinforced kerb units.
F_o' = 25 MPa. Max. agg. size = 20 mm.
2. FOR LAYOUT OF APPROACH JETTY REFER
DRG. CTG1122/5/003 & 004.

THE CORPORATION OF THE TREASURER OF QUEENSLAND																											
NO.	DATE	AMENDMENTS																									
A	29.11.76	BAY NOS 18 & 19 NOW EXPANSION TYPE.																									
B	15.12.77	Drawing amended.																									
<table border="1"> <tr> <th colspan="2">THE SUGAR BOARD BULK SUGAR TERMINALS</th> <th>DRAWING I/E</th> </tr> <tr> <td>SCALES</td> <td>As shown</td> <td></td> </tr> <tr> <td>DRAWN</td> <td>Whealy</td> <td></td> </tr> <tr> <td>PASSED</td> <td>Whealy</td> <td></td> </tr> <tr> <td>DATE</td> <td>28.6.76</td> <td></td> </tr> <tr> <td colspan="2">B.S.T. LUCINDA - STAGE III OFFSHORE LOADING FACILITY. ALTERNATE PRECAST ROADWAY. GENERAL ARRANGEMENT.</td> <td>CTG1122/5/030</td> </tr> <tr> <td colspan="3">MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST PAULS TERRACE, PORTSMOUTH VALLEY QLD 4006 PH 33 7121</td> </tr> <tr> <td colspan="3">IN ASSOCIATION WITH CSR LIMITED</td> </tr> </table>				THE SUGAR BOARD BULK SUGAR TERMINALS		DRAWING I/E	SCALES	As shown		DRAWN	Whealy		PASSED	Whealy		DATE	28.6.76		B.S.T. LUCINDA - STAGE III OFFSHORE LOADING FACILITY. ALTERNATE PRECAST ROADWAY. GENERAL ARRANGEMENT.		CTG1122/5/030	MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST PAULS TERRACE, PORTSMOUTH VALLEY QLD 4006 PH 33 7121			IN ASSOCIATION WITH CSR LIMITED		
THE SUGAR BOARD BULK SUGAR TERMINALS		DRAWING I/E																									
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DATE	28.6.76																										
B.S.T. LUCINDA - STAGE III OFFSHORE LOADING FACILITY. ALTERNATE PRECAST ROADWAY. GENERAL ARRANGEMENT.		CTG1122/5/030																									
MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST PAULS TERRACE, PORTSMOUTH VALLEY QLD 4006 PH 33 7121																											
IN ASSOCIATION WITH CSR LIMITED																											
CSR REFERENCE NO.																											



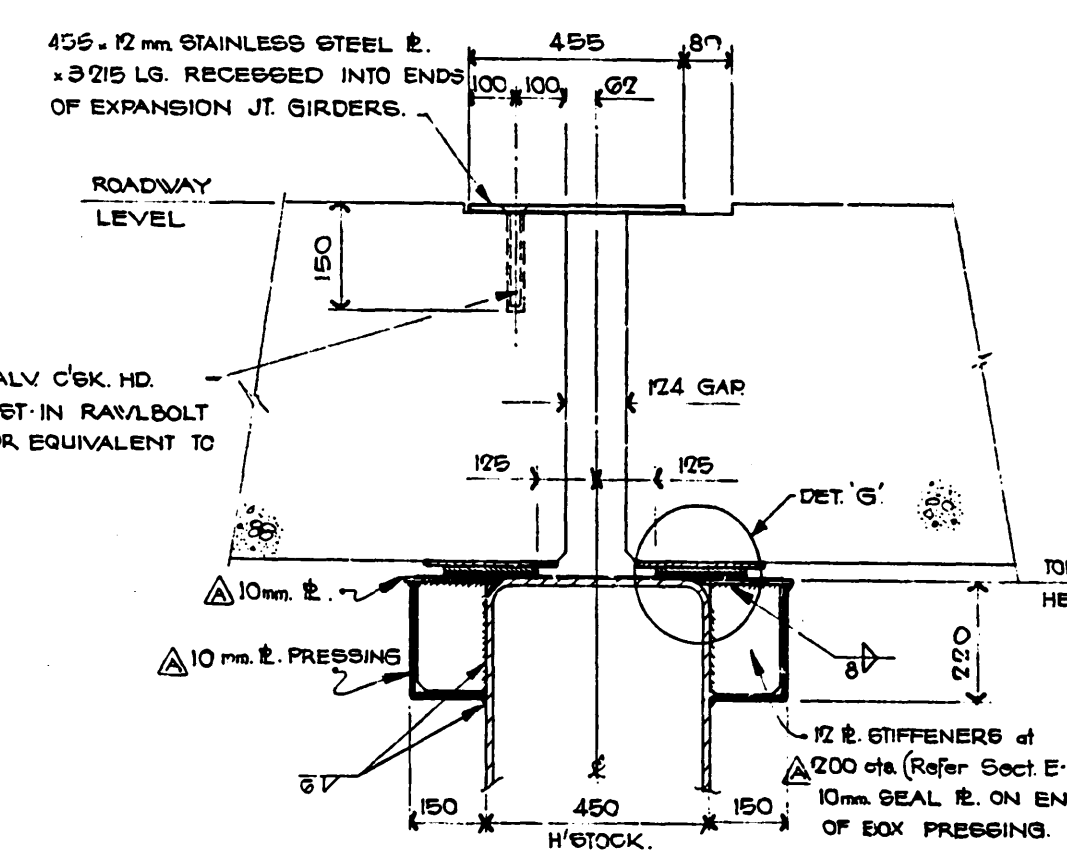
THE CORPORATION OF THE TREASURER OF QUEENSLAND		
NO.	DATE	AMENDMENTS
A	10-1-77	Drawing Revised.
B	14-2-77	DETAILS SHEAR KEY WELD ADDED.
C	10-5-77	DETAILS EDGE ANGLES ADDED.
D	10-5-77	SITE WELDING OF 152L AMENDED TO SHOP WELDING TO SUIT CONTRACTOR.
CSR REFERENCE NO.		

THE SUGAR BOARD BULK SUGAR TERMINALS		
SCALES 120, 110 DRAWN J.T. PASSED DATE JAN. '77	B.S.T LUCINDA STAGE III APPROACH JETTY STD. ROADWAY GIRDER DETAILS	DRAWING NO. CTQ1122/S/031
MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST. PAULS TERRACE, FORTITUDE VALLEY, QLD 4006 PH 32 77 73 IN ASSOCIATION WITH		CSR LIMITED

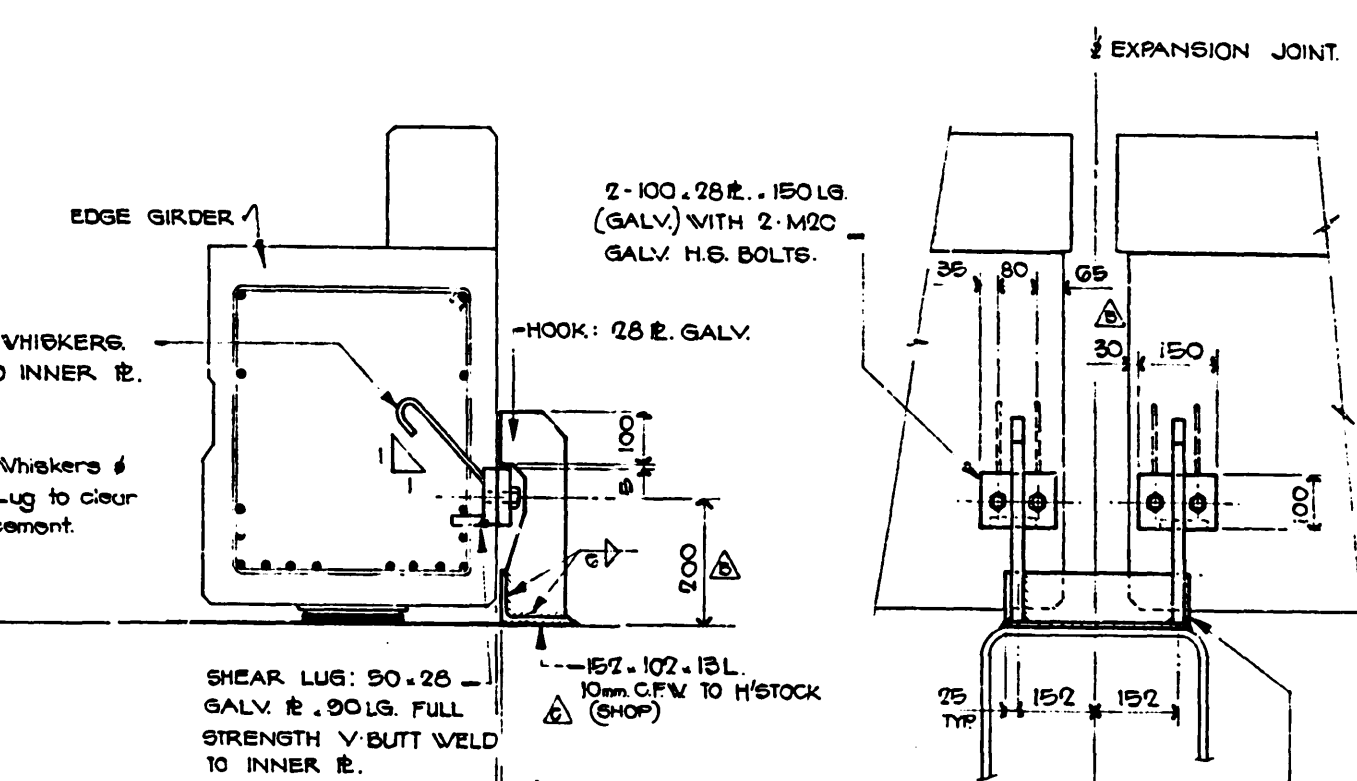


Technical drawing of a kerb anchor plate. The drawing shows a rectangular plate with a central rectangular hole. Dimensions are given in millimeters (mm). The overall width is 500 mm. The overall height is 200 mm. The central hole has a width of 380 mm and a height of 130 mm. The plate has a thickness of 12 mm. The central hole has a depth of 25 mm. The plate is labeled "KERB ANCHOR PLATE" and "OUTSIDE". The drawing includes a scale bar from 0 to 500 mm.

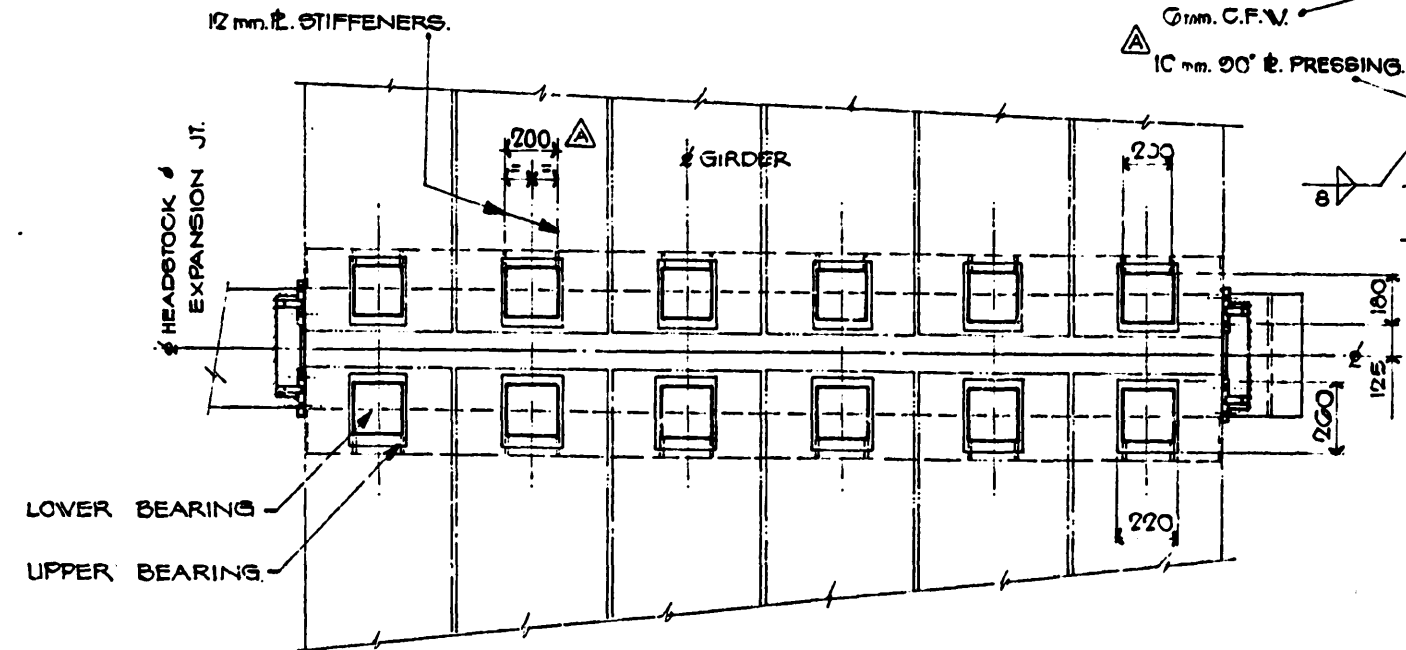
SECTION C-C



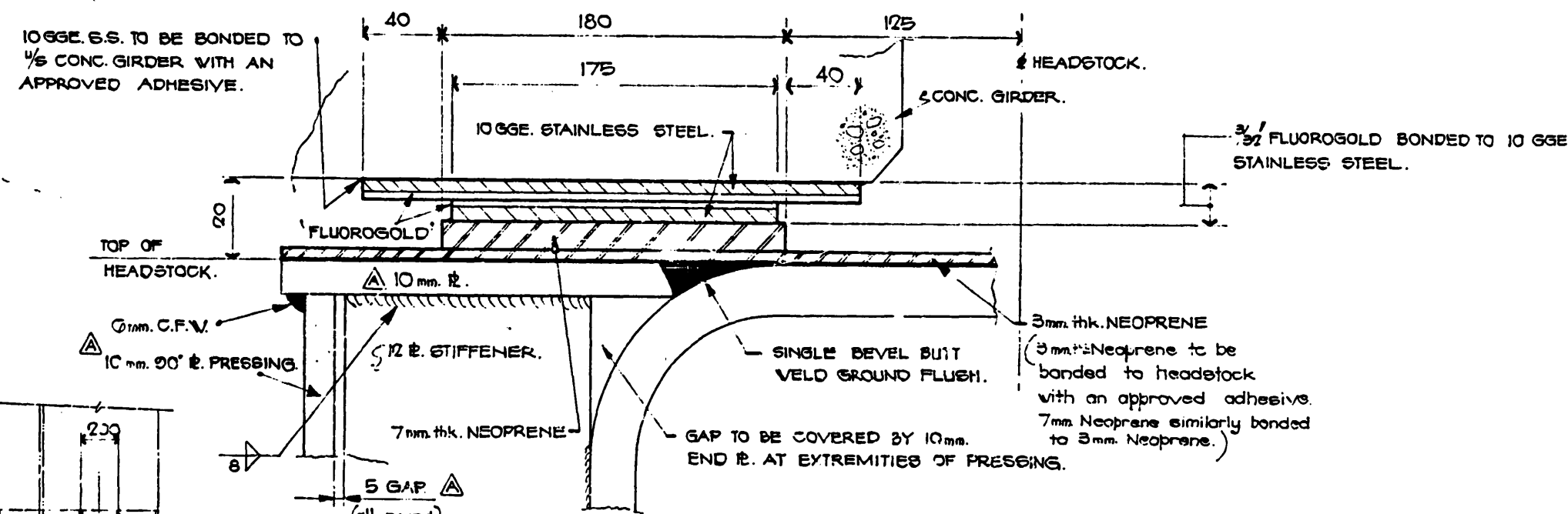
SECTION D-D



DETAIL 'F'



SECTION E-E



DETAIL 'G'

NOTES:

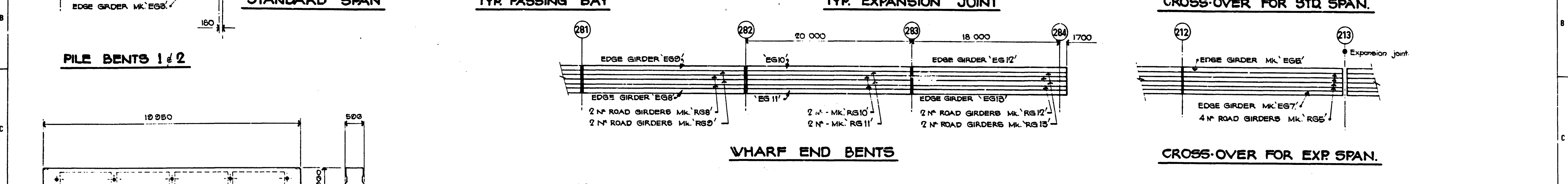
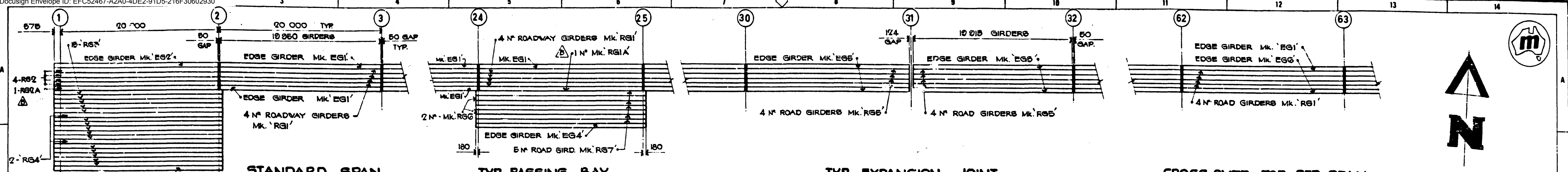
- REFER DRG. CTQ 1127/5/031 FOR NOTES 1 to 4.
- REFER DRG. CTQ 1127/5/001 FOR EXPANSION JT. LOCATION.

ELEVATION OF EXPANSION JOINT.



THE CORPORATION OF THE TREASURER OF QUEENSLAND		
NO	DATE	AMENDMENTS
A	14-2-77	Bearing support amended. Stiffener c/s altered.
B	10-3-77	Detail 'F' dimensional change.
C	10-5-77	Detail 'H' added. Detail 'F' amended.
D	10-5-77	Site Welded IER L amended to Shop Welded to suit Contractor.
E	10-8-77	Pile Height amended on headstock.
CSR REFERENCE NO.		

THE SUGAR BOARD BULK SUGAR TERMINALS		
SCALES 1:20, 1:10 DRAWN J.T. PASSED DATE JAN '77	B.S.T. LUCINDA STAGE III APPROACH JETTY EXPANSION JOINT DETAILS	DRAWING NO. CTO1122/S/032
MACDONALD WAGNER & ARDLE PTY. LTD. CONSULTING ENGINEERS 313 ST PAUL'S TERRACE PORTLAND QUEENSLAND 4000 PH 52 7122		IN ASSOCIATION WITH CSR LIMITED



File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.
2 to 5	4.4 = 10	156 (PB)	1.5 = 5	168, 180	2.4 = 8	17	1
8 to 17	10.4 = 40	157 to 161	5.4 = 20	186, 187	2.4 = 8	32	1
20 to 23	4.4 = 10	164 to 171	8.4 = 32	204, 205	2.4 = 8	47	1
24 (PB)	1.5 = 5	174 to 179	6.4 = 24	212, 213	2.4 = 8	62	1
25 to 29	5.4 = 20	182 to 187	6.4 = 24	220, 221	2.4 = 8	77	1
32 to 41	10.4 = 40	190 to 195	6.4 = 24	230, 231	2.4 = 8	92	1
44 to 53	10.4 = 40	196 & 199	2.4 = 8	242, 243	2.4 = 8	107	1
56 to 59	4.4 = 10	200 (PB)	1.5 = 5	254, 255	2.4 = 8	122	1
60 (PB)	1.5 = 5	201 to 203	3.4 = 12	266, 267	2.4 = 8	137	1
61 to 65	5.4 = 20	206 to 211	6.4 = 24	278, 280	2.4 = 8	152	1
68 to 77	10.4 = 40	214 to 219	6.4 = 24	150, 151	2.4 = 8		
80 to 89	10.4 = 40	222 to 229	8.4 = 32				
92 to 101	10.4 = 40	232 to 241	10.4 = 40				
104 to 107	4.4 = 10	244 to 247	4.4 = 10				
108 (PB)	1.5 = 5	248 (PB)	1.5 = 5				
109 to 113	5.4 = 20	249 to 253	5.4 = 20				
116 to 125	10.4 = 40	256 to 258	10.4 = 40				
128 to 137	10.4 = 40	268 to 278	11.4 = 44				
140 to 149	10.4 = 40						
152 to 155	4.4 = 10						

File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.
2 to 5	4.2 = 8	116 to 121	6.2 = 12	227 (co)	1.1 = 1	186, 189	2.2 = 4
8 to 10	3.2 = 8	122 (co)	1.1 = 1	228 & 229	2.2 = 4	196, 197	2.2 = 4
17 (co)	1.1 = 1	123 to 125	3.2 = 6	232 to 241	10.2 = 20	204, 205	2.2 = 4
20 to 23	4.2 = 8	128 to 136	3.2 = 18	244 to 247	4.2 = 8	212	1.1 = 1
24 (PB)	1.1 = 1	137 (co)	1.1 = 1	248 (PB)	1.1 = 1	220, 221	2.2 = 4
25 to 29	5.2 = 10	140 to 149	10.2 = 20	249 to 253	5.2 = 10	230, 231	2.2 = 4
32 (co)	1.1 = 1	152 (co)	1.1 = 1	256	1.2 = 2	242	1.1 = 1
33 to 41	8.2 = 18	153 to 155	3.2 = 6	257 (co)	1.1 = 1	254, 255	2.2 = 4
44 to 46	3.2 = 6	156 (PB)	1.1 = 1	258 to 265	8.2 = 18	266, 267	2.2 = 4
47 (co)	1.1 = 1	157 to 161	5.2 = 10	268 to 271	4.2 = 8	278, 280	2.2 = 4
48 to 53	5.2 = 12	164 to 166	3.2 = 6	272 (co)	1.1 = 1	213	1.2 = 2
56 to 59	4.2 = 8	167 (co)	1.1 = 1	273 to 278	6.2 = 12	243	1.2 = 2
60 (PB)	1.1 = 1	168 to 171	4.2 = 8			150, 151	2.2 = 4
61	1.2 = 2	174 to 179	6.2 = 12				
62 (co)	1.1 = 1	182 (co)	1.1 = 1				
63 to 65	3.2 = 6	183 to 187	5.2 = 10				
68 to 70	2.2 = 18	190 to 195	4.2 = 8				
77 (co)	1.1 = 1	196	1.1 = 1				
80 to 89	10.2 = 20	198 to 199	1.2 = 2				
92 (co)	1.1 = 1	201 to 203	3.2 = 6				
93 to 101	9.2 = 18	206 to 211	6.2 = 12				
104 to 106	3.2 = 6	214 to 219	6.2 = 12				
107 (co)	1.1 = 1	222 to 226	5.2 = 10				
108 (PB)	1.1 = 1						
109 to 113	5.2 = 10						

NOTES:

- FOR ALL REINFORCEMENT DETAILS - REFER DRG. N° CTQ 1122/5/031
- FOR GENERAL LAYOUT OF APPROACH JETTY - REFER DRG. N° CTQ 1122/5/003 & 004.
- FOR REMAINING GIRDER DETAILS - REFER SHEET 2 - DRG. N° CTQ 1122/5/034.

THE CORPORATION OF THE TREASURER OF QUEENSLAND			
THE SUGAR BOARD			
BULK SUGAR TERMINALS			
NO.	DATE	AMENDMENTS	
A	15-2-77	Kerb Anchor Bars in Edge Girders added.	
B	15-3-77	No. Req'd amended for RG1 & RG1A added.	

SCALES N.T.B.

DRAWN J.T.

PASSED

DATE JAN. '77

B.S.T. LUCINDA - STAGE III

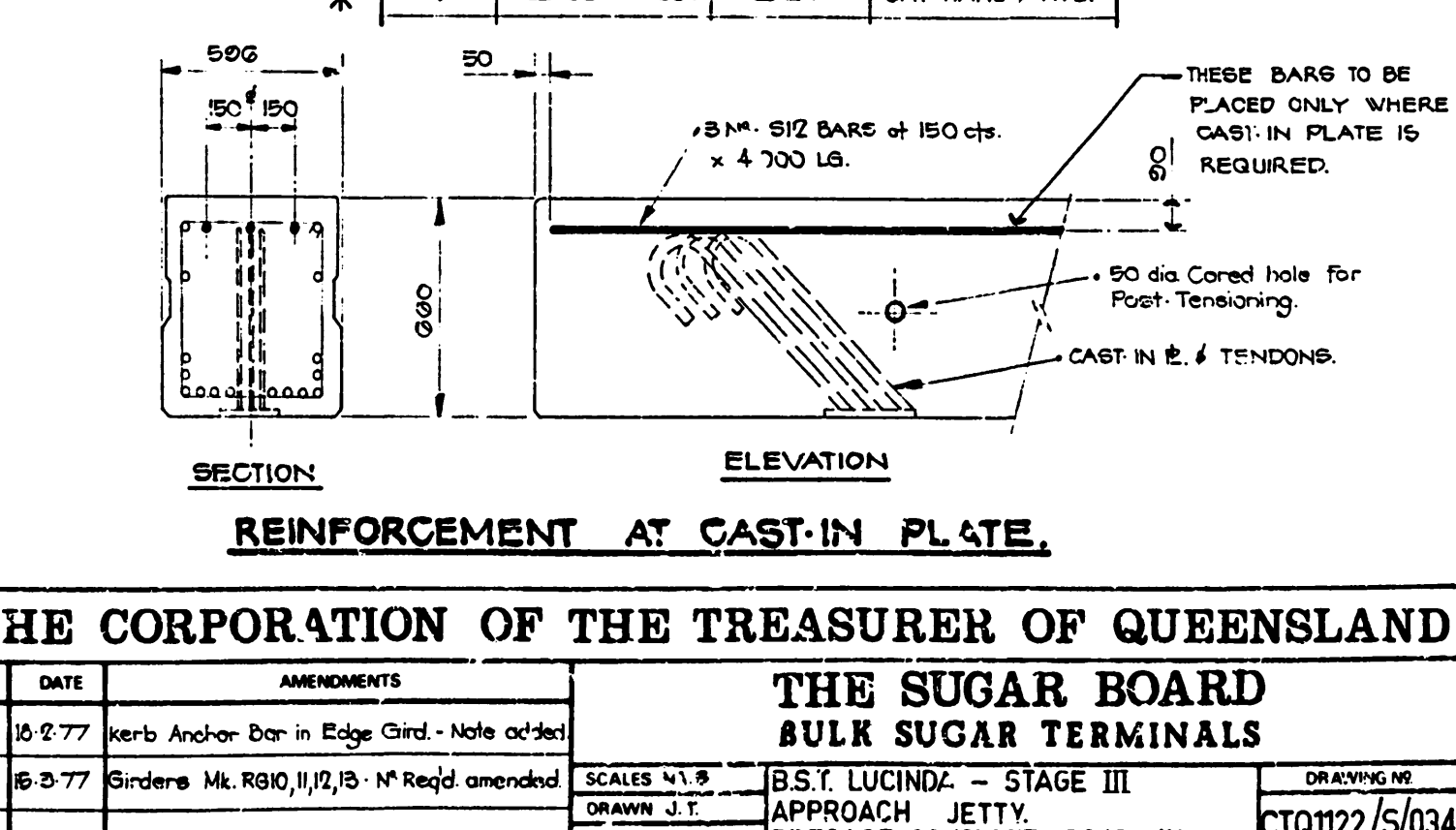
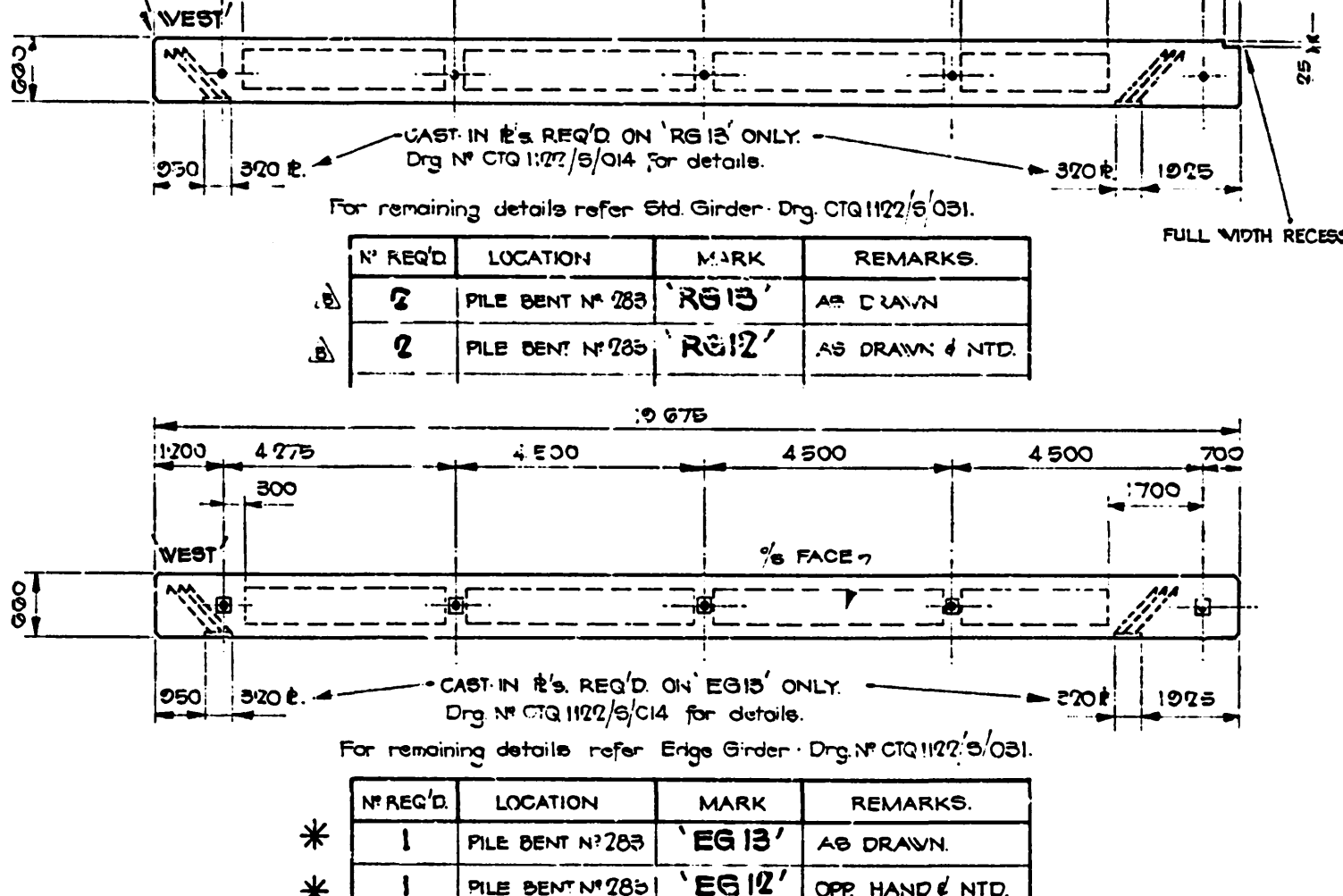
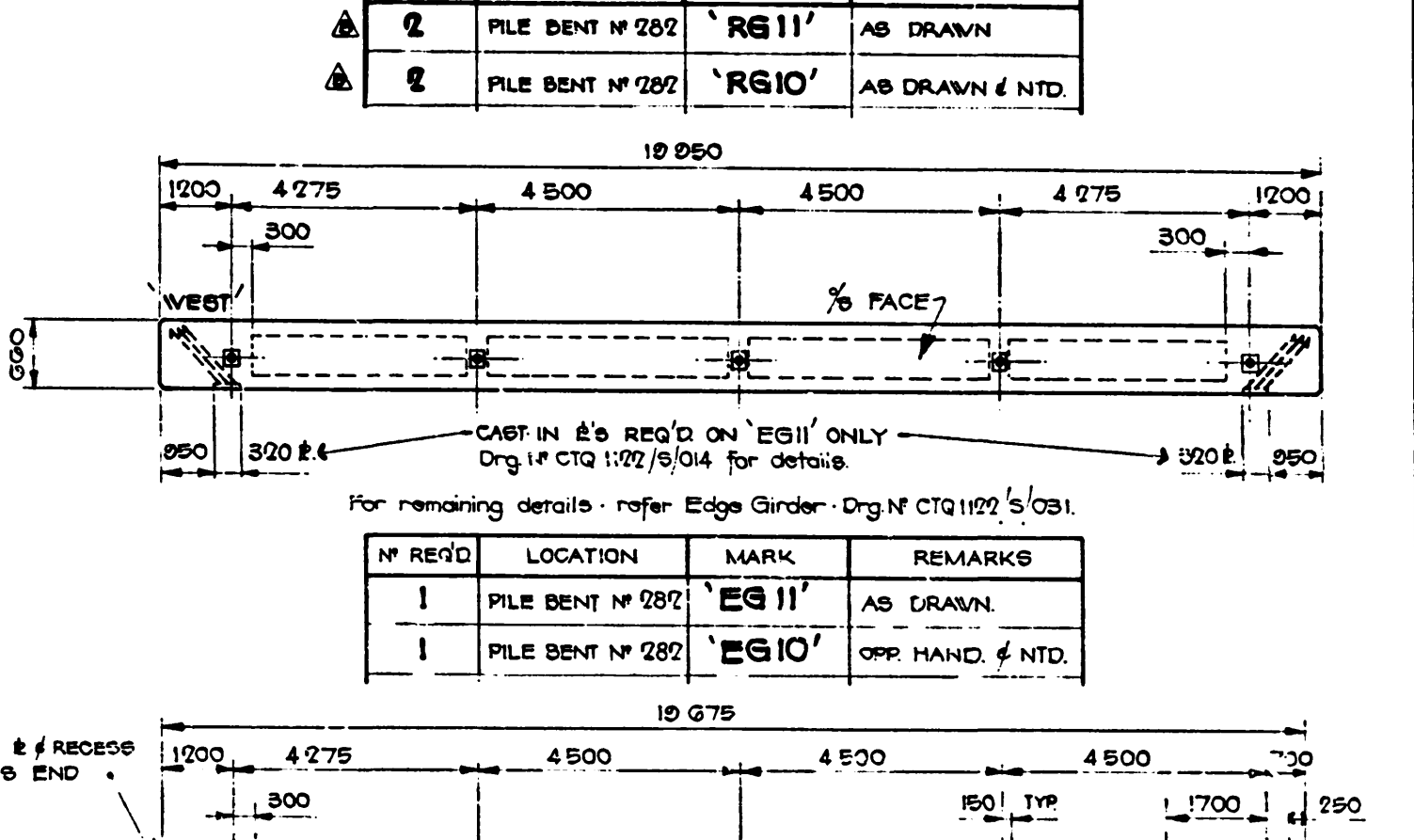
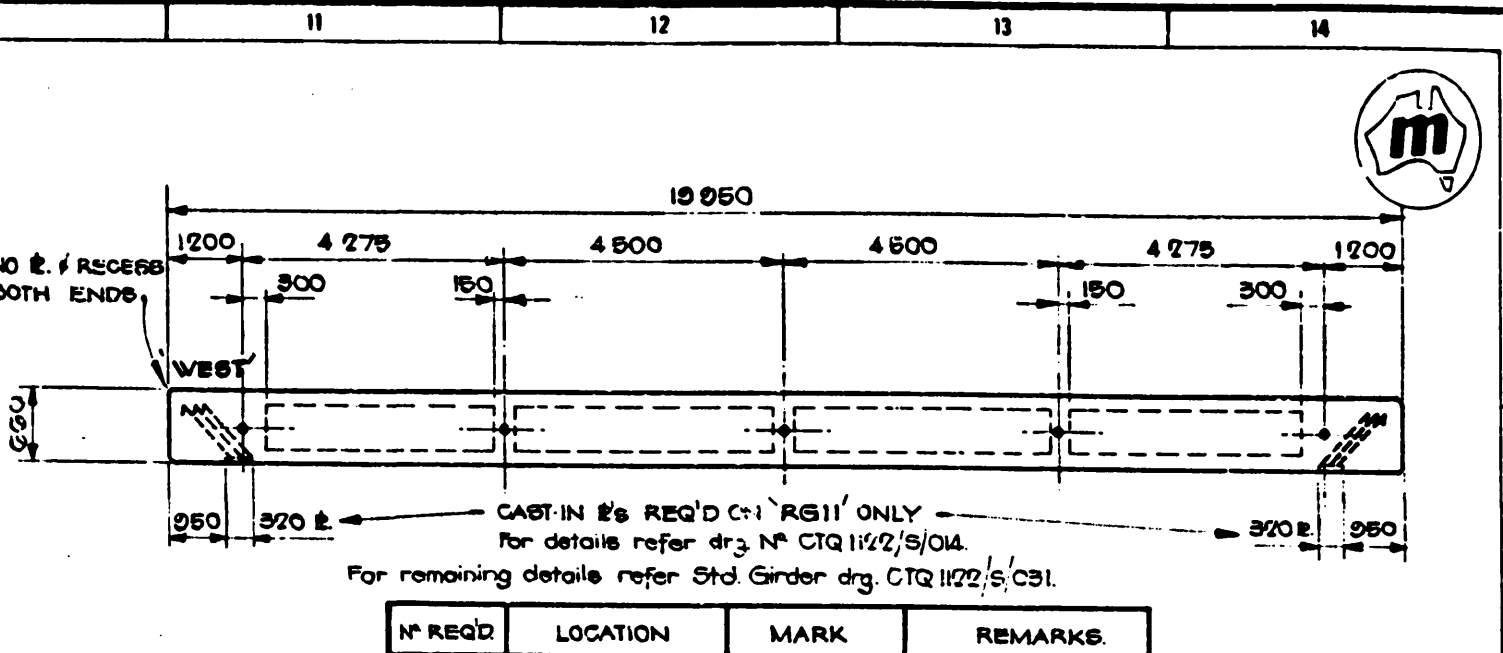
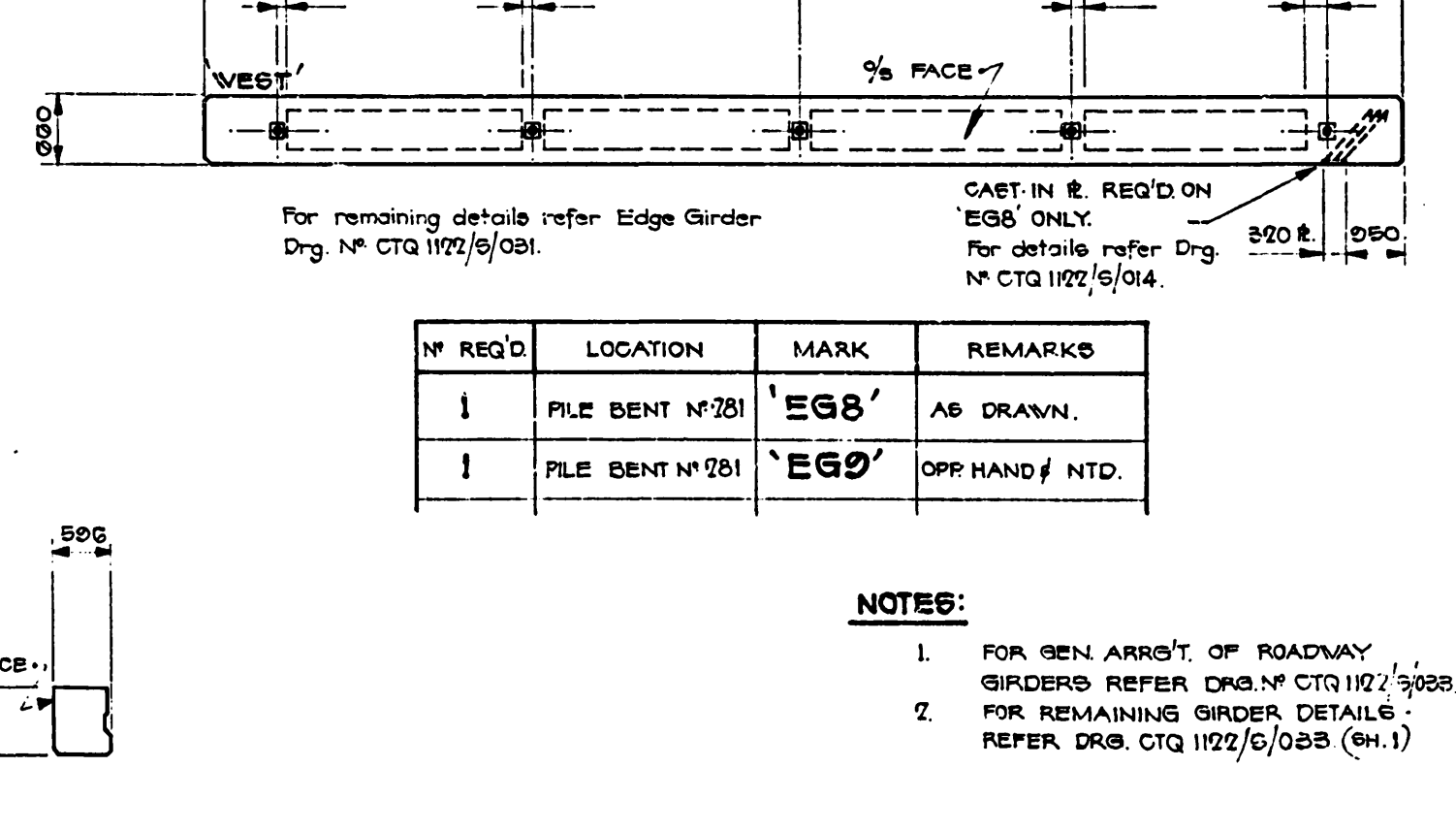
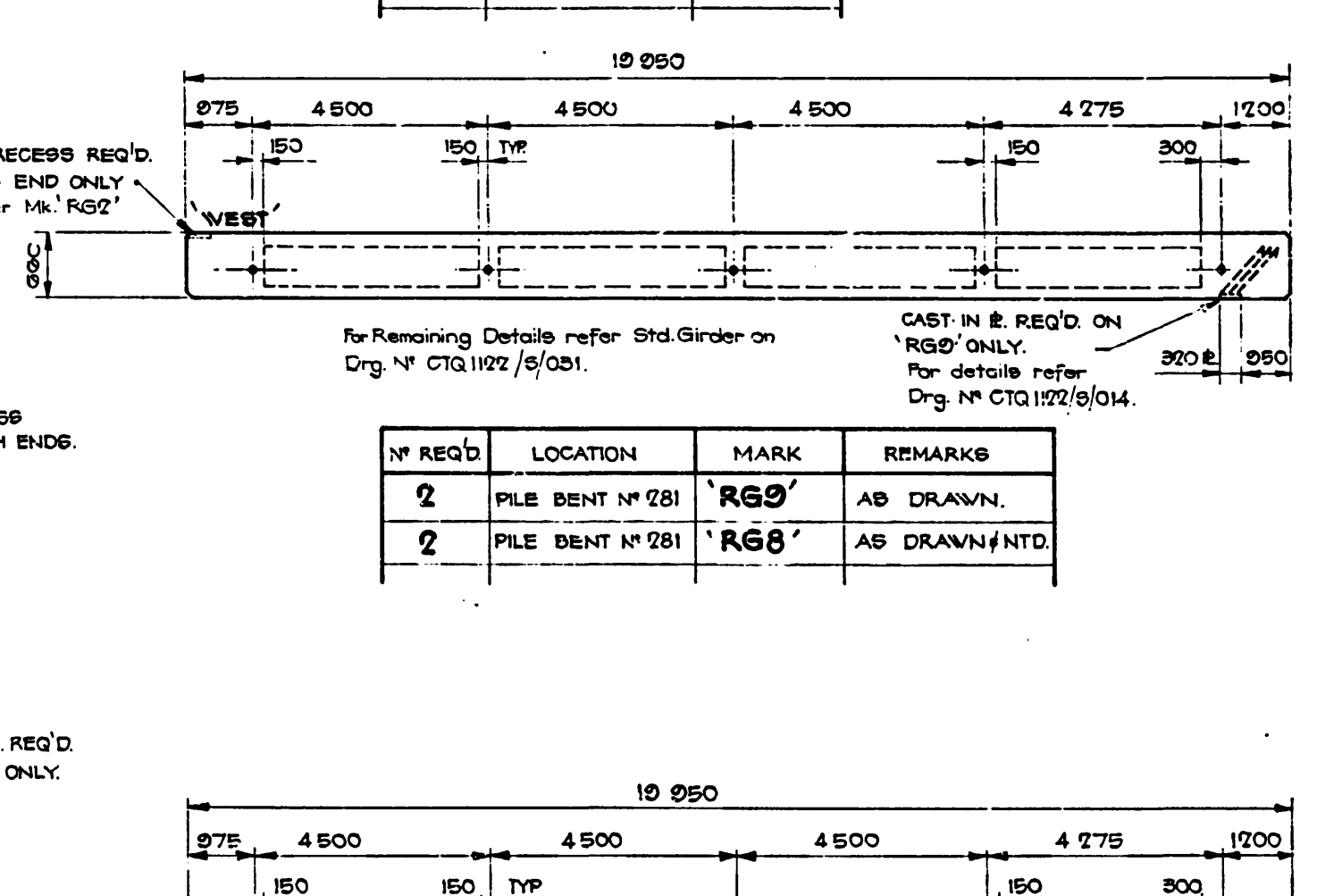
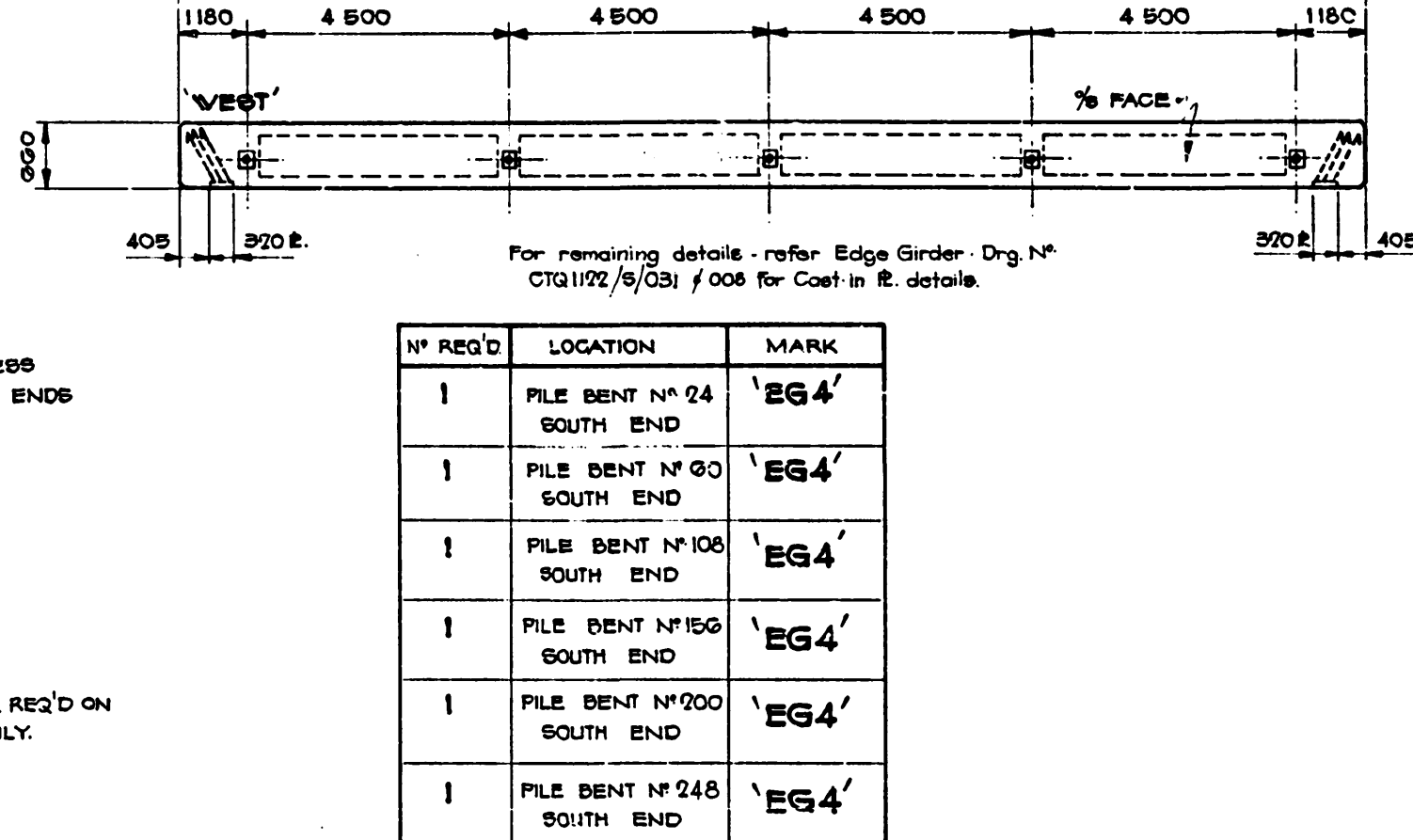
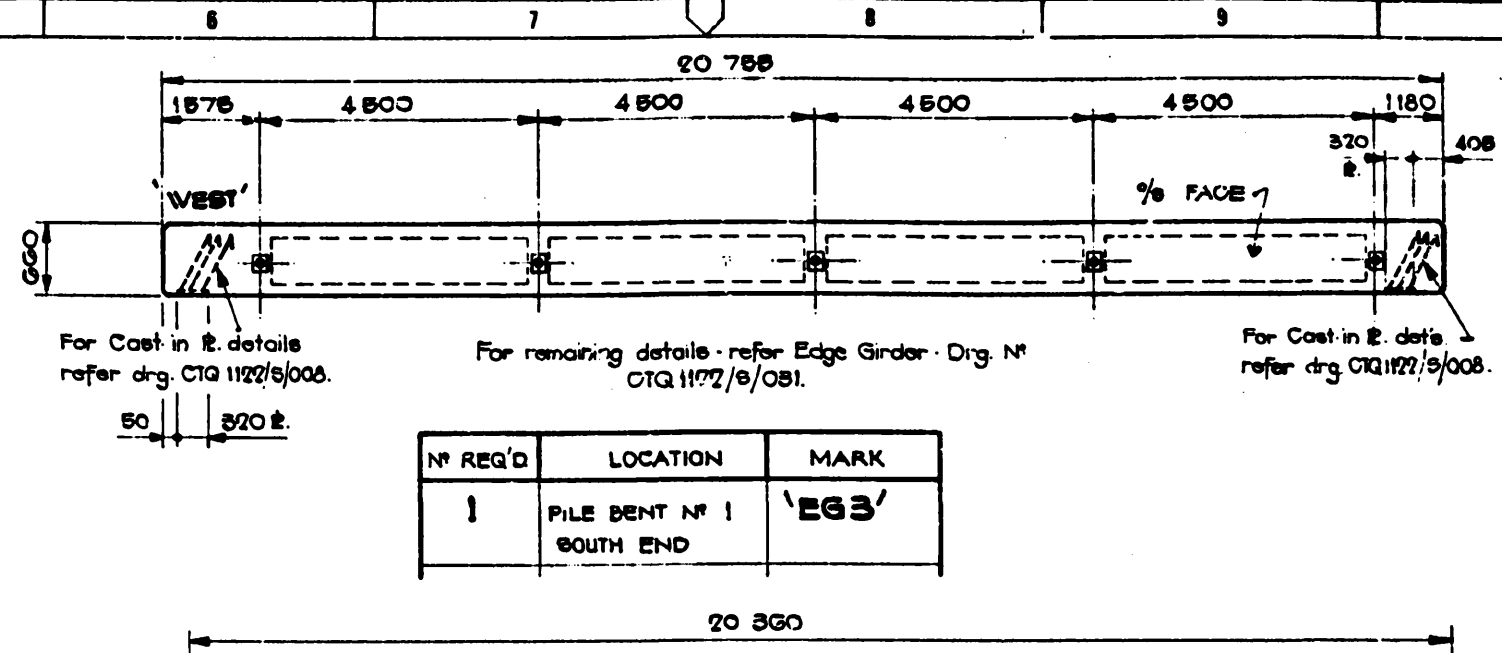
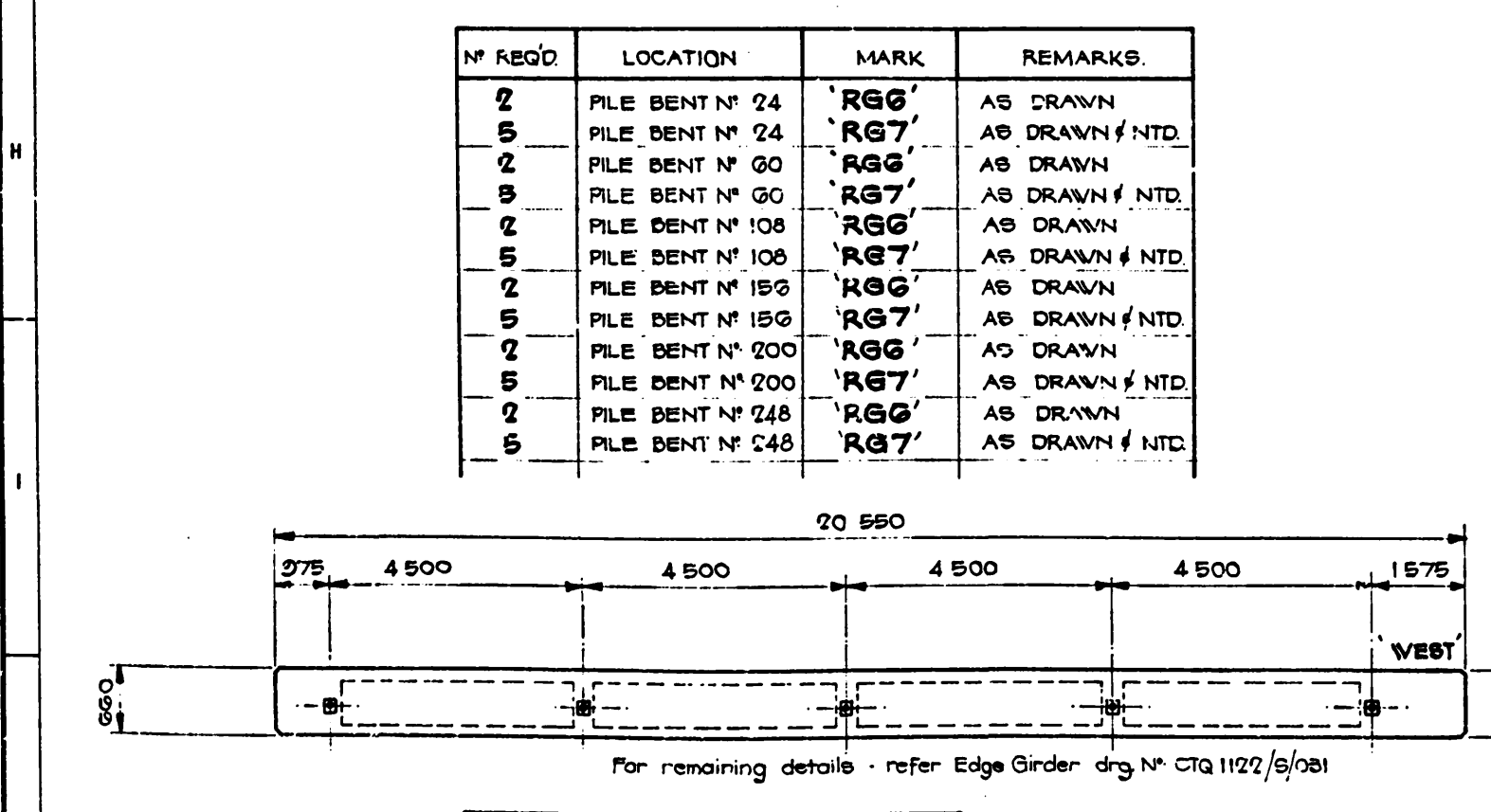
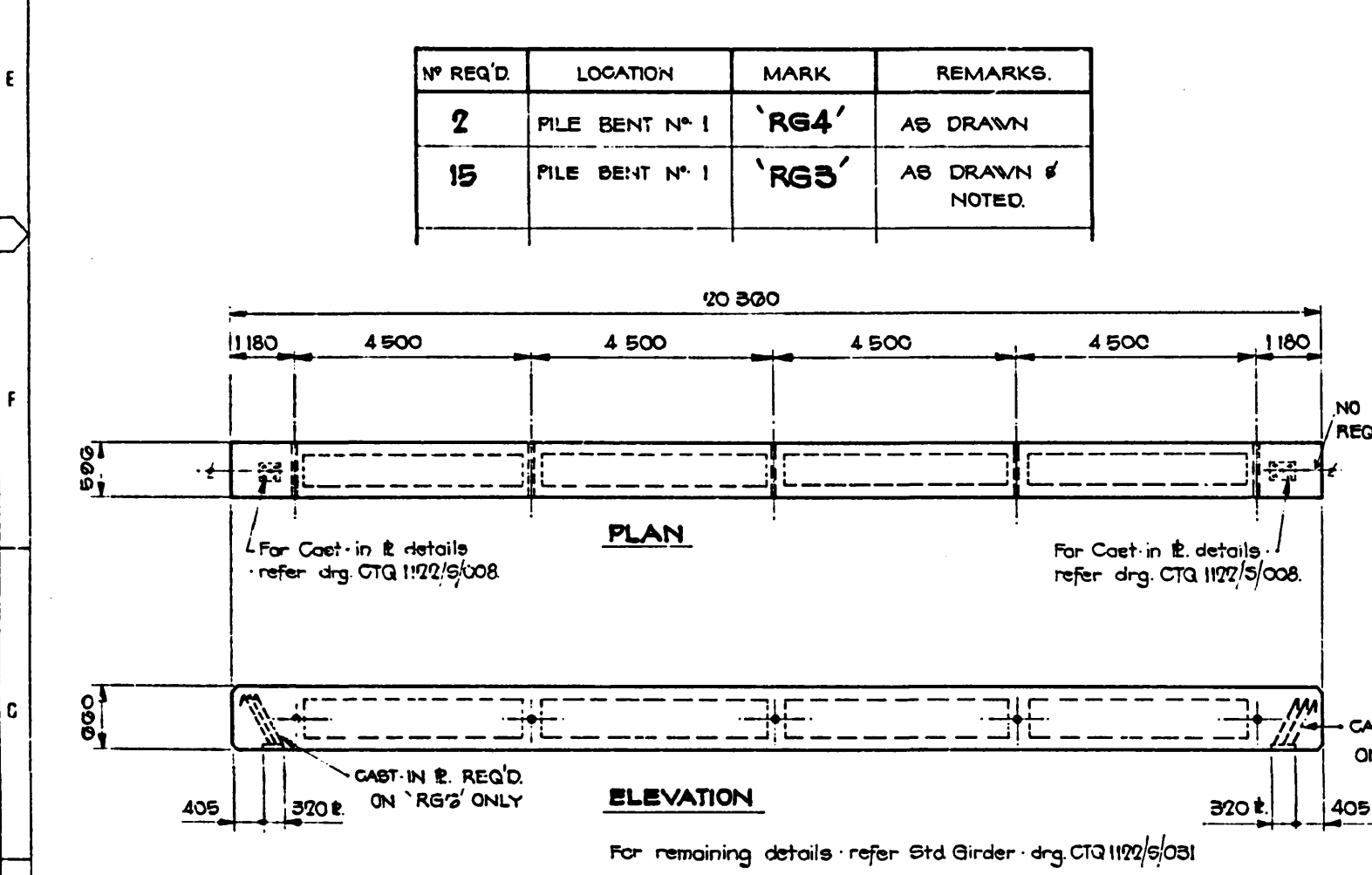
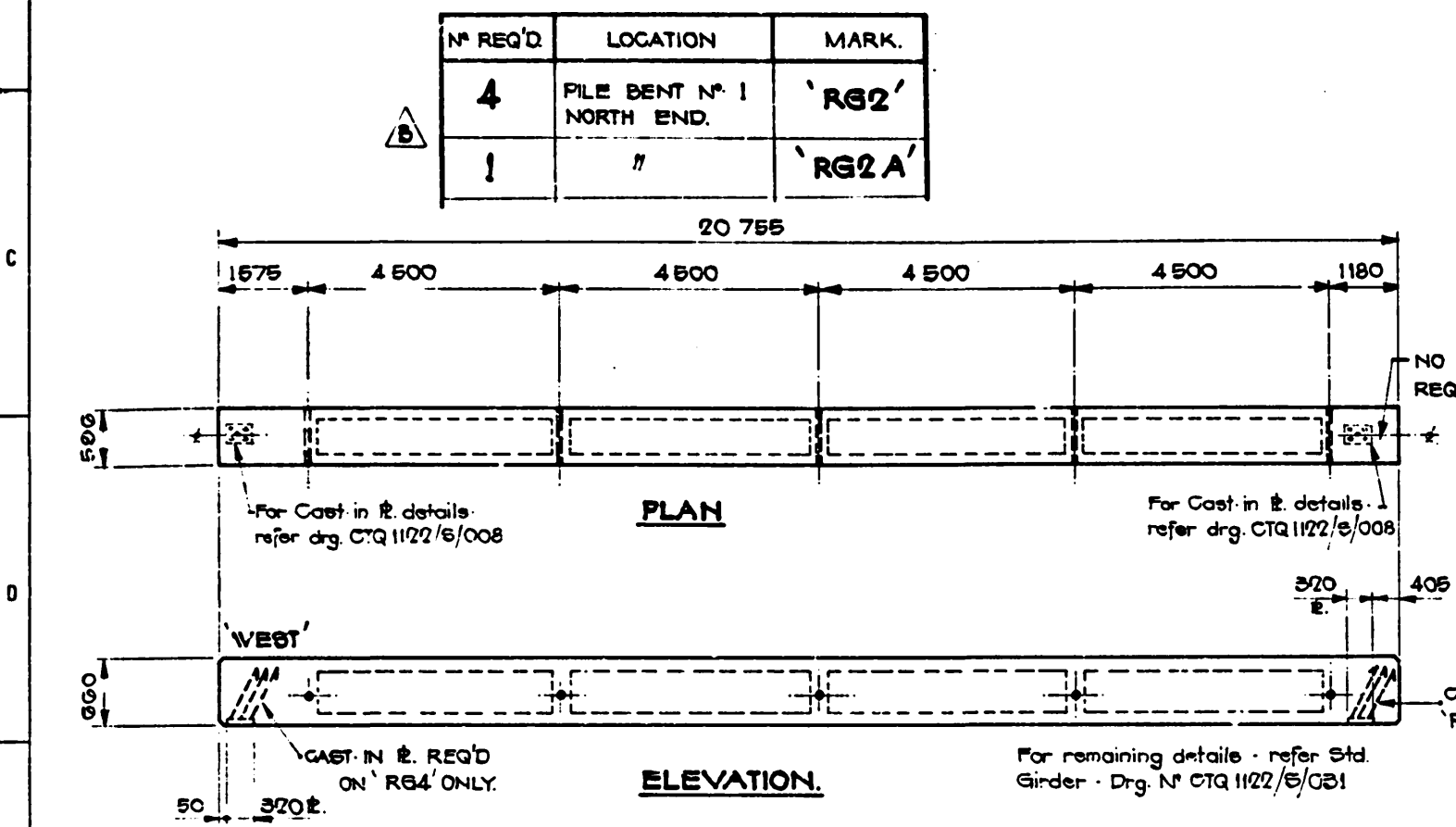
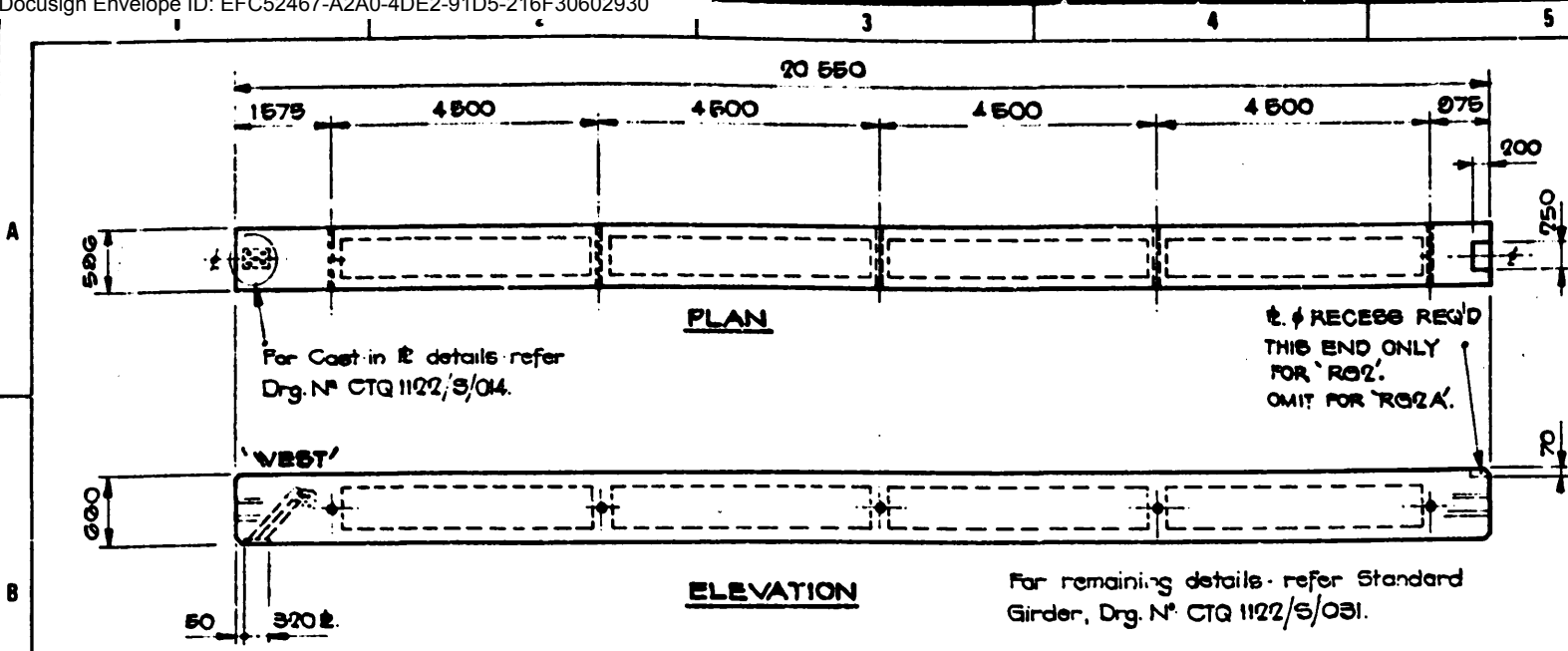
APPROACH JETTY

PRECAST CONCRETE ROADWAY

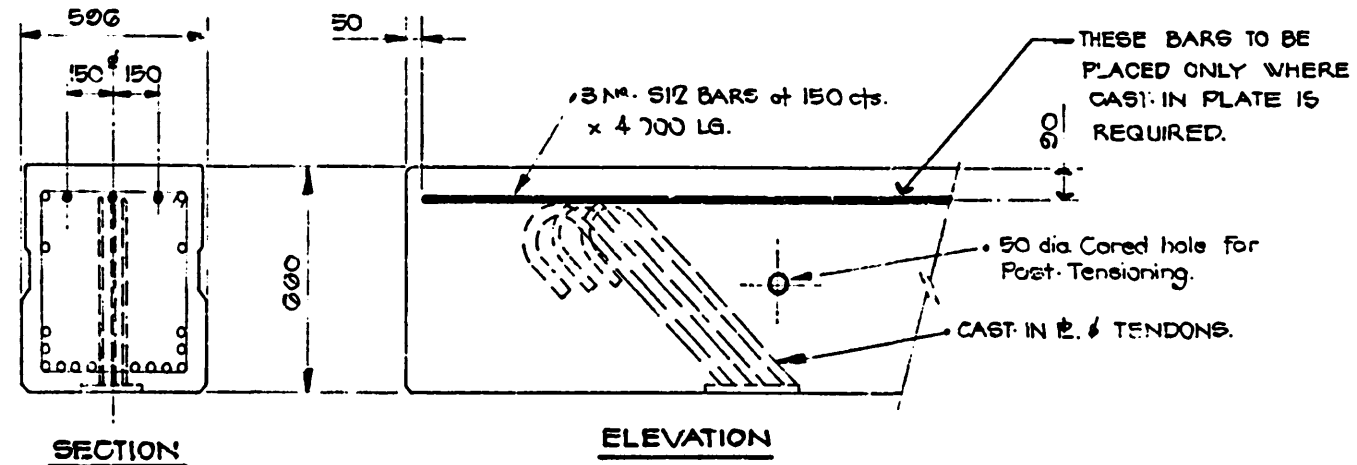
GIRDER ARRGT. & DETAILS. (SHEET 1)

MACDONALD WAGNER & PRIDDLE PTY. LTD.

CONSULTING ENGINEERS 313 ST PAULS TERRACE PORTLAND VALLEY QLD 4806 PH 31 7121



REINFORCEMENT AT CAST-IN PLATE.



- NOTES:**
- FOR GEN. ARR'G. OF ROADWAY GIRDERS REFER DRG. N° CTQ 1122/S/030.
 - FOR REMAINING GIRDER DETAILS REFER DRG. CTQ 1122/S/033 (SH.1)

* EDGE GIRDERS 'EG12' & 'EG15' TO HAVE KERB ANCHOR BARS ON EAST END POSITIONED TO ALLOW FOR SPECIAL KERB UNIT. - DRG. N° CTQ 1122/S/030.

THE CORPORATION OF THE TREASURER OF QUEENSLAND			
N°	DATE	AMENDMENTS	
A	18-2-77	Kerb Anchor Bar in Edge Gird. - Note added	
B	15-3-77	Girders Mk. RG10, 11, 12, 13 - N° Req'd amended	

THE SUGAR BOARD		BULK SUGAR TERMINALS	
Scales 1/10		Scales 1/10	
DRAWN J.T.		DRAWN J.T.	
PASSED		PASSED	
DATE FEB '77		DATE FEB '77	

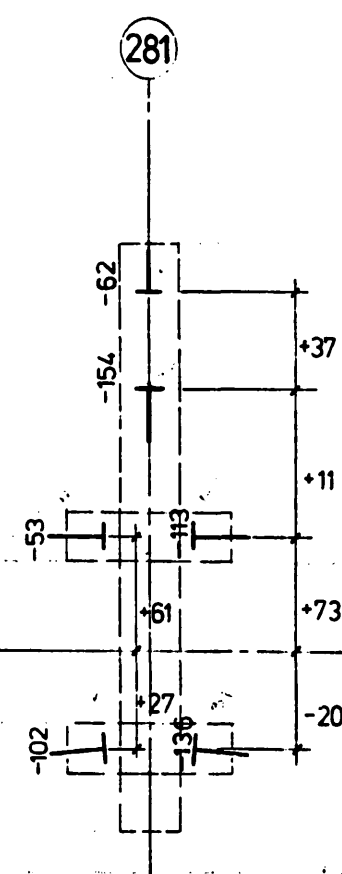
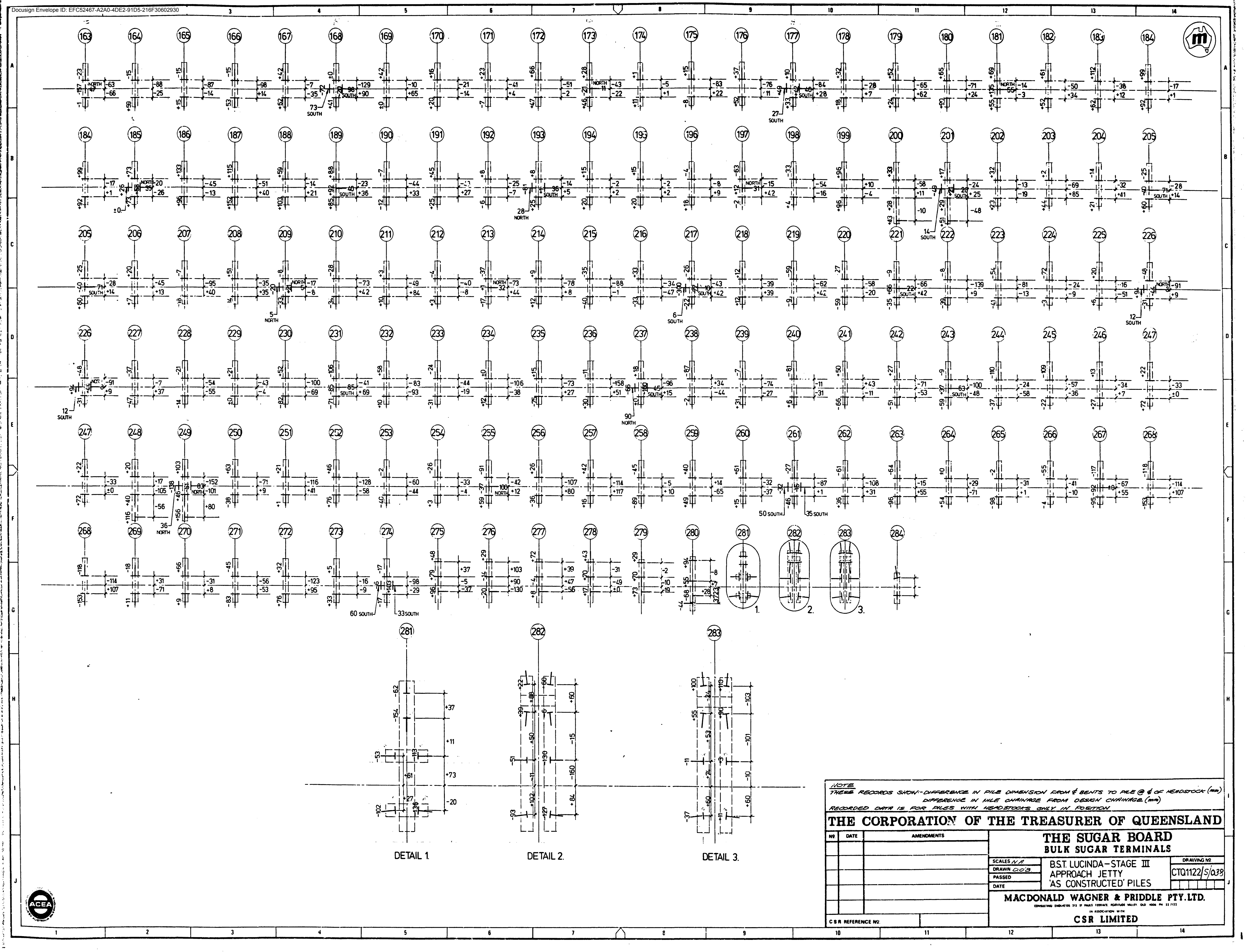
MACDONALD WAGNER & PRIDDLE PTY. LTD.

OPERATING ENGINEERS 213 ST PAUL'S TERRACE, PORTHURST VALLEY, 310 4000 PH 31 7172

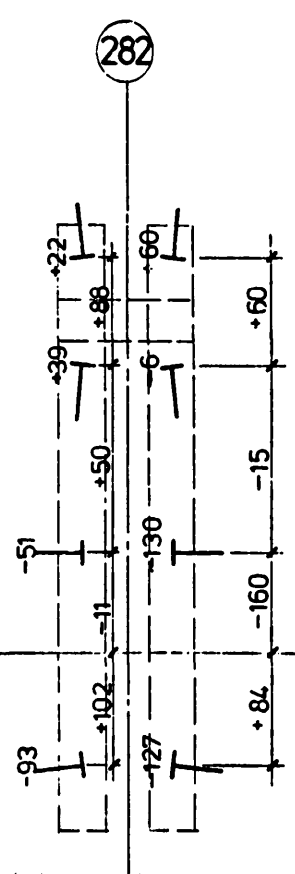
IN ASSOCIATION WITH

CSR LIMITED

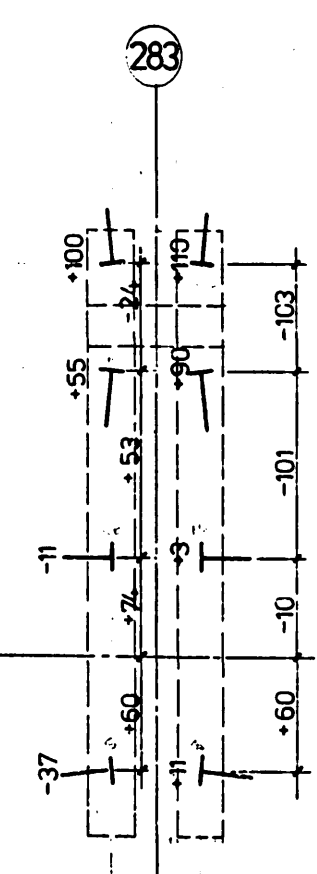
CSR REFERENCE N°.



DETAIL 1



DETAIL 2



DETAIL 3

NOTE
THESE RECORDS SHOW:- DIFFERENCE IN PILE DIMENSION FROM BENTS TO PILE @ & OF HEADSTOCK (mm)
DIFFERENCE IN PILE CHAIRING FROM DESIGN CHAIRING (mm)
RECORDED DATA IS FOR PILES WITH HEADSTOCKS ONLY IN POSITION

THE CORPORATION OF THE TREASURER OF QUEENSLAND

NO	DATE	AMENDMENTS

THE SUGAR BOARD
BULK SUGAR TERMINALS

SCALE: 1/1
DRAWN: GCS
PASSED:
DATE:

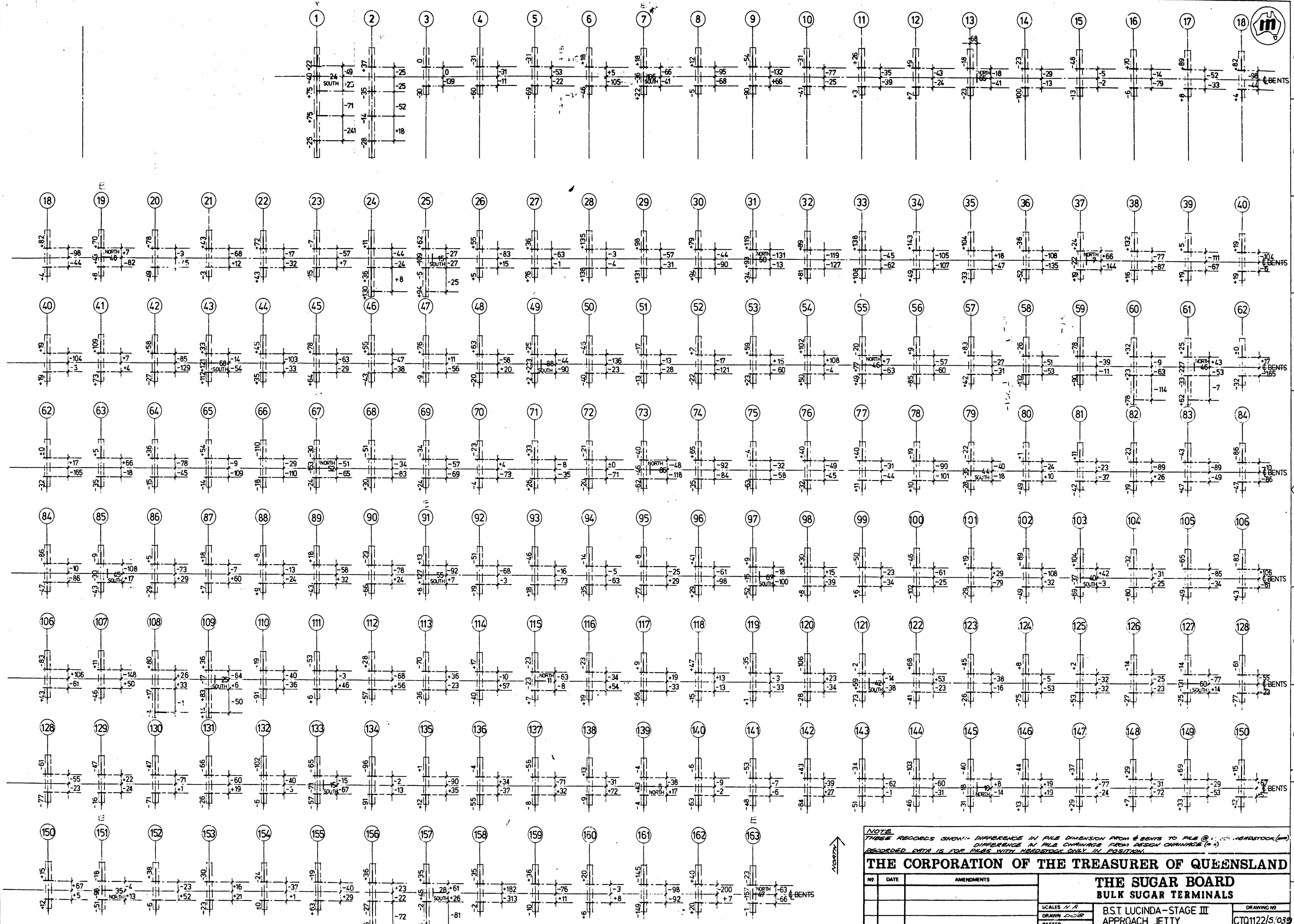
B.S.T. LUCINDA - STAGE III
APPROACH JETTY
'AS CONSTRUCTED' PILES

MACDONALD WAGNER & PRIDDLE PTY. LTD.
CONSULTING ENGINEERS 713 ST PAULS TERRACE, PORTLAND VALLEY QLD 4006 PH 32 7122
IN ASSOCIATION WITH
CSR LIMITED

DRAWING NO:
CTQ1122/S/039

CSR REFERENCE NO.:





DESIGN R.L.'S:-
TOP OF HEADSTOCKS 7320
PILE HEAD 7058
ANCHOR PILE HEAD 6258
E.T. " " 7008

North

NOTE:
THESE RECORDS SHOW:- DIFFERENCE IN PILE DIMENSION FROM BENTS TO PILE @ 10' HEADSTOCK (mm)
DIFFERENCE IN PILE DIMENSIONS FROM DESIGN CHAINAGE (m)
RECORDED DATA IS FOR PILES WITH HEADSTOCK ONLY IN POSITION.

THE CORPORATION OF THE TREASURER OF QUEENSLAND

NO	DATE	AMENDMENTS

CSR REFERENCE NO. _____

THE SUGAR BOARD
BULK SUGAR TERMINALS

SCALE: 1/4" = 1'-0"
DRAWN: DGB
PASSED: _____
DATE: _____

BST LUCINDA-STAGE III
APPROACH JETTY
'AS CONSTRUCTED' PILES

DRAWING NO. CTQ1122/5/039

MACDONALD WAGNER & PRIDDLE PTY. LTD.
CONSULTING ENGINEERS 213 B PAVIL TERRACE PORTLAND VALLEY QLD 4006 PH 33 7173
IN ASSOCIATION WITH
CSR LIMITED

Handover of one jetty 'spider' unit at completion of contract

The Contractor shall hand over one of the Jetty 'spiders' to the Principal at the completion of the contract. The unit handed over shall be of the Contractor's choice. Any additions made to the unit since the initial design and approval may be removed by the Contractor or left in place.

The unit handed over shall be in an as-is condition. Upon handover, the unit does not form part of 'The Works' as per Clause 1 in the General Conditions of Contract. Upon handover, no warranty, guarantee or defects liability period applies. The Principal assumes all risk with respect to further use of the unit for any purpose.

Annexure Part E – Methodology

RESPONSE SCHEDULES SCHEDULE E – METHODOLOGY

Schedule E1 – Methodology

Provide a statement of the Tenderer's proposed arrangements, procedures and methodologies for carrying out and completing WUC. This should include:

1. *An overview of the methodology proposed by the Tenderer for the carrying out and completion of WUC;*

GENERAL:

Advanced Aqua Blasting (AAB) will have on site depending on work schedule

- 1 x AAB supervisory staff
- 1 x APP Group Quality Assurance person
- 2 x Ingham Agencies (IA) welding teams consisting of 2 men each with the relevant qualifications
- 1 x AAB abrasive blasting and painting team consisting of 3 men
- 1 x AAB Denso Seashield installation team consisting of 3 men
- 1 x Benchmark Scaffolding (BM) scaffold team consisting of 1 supervisor and 4 scaffolders

All AAB team members will be interchangeable depending on skills.

SCAFFOLD

AAB will seek permission to **work on Sundays** during scaffold erection / dismantling to minimize traffic interruptions. Scaffolders will work their 7 shifts including weekends. Their RDO will be weekdays.

Start the erection of 6 scaffolds from PRIORITY A PILEBENT 257.

It is anticipated that 5 pilebents will be fully scaffolded at any one time with work proceeding on them, and with 1 being dismantled and 1 being erected. All scaffolds build plans will be engineered.

The scaffolding will then proceed in numerical order, priority A and B to the last pilebent 281.

AAB will then relocate to pilebent 256 and proceed inwards to complete pilebents priority A and B in numerical order.

The scaffold will be erected so that access to the headstock and piles is on the southern side of the roadway and not impede roadway spider movement.

Initial access will also be from removing the gallery underpans. Replaced during shipping.

The scaffold will be erected so that the roadway spiders can travel along the roadway.

The headstock scaffolding will be erected to facilitate encapsulation and to allow the removal of the petroleum tape and masking around the gallery feet and the removal of any tape / jacket system down to > 600mm on the piles to allow for abrasive blasting and painting. The scaffold will also be built to allow removal of any part of the structure in contact with the headstock to be removed for abrasive blasting and painting.

****See attached scaffold design****

HEADSTOCKS / PILES – similar methodology for all PRIORITY A and B pile bent (Appendix D)

1) Welding repairs

On the completion of the scaffold on pile bent 257. AAB will

- a) Position spider 1W (1W is the welding equipped spider).
- b) High pressure water wash.
- c) Start steel thickness inspections and recordings as per **Contract Technical Specification 4.1.2 and the HOLD and WITNESS Points** in the Inspection and Test Plan (ITP).
- d) Steel repairs on both headstock and piles. (Specification and drawing APPENDIX J).
- e) Replacement of deck unit keeper (QSL 2501-JPHR-G01 APPENDIX A).
- f) After all welding repairs are completed, repeat the above step for the next scaffolded pile bent

2) Denso / Blast and Paint

On the completion of all welding repairs.

- a) Encapsulation and removal / protection of Denso wrapping around gallery feet and removal any tape/wrapping to >600mm down the piles
- b) Position spider 2B (2B is the abrasive blasting and painting equipped spider)
- c) Abrasive blast and paint
- d) Collect waste for transportation offshore and remove encapsulation
- e) Set up the spider 2B on the next scaffolded pilebent.
- f) Denso Seashield jacket fitting **Contract Technical Specification 2.2.3 and the HOLD and WITNESS Points** in the Inspection and Test Plan (ITP).

NOTE: The second coat of protective coating may be applied using QSL gallery air.

Scaffold to remain for the jacket fitting if required

*****See attached Drawings for Jetty Spider*****

2. ***Approach with regards to mobilisation for the works over a multi-year program;***

All scaffold and the 2 roadway spiders will remain at the QSL contractors yard during the demobilisation.

The 40' smoko and office donga's will remain at the QSL contractors yard during the demobilisation.

AAB staff on demobilisation will be re employed on other works.

Ingham Agencies (IA) staff on demobilisation will be re employed on other works.

AAB and IA'S staff are local, so they will be available for mobilisation each year.

Benchmark Scaffolding (BM) staff on demobilisation will return home.

All plant will be returned to AAB's Ingham workshop for servicing and under cover storage.

**** See attached drawings - proposed allocated space at the QSL contractors' yard***

3. ***Methodology for managing periods of inclement weather with respect to avoiding delays and maintaining productivity.***

AAB will present to the Principal a "*Stand down agreement*" for consideration. This has worked well in the past tenders and has the following main points:

During periods of Strong wind warning from the BOM, work will cease on the pilebents and staff will work on the C7B Conveyor gallery as per Contract Technical Specification 2.3

It is envisaged that some staff will return to work at the AAB and IA's workshop if the standdown is of a prolonged time frame.

CYCLONES

It is anticipated that no work will be ongoing during the normal cyclone season December through to April. If the BOM issues a cyclone alert all outside of this time frame all efforts will be made to secure the site.

During cyclone season (1 November to 30 April), AAB will maintain an appropriate cyclone readiness procedure with tasks at each cyclone level of green, yellow and orange in accordance with the Port of Townsville categories.

4. ***Methodology for managing jetty roadway access and details of how it proposes to co-ordinate and work with the Principal's staff and with other contractors:***

- (a) ***during the works;***
- (b) ***during shipping activities; and***
- (c) ***during major maintenance or construction activities on the wharf.***

(a). ROADWAY

AAB will attend or convey to QSL local staff the daily scope of works particularly road closures and the likely impact on QSL activities. This will convey the passing bay being used and the times of the 3hr closures if any.

AAB anticipates that scaffold erection and dismantling will have the most impact on roadway access.

AAB will use weekends for this activity whenever possible. Built scaffold will have no impact on vehicle roadway travel.

The spider design allows unimpeded normal vehicle passing on the roadway.

AAB will have a ***Traffic Management Plan*** which includes signage at the start and end of the work with speed limits, mobile contact numbers and names if access is required outside of the prearranged times. 2 way communication will also be used.

(b). SHIPPING

All removed gallery underpans will be replaced for a shipping event.

An inspection before shipping as well as normal end of day inspections will be carried out to ensure nothing impedes the operation of the belt and no material is left on the belt during gallery work.

QSL and other contractors staff have priority during shipping. AAB will have a Maritime Safety Identification Card (MSIC) holder on site during shipping.

(c). OTHER ACTIVITIES

AAB sees no issues with QSL engaged contractors or visitors such as CSIRO staff travelling through the designated work site. The worksite will be designated by signage and kept as compact as possible. This will be aided by the work methodology of working on consecutive headstocks.

5. ***defect rectification management proposed;***

AAB will submit a detailed Quality Plan as well Inspection and Test Plans (ITP's). that met its requirements as outlined in the Contract General Specification S9.2. This will include defect identification and repair.

This is outlined in detail in SCHEDULE H3 Quality Management.

6. ***provide a summary of any cost saving initiatives or opportunities that it has identified or recommends;***

AAB will employ local people and trades whenever possible to eliminate accommodation costs.

Advanced Aqua Blasting and Ingham Agencies are local companies with a high profile in the community.

AAB will purchase plant and equipment new where ever possible to minimize the possibility of *break downs over the contract life span*.

7. ***provide a summary of innovative work procedures or any other innovation that it recommends or offers during the course of carrying out and completion of WUC; and***

The use of 2 roadway spiders to maximise clearance for vehicles designed by Doug Lee from NQ Engineering, RPEQ certified and built locally by Ingham Agencies,. These spiders will carry equipment for welding, abrasive blasting and painting. They are steerable and towable.

AAB proposed to have on site 2 x 6 seater Polaris vehicles. These will transport all staff engaged on the contract from the staff donga at the contractor yard after sign on, Daily pre start to the work site where they will remain on the assigned passing bay. The use of 2 vehicles allows flexibility in lunch time travel to the off shore crib room, bringing people in and out etc.

AAB will mark the roadway in the work area with yellow paint to identify the 2m from the gallery wall where Personal Flotation Devices (PFD) are not required to be worn. The remaining area is considered an exposed edge and PFD wearing is mandatory. The identification line will be extended as the work site moves.

8. ***details of how it proposes to co-ordinate and work with the Principal's staff and with other contractors which might be retained by the Principal on the same Site***

(a) for the duration of WUC; and

AAB will hold a formal induction for all AAB and contractors staff with QSL, STL invited to attend.

At this induction all staff will also do the QSL Lucinda BST site specific induction.

This induction will be only for staff working on the contract and will safety oriented. QSL and other contractors staff need not be inducted unless working in AAB's designated work site. People needing the full AAB site specific induction will normally be inducted on a Monday at the AAB site office in the contractors yard.

AAB will provide an escort within reason for small tasks within the AAB worksite.

AAB will attend or convey to QSL local staff the daily scope of works particularly road closures and the likely impact on QSL activities.

(b) in respect to any shut downs of the Principal's facilities which may occur during the course of carrying out and completing WUC."

AAB will be using main gate passes on a limited basis. Entry to QSL will be through the gate in the contractors yard so main gate issues are not a problem.

AAB will be using QSL gallery water and air as needed. If interruption to these services occur AAB will make alternate arrangements.

AAB will use the off shore crib room while the work site is closest to the wharf. AAB will require electrical power for fridges, micro waves etc. AAB will provide a generator for any prolonged outage.

Please nominate proposed working hours to be inserted in Item 42 of the Contract. Where night works are nominated, please describe the proposed works and include a methodology for managing risks associated with night works.

AAB will seek permission to **work on Sundays** during scaffold erection / dismantling to minimize traffic interruptions. Scaffolders will work their 6 shifts including weekends. Their RDO will be a week day. Normal work hours 0600 – 1700.

For AAB and IA staff – Normal work hours 0600 – 1700 Monday / Saturday with Sundays as the RDO. This may alter to accommodate favorable tide cycles for pile jacket fitting.

AAB will also make provision for seeking approval for a night shift during favorable tide cycles for welding repairs and or Seashield jacket fitting.

The following form part of the Methodology

Source	Topic	Description
RFI01+Interview responses	QA/QC Representative role implementation	Ben Eames from APP Group will be utilised to provide and set up the Dashpivot system for Quality Assurance record management and other management and reporting functions. Refer to further description of the proposed system in slides from tender interview in annexure Part I - Key Personnel. Rod Cockle will commence in a QA/QC officer role until a QA/QC person can be permanently added to AAB Staff. The permanent person is likely to be Wade Mitchell, who will also assist in scheduling, safety and other project management tasks.
RFI01 + Interview responses	Online quality reporting tool	AAB will utilise Dashpivot software system and upload data daily during weekdays. Refer to further description of the proposed Dashpivot system in slides from tender interview in annexure Part I - Key Personnel.
RFI01	Use of Polaris vehicles or similar	Principal RFI: Schedule E methodology notes the use of Polaris vehicles. QSL has a speed limit of 20km/h for these vehicles in accordance with state regulations. Please confirm that AAB has reviewed the applicable state regulations and agrees to implement any/all safety measures, as required, in the use of Polaris vehicles along the jetty/wharf. Contractor Response: AAB understand the state regulations and have taken that into considerations. ATV to be governed to 20K/hour. All doors to be solid hinged. AAB choose this as a transportation method to minimize vehicle footprint onsite and allow more flowing worksite for breaks, etc.
RFI01	Conveyor gallery access	Principal RFI: Please indicate how workers will access the conveyor gallery wrapping above the belt. Contractor Response: AAB to have engineered and approved platforms with access ladders that straddle the conveyor system. Design to be approved by QSL to prior to fabrication. See attched proposal design. Platform to be made from aluminium. Bolt on ladders for ease of handling and dismantling in a shipping event. 3 x to be made so multiple work fronts can be accessed. Platform to be 1200mm wide.
RFI02	Expansion joint scaffold access	Principal RFI: Could you please re-examine and explain pricing for item 2.9.1 (access for expansion joints). As the price is much higher than access for similar pilebents, we would like to understand the additional effort, modified design and inclusions that drive this price increase. Contractor Response: The variance in pricing is due to a design change required to support the scaffolding. Benchmark proposes to weld a steel member to the end of the headstock to provide support for the southern end of the scaffold. The quoted price includes: 1) Labour hours for a boilermaker to complete the welding works; 2) Temporary scaffolding to enable safe access for the boilermaker to carry out the welding; and 3) Subsequent installation of the scaffolding to access the headstock.
RFI05	Embedding of learnings from safety incidents	Principal RFI: please detail how AAB plan to embed past incident learnings from all projects into their daily WHS practices to prevent recurrence of similar injuries, such as manual handling strains, eye injuries from foreign bodies, and lacerations. Contractor Response: All AAB Incident investigation and resolution is around the "Hierarchy of Controls" . During this process, AAB will make what ever change is warranted regardless of the inconvenience and the cost as AAB puts worker safety first. To have Incident reporting, you must first have an Incident reporting culture and forums to encourage this. AAB holds weekly toolbox meetings at AAB workshop with items on the agenda covering incidents or nearmisses, safety shares and any recommended changes to SWMS. These are also shared between site works to allow all staff to be informed of any issues or changes. Through this we have implemented changed like Foam Backed Glasses to be worn through out identified scopes of work. For the LBST Jetty Remediation Project, AAB will hold a monthly safety toolbox BBQ (additional to weekly toolbox) with workers being awarded for Hazard Identification, safety culture and take 5 inputs. On the agenda AAB will find incidents and challenges in industries similar to ours to find learnings we could use to implement changes to our WHS Practices. AAB values Third Party Audits to review our processes and will make suggested changes as soon as practicable and ensure compliance and reporting feedback .
RFI08	Recovery of loose and flaking paint during preparation	Principal RFI: Please provide brief details of how AAB will minimise loose and flaking paint that is removed during waterblasting of headstocks and piles from entering the environment Contractor Response: The headstock scaffold will be fully encapsulated on all sides with fire resistant powerclad before water washing, with geofab on the floor to allow water to be strained before release. After water washing the geofab, removed loose coating and rust will be removed, the powerclad on the sides will remain. After welding repairs, powerclad or similiar will be placed on the scaffold floor to complete encapsulation. It will not be a negative pressure encapsulation so a small amount of nuisance dust will escape during abrasive blasting. This is permitted under AAB'S IEMS. The removed coating and spent garnet will be collected and transported inshore for collection by MAMS Environmental services on an as need basis. The majority of loose coatings and rust on the piles will be removed manually during the steel inspection process before water washing and will be collected in a funnel type arrangement. This funnel will stay inplace during water washing of the piles allowing strained water to pass through a geofab filter material. The removed loose coating and rust will also be collected and transported inshore for collection by MAMS Environmental services. This collection funnel will be installed above the barnacle line on the pile.

CONCEPT NOTES:

1. This is a proof of concept drawing only. RPEQ licenced engineer Doug Lee who is familiar with the JMV concept and requirements at Lucinda sugar terminal has attended site with Advanced Aqua Blasting and confirms that development of this concept design is feasible and if progressed will be able to be detail designed, certified and manufactured for use in accordance with all required QLD regulation, AS standards, and QSL (Lucinda site) standards/requirements.
2. This is a concept for a load rated platform trailer that can be towed into position along the jetty, it is intended to be loaded with painting equipment by Advanced Aqua Blasting for the purpose of jetty refurbishment.

MATERIAL NOTES:

1. Structural beams 250UB
2. Drawbar 75x6 SHS
3. Platform, stairs, handrails all Webforge Monowills system typical
4. Solid tyres 8.25-15, no suspension, no service brake, park brake only
5. Linkages / hinges / hydraulics detail design to be developed

FABRICATION NOTES

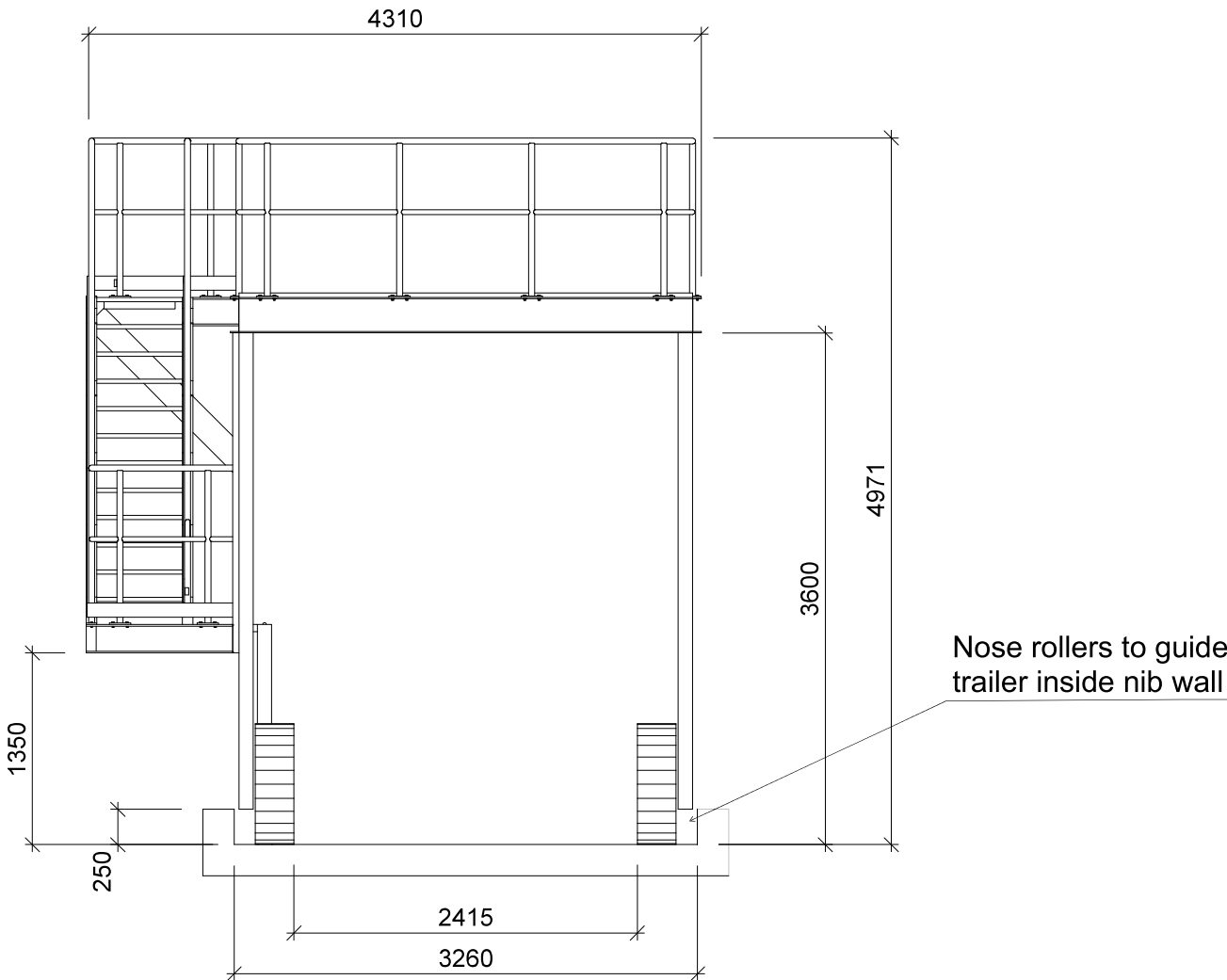
1. The main portals are fully welded
2. Ancillary equipment can be bolted or be removable as required
3. The drawbar is fully welded mounted on special swivelling/locking hinges
4. Painting to owners specification

DIMENSIONAL NOTES:

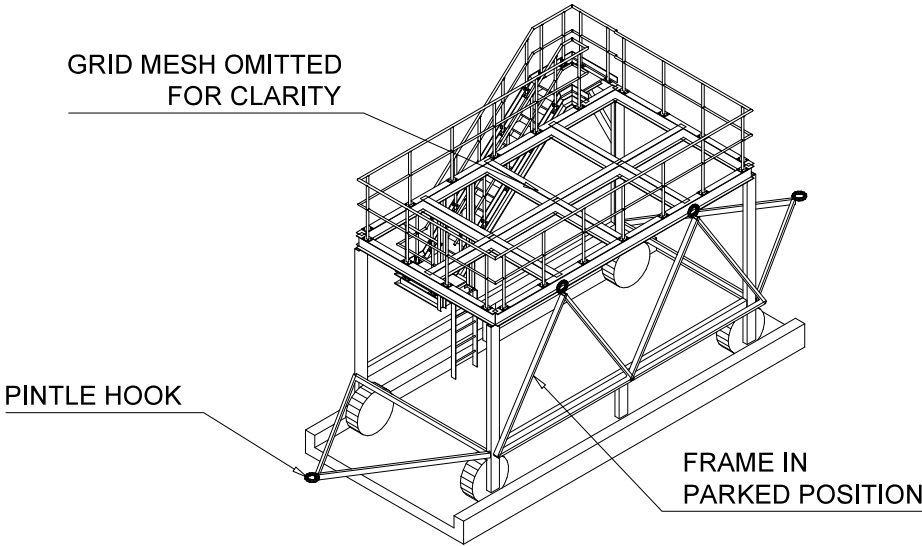
1. Trailer platform length between 4000 & 6000 mm
2. Under trailer clearances to be confirmed by emergency service vehicle requirements.
3. Exact location of point loads (welders, blast pots, garnet storage etc) to be specified

CAPACITY NOTES:

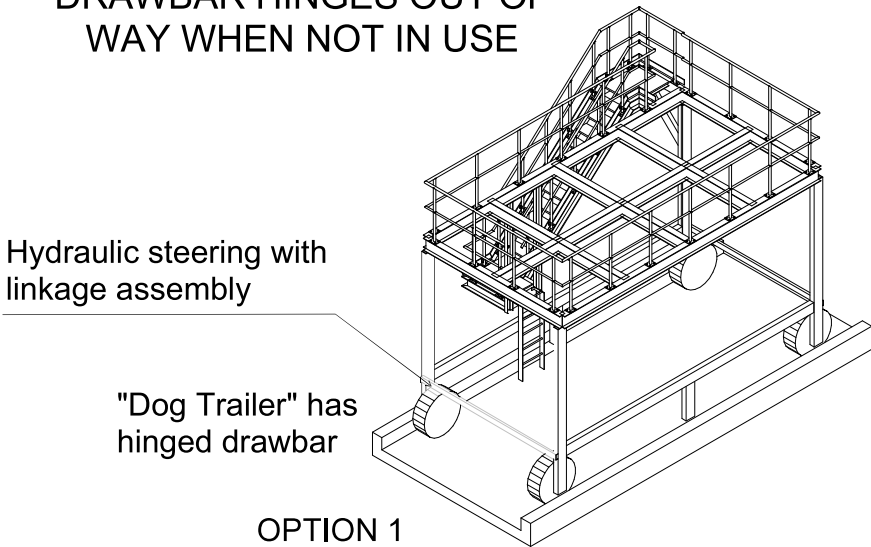
1. Aggregate Trailer Mass = 3500 kg
2. Tare mass estimated = 1800 kg
3. The load platform is rated to hold payload up to 3500kg
4. Tow vehicle must be rated to pull trailer, an unbraked, loaded trailer may not be suitable for towing with light vehicle ute. If towing with ute electric brakes to be fitted at higher laden weights.



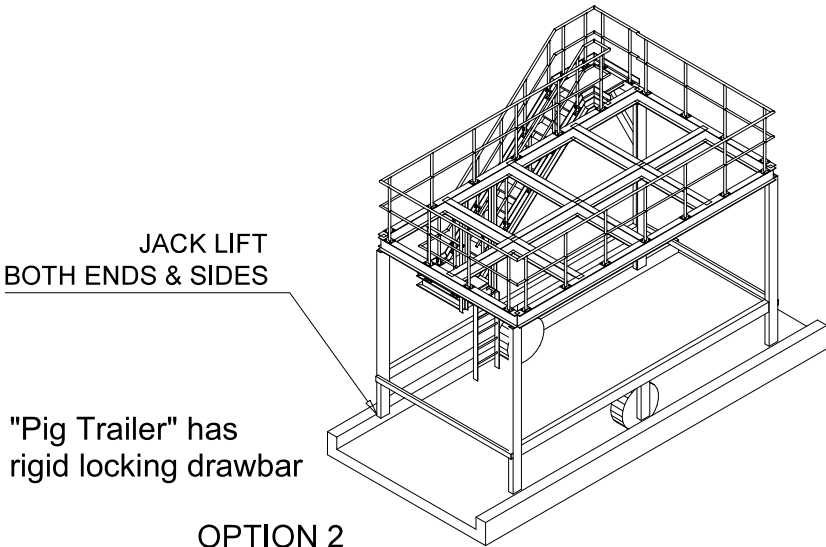
FRONT VIEW



DRAWBAR TYPICAL
DRAWBAR HINGES OUT OF
WAY WHEN NOT IN USE



OPTION 1
4 WHEELED STEERING TRAILER



OPTION 2
2 WHEELED TRAILER
WITH JACKS

CONCEPT REVIEW

DRAWN TO AS1100 TECHNICAL DRAWING STANDARDS ALL DIMENSIONS IN
SCALE: As Shown
COPYRIGHT © This drawing remains the property of North Queensland Engineering Pty Ltd and shall not be passed on to any third party without prior written approval from North Queensland Engineering Pty Ltd.

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Queensland
Engineering

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Job No.: 240723
Client : ADVANCED AQUA BLASTING
Project : JETTY SPIDER

Mark No :		3RD ANGLE PROJECTION
Revision Date : 14.11.2024	Scale : 1:125,1:50	
Revision : A	Drawn : G.RICHTER	
DRG No. : 240723 -A-	001	A3

LUCINDA BULK SUGAR TERMINAL TYPICALS BENCHMARK

GENERAL NOTES

- IF IN DOUBT, PLEASE ASK,
- DO NOT OBTAIN DIMENSIONS BY SCALING FROM THESE DRAWINGS.
- VERIFY ON SITE ALL SET OUT DIMENSIONS SHOWN ON THESE DRAWINGS.
- ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE ON THESE DRAWINGS.
- ALL PERMANENT STRUCTURE DIMENSIONS ARE TO BE CHECKED & VERIFIED PRIOR TO ERECTION OF SCAFFOLD. THE SCAFFOLD DESIGN IS BASED ON THE DIMENSIONS SHOWN ON THESE DRAWINGS. RE-DESIGN MAY BE REQUIRED WHERE ACTUAL DIMENSIONS DIFFER.
- NO ALTERATIONS TO THE DESIGN SHOWN ON THESE DRAWINGS ARE PERMITTED WITHOUT THE PRIOR APPROVAL OF THE DESIGNER. SUPPORT LAYOUT MAYBE RE-ARRANGED TO SUIT SITE CONDITIONS, PROVIDING THAT THE MAXIMUM BAYS SIZES AND LEG LOADS DO NOT EXCEED THE VALUES INDICATED IN THE NOTES.
- THE DESIGN DEFINED ON THESE DRAWINGS IS VALID FOR THE NOMINATED PROJECT & STRUCTURE ONLY.
- THE SCAFFOLD SHOWN IN THESE DRAWINGS HAS GENERALLY BEEN DESIGNED IN ACCORDANCE WITH THE RELEVANT SECTIONS OF THE FOLLOWING AS AUSTRALIAN STANDARDS & DOCUMENTS:
 - AS/NZS 1170.0:2002 STRUCTURAL DESIGN ACTIONS PART 0: GENERAL PRINCIPALS
 - AS/NZS 1170.2:2021 STRUCTURAL DESIGN ACTIONS PART 2: WIND ACTIONS
 - AS/NZS 1576,1:2019 SCAFFOLDING PART 1: GENERAL REQUIREMENTS
 - AS 4100:2020 STEEL STRUCTURES
 - AS 4576:2020 GUIDELINES FOR SCAFFOLDING
 - RELEVANT SCAFFOLD CODE OF PRACTICE
- THE DESIGN HAS BEEN BASED ON THE USE OF ACROW FORMWORK & SCAFFOLDING EQUIPMENT ONLY. OTHER EQUIPMENT MAY HAVE A DIFFERENT SAFE WORKING LOAD AND THUS MAY NOT BE ADEQUATE IN THE APPLICATION SHOWN ON THIS DRAWING.
- IRRESPECTIVE OF ANY RECOMMENDATIONS SHOWN ON THE DRAWINGS, THE USER IS RESPONSIBLE FOR THE STRUCTURAL ADEQUACY OF MATERIAL OF THEIR OWN SUPPLY AND FOR ENSURING THAT THE APPROVED WORKING LOAD LIMITS OF ACROW FORMWORK & SCAFFOLDING EQUIPMENT ARE NOT EXCEEDED.
- THE CLIENT MUST ENSURE THAT THE PERMANENT STRUCTURE HAS ADEQUATE STRENGTH TO WITHSTAND THE IMPOSED LOADS NOTED. ALL THE LOADS MUST BE TAKEN BY THE PERMANENT STRUCTURE.
- THE CLIENT IS TO ENSURE THAT THE SUPPORTING GROUND (WHERE APPLICABLE) HAS ADEQUATE CAPACITY TO SUPPORT THE LOADS FROM THE SCAFFOLD WITHOUT EXCESSIVE SETTLEMENT INCLUDING DIFFERENTIAL SETTLEMENT.
- ALL TYPICAL DETAILS ARE TO BE USED AS EXAMPLES ONLY AND IF SPECIAL OR DIFFERENT DETAILS ARE NOTED IN THE CONSTRUCTION DRAWINGS VARY. TYPICAL DETAILS ARE TO BE DISREGARDED.
- ALL SCAFFOLD SHALL BE INSPECTED & STRUCTURE MUST BE CHECKED BY CLIENT PERIODICALLY AND AFTER INCLEMENT WEATHER OR MAIN LOADING EVENT TO ENSURE THAT ALL COMPONENTS AND CONNECTIONS ARE TIGHT AND THE INTEGRITY OF THE STRUCTURE IS INTACT.
- LOADS SHOWN ARE UNFACTORED WORKING LOADS.

SCAFFOLD NOTES

- THE STABILITY OF THE SCAFFOLD MUST BE MAINTAINED AT ALL TIMES, AND IF NECESSARY, ADDITIONAL BRACING AND ANCHORING MUST BE PROVIDED.
- IMPACT LOADS HAVE NOT BEEN ALLOWED FOR IN THIS DESIGN. IT SHALL REMAIN THE CLIENTS RESPONSIBILITY TO ENSURE THAT SUFFICIENT PRECAUTIONS ARE TAKEN TO AVOID IMPACT LOADS ON THE SCAFFOLD DURING USE AND CRANESAGE & VEHICULAR OPERATIONS.
- THIS DESIGN HAS BEEN PREPARED FROM THE INFORMATION SUPPLIED TO THE DESIGNER BY, OR ON BEHALF OF THE SITE TEAM. THE SITE TEAM MUST CHECK THAT WE HAVE CORRECTLY INTERPRETED THEIR REQUIREMENTS AND THAT WHERE APPLICABLE, ALL LOADINGS, DIMENSIONS, DETAILS, ERECTION AND SEQUENCES ETC, ARE AS REQUIRED AND PRACTICABLE.
- ALL ERECTION WORK IS TO BE CARRIED OUT IN A SAFE TRADESMANLIKE MANNER AND BE IN ACCORDANCE WITH ACROW FORMWORK & SCAFFOLDING GUIDELINES, AS1576, AS4576, LOCAL AUTHORITY REQUIREMENTS AND OCCUPATIONAL HEALTH AND SAFETY REGULATIONS.
- TIES OR OTHER COMPONENTS ARE NOT TO BE MOVED OR TAMPERED WITH IN ANY WAY EXCEPT BY ACROW FORMWORK & SCAFFOLDING OR OUR NOMINATED SUB-CONTRACTORS.
- UNLESS NOTED OTHERWISE ON DRAWING THE DISTANCE BETWEEN BRACED BAYS IS NOT TO EXCEED FIVE BAYS.
- UNLESS NOTED OTHERWISE ON THE DRAWING, ALL RETURNS OR BREAKS IN SCAFFOLD ARE TO BE TUBE TIE, COMPLETE WITH PLANKING, GUARDRAILS OR MESH INFILL AS REQUIRED ON ADJACENT SCAFFOLD.
- UNLESS NOTED OTHERWISE ON THE DRAWING, SCAFFOLD MUST BE A MAXIMUM OF 225mm FROM THE WORKING FACE UNLESS DIRECTED OTHERWISE BY LOCAL SAFETY AUTHORITIES. GAPS TO WORKING FACE GREATER THAN 225mm TO BE FILLED AS REQUIRED ON SITE.
- INDEPENDENT SCAFFOLD BAYS TO BE LINKED USING SCAFFOLD TUBE AND FITTING AT LEDGER POINTS NO GREATER THAN 2.00m VERTICALLY.
- UNLESS NOTED OTHERWISE ON DRAWING PLATFORM BRACKETS ("HOP UPS") ARE NOT TO BE LOADED WITH MATERIALS. THEY ARE TO BE USED FOR ACCESS BY PERSONNEL ONLY.
- BELOW SYMBOLS SHOWN ON THE DRAWINGS INDICATE LAPS AND RAKERS/TIES.

LAP DETAIL MARKER INDICATES SCAFFOLD TUBES, FITTINGS & LAPPING BOARDS TO JOIN ADJACENT SCAFFOLDS. TUBE TIE AT 2000mm CENTERS VERTICALLY (INTERNAL & EXTERNAL STD)



TIE DETAIL MARKER INDICATES SCAFFOLD TIES FIXED TO SLAB EDGE, THROUGH WINDOW OPENINGS AND TO THE SIDE OF EDGE BEAMS. HORIZONTAL DISTANCE BETWEEN TIES AS SHOWN AND VERTICAL DISTANCE BETWEEN TIES MUST NOT EXCEED 400mm UNDO. TIES TO BE TIED TO BOTH STANDARDS UNDO.



TIES AND BRACING

THE PRINCIPAL CONTRACTOR IS RESPONSIBLE FOR ENSURING ALL SCAFFOLD STRUCTURES REMAIN ADEQUATELY TIED AND BRACED TO CARRY THE TIE FORCES AND ENSURE STABILITY. TIES OR HOP-UPS ARE NOT PERMITTED TO BE MOVED BY OTHER TRADES REGARDLESS IF THEY HOLD A VALID SCAFFOLD TICKET WITHOUT CONSULTING AFS.

- CONSULT ACROW FORMWORK & SCAFFOLDING FOR SCAFFOLD TIE AND BRACING REQUIREMENTS FOR SCAFFOLD EXCEEDING 50m TO THE TWP.
- REFER DRAWING FOR REQUIREMENTS REGARDING DOUBLED LEGS OR 1m LIFTS IF REQUIRED.
- NO JOINTS IN STANDARDS ABOVE TOP TIE IS ALLOWED UNLESS PINNED OR SPLICED AS PER THE TYPICAL DETAIL.
- PRESCRIBED CONTAINMENT/LINING NOT TO EXTEND MORE THAN 2m PAST LAST TIE. UNO.
- TIES ARE TO BE INSTALLED AS PER STANDARD TIE DETAILS SHOWN IN THESE DRAWINGS. ALL TIE LOCATIONS MAY UTILISE ANY OF THESE DETAILS TO SUIT THE PARTICULAR TIE POINT IF APPLICABLE.
- TIES MUST BE INSTALLED WITHIN 300mm OF A CONNECTED NODE OR SCAFFOLD LEG TO BE DOUBLED AS PER TYPICAL SINGLE LEG TIE DETAIL IN MIDDLE OF STANDARD.
- TUBE IN SCAFFOLD COUPLERS MUST PROTRUDE BY AT LEAST 10mm OUT OF THE COUPLER. IT MAY ALSO BE PRUDENT TO WRAP END OF TIES IF THE SCAFFOLD TUBE EXTENDS INTO THE WALKWAY AREA.
- IF WIND SPEEDS ARE LIKELY TO REACH OR EXCEED 100km/hr, PRESCRIBED CONTAINMENT SHEETING TO BE REMOVED.
- SCAFFOLD BOARDS SHALL BE STRAPPED DOWN OR REMOVED IF WIND SPEEDS ARE FORECAST TO EXCEED 60km/hr.
- REFER TO UNIMESH FR2 INSTALLATION GUIDE FOR INSTRUCTIONS ON HOW TO ATTACH CONTAINMENT TO PERIMETER SCAFFOLD.
- WHERE ANCHORS INTO CONCRETE ARE REQUIRED, THEY SHALL BE SET AS PER MANUFACTURERS SPECIFICATION CENTRALLY INTO MIN. 200mm THICK SLAB @ 25MPa MIN.

DESIGN LOADS

- THE SCAFFOLD HAS BEEN DESIGN ED USING THE FOLLOWING PARAMETERS:
SCAFFOLD INTENDED USE - ACCESS AND EDGE PROTECTION
TYPE OF SCAFFOLD - RINGLOCK (AT-PAC)

MAX. SCAFFOLD HEIGHT TO TOP DECK = xx m
NO. OF DECK LEVELS = x
NO. OF WORKING LEVELS AT ANY ONE TIME = x
MAX. ADJUSTABLE BASE JACK EXTENSION = 450 mm
MAX. LEG LOAD (EXCL. STAIRS) = xx kN
MAX. WORKING TIE LOAD = 6 kN
- THE SCAFFOLD DUTY RATING, MAXIMUM LOAD APPLIED UNIFORMLY OVER THE ENTIRE BAY & MAXIMUM NUMBER OF DECKS TO BE USED AT ANY ONE TIME SHALL BE IN ACCORDANCE WITH TABLE 1.0.

TABLE 1.0: DUTY RATING		
RATING	MAX BAY LOAD	MAX NO LEVELS
LIGHT DUTY	225 KG	0
MEDIUM DUTY	450 KG	0
HEAVY DUTY	675 KG	0

- LOADING BAYS UNIFORMLY DISTRIBUTED LOAD = 5 kPa
- STAIRS UNIFORMLY DISTRIBUTED LOAD = 2.5 kPa
- WIND LOADS CALCULATED IN ACCORDANCE WITH AS/NZS 1170.2 USING THE FOLLOWING PARAMETERS:
 - IMPORTANCE LEVEL 2 (CONSTRUCTION)
 - REGION (A / B / C)
 - TERRAIN CATEGORY 2
 - V_{ref} (ULTIMATE) = [41 / 48 / 56] m/s (WITHOUT HOARDING), 27.8 m/s (WITH HOARDING)

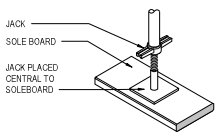
CERTIFICATION

- IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN DESIGN CERTIFICATION IN ACCORDANCE WITH THE RELEVANT CODE OF PRACTICE. ACROW FORMWORK & SCAFFOLDING CAN PROVIDE DESIGN CERTIFICATION UPON REQUEST.
- UNLESS DRAWINGS HAVE BEEN ACCOMPANIED BY A DESIGN CERTIFICATE, IT IS TO BE ASSUMED THE DRAWINGS ARE NOT CERTIFIED AND CONSTRUCTION WORK SHOULD NOT PROCEED.

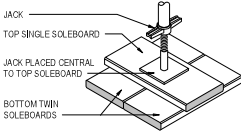


acrow.
ENGINEERING & CONSTRUCTION

DRAWING REFERENCE		
SHEETS START-END	DESCRIPTION	
1 9	TITLE SHEETS & TYPICAL DETAILS	
10 99	GENERAL ARRANGEMENTS	
100 199	SECTIONS & ELEVATIONS	
200 299	PERSPECTIVES	
300 399	STANDARD DETAILS	



SINGLE TIMBER SOLEBOARD
MAXIMUM DESIGN LEG LOAD - 4kN
MINIMUM REQUIRED GROUND BEARING CAPACITY - 4kPa



TRIPLE TIMBER SOLEBOARD
MAXIMUM DESIGN LEG LOAD - 4kN
MINIMUM REQUIRED GROUND BEARING CAPACITY - 4kPa

TIMBER SOLEBOARD SELECTION REQUIREMENTS:

- THE CLIENT MUST ENSURE THAT THE SUPPORTING FOUNDATION HAS ADEQUATE BEARING CAPACITY TO SUPPORT THE LOAD IMPOSED BY THE SCAFFOLD. THIS SHOULD BE SUPPLIED TO THE CLIENT BY MEANS OF A REPORT BY A ENGINEER COMPETENT IN GEOTECHNICAL ENGINEERING OR WRITTEN CONFIRMATION THAT THE SUPPORTING GROUND OR STRUCTURE IS ADEQUATE FOR THE LOADS APPLIED AS SHOWN IN THE DESIGN.
- THE CLIENT SHALL ENSURE THAT THE SCAFFOLD IS ERECTED ON A LEVEL AND STABLE BEARING SURFACE FOR ALL SOLEBOARD LOCATIONS.
- SOLEBOARDS SHALL BE MINIMUM 225mm WIDE x 38mm THICK x 500mm LONG PER SOLEBOARD.

TYPICAL DETAIL - SOLEBOARD LAYOUTS

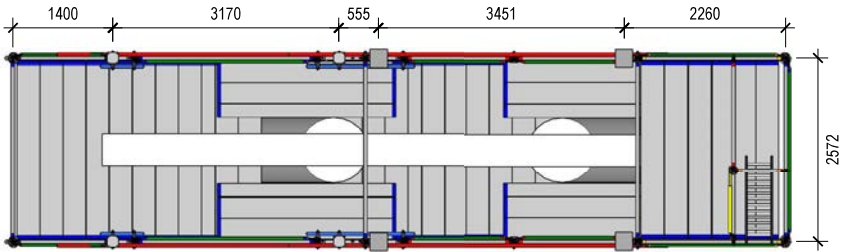
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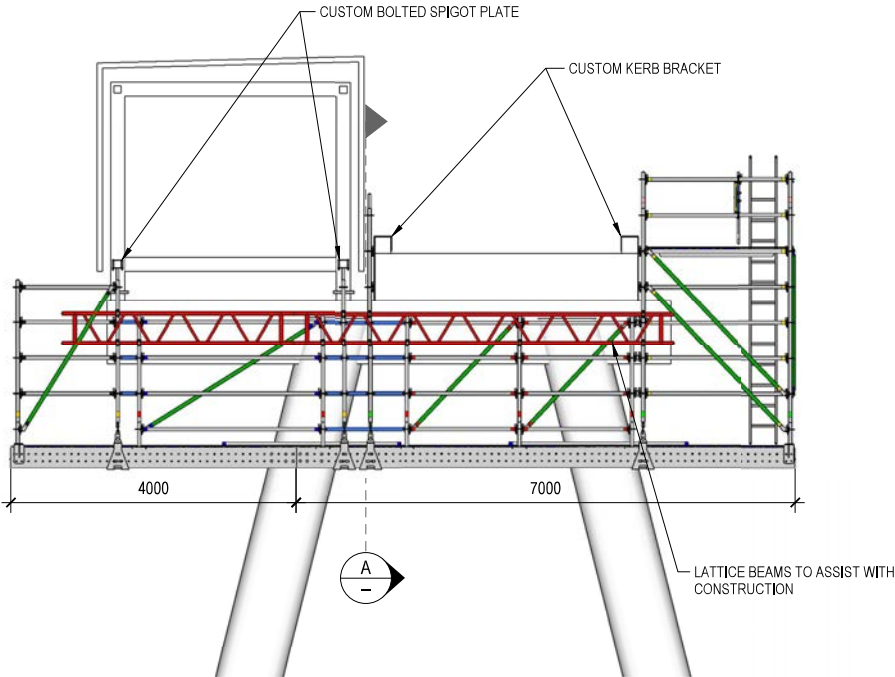
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REV	DATE	DESCRIPTION		DRAWN/CHKD	
CLIENT		DRAWING TITLE			
BENCHMARK		TITLE PAGE & GENERAL NOTES			
PROJECT		LBST TYPICALS		JOB NO. 250218	DWG. 01
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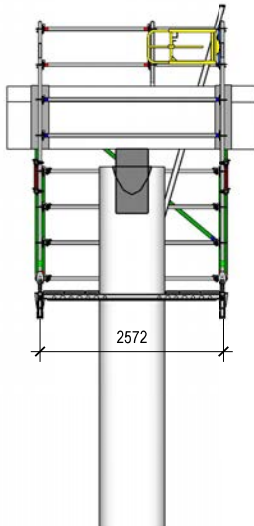
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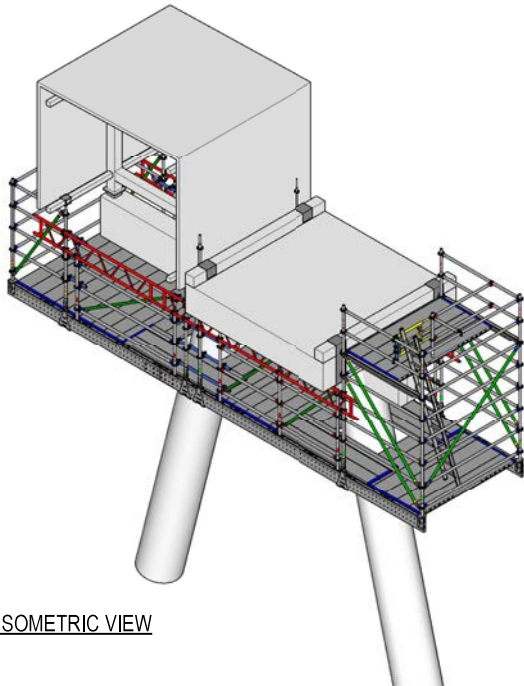
PLAN VIEW
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ELEVATION
SCALE: 1:75 @ A3



SECTION
SCALE: 1:75 @ A3



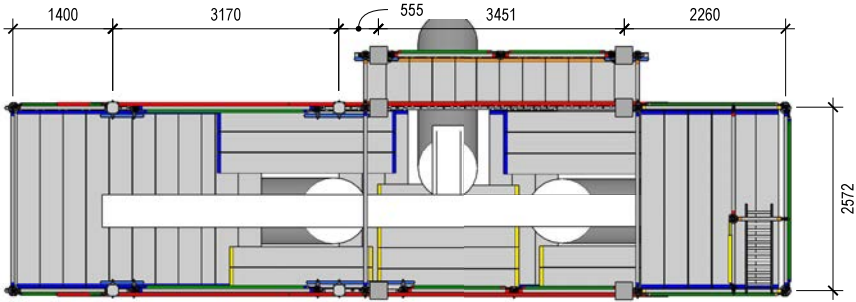
ISOMETRIC VIEW

ALL SCAFFOLD TO BE RINGLOCK

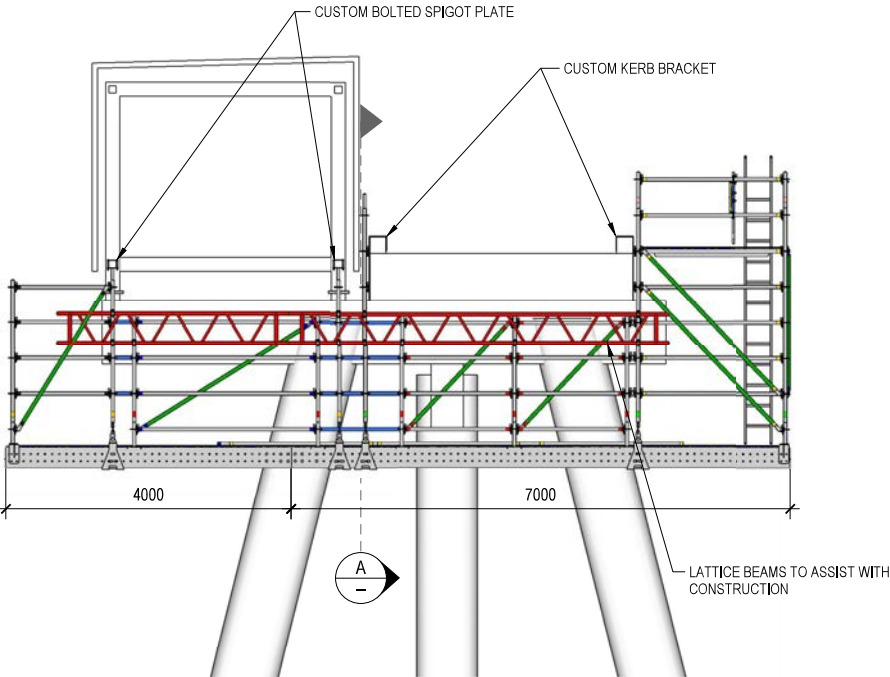
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REV	DATE	DESCRIPTION	DRAWN CHKD	
CLIENT		DRAWING TITLE		
BENCHMARK		TYPICAL BAY		
PROJECT		JOB NO.		DWG.
LBST TYPICALS		250218		10
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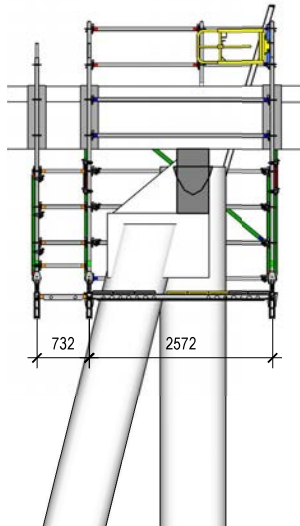
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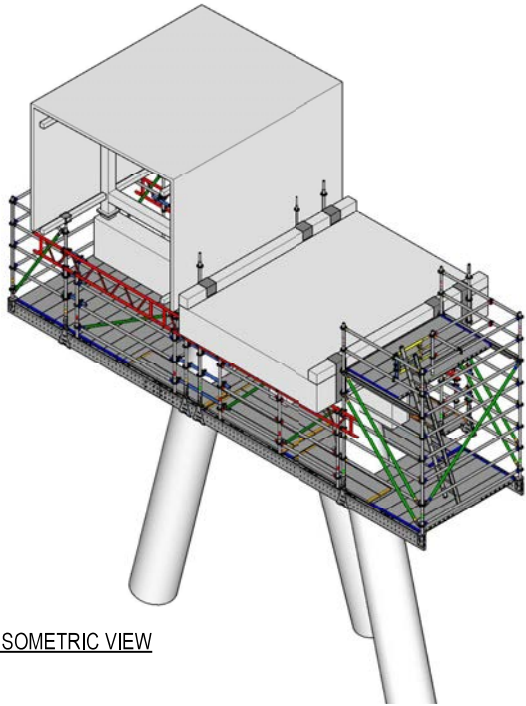
PLAN VIEW
SCALE: 1:75 @ A3



ELEVATION
SCALE: 1:75 @ A3



SECTION
SCALE: 1:75 @ A3



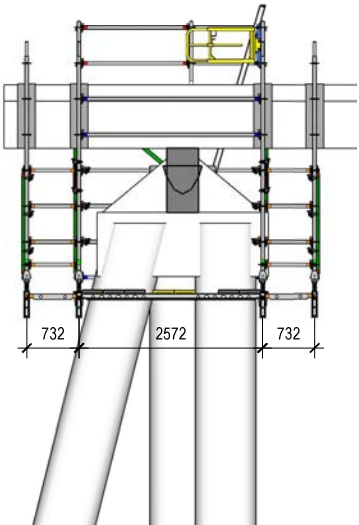
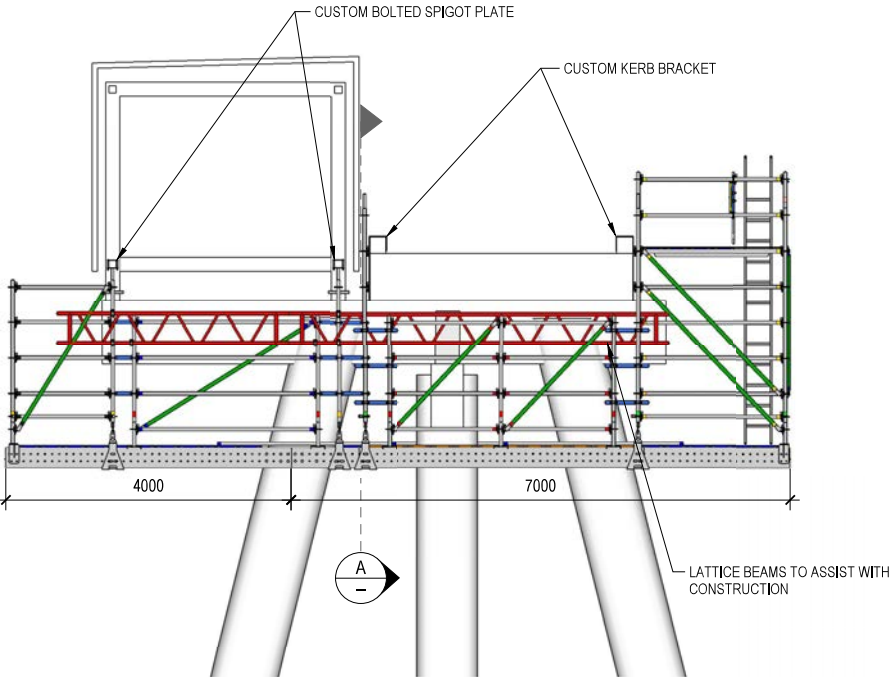
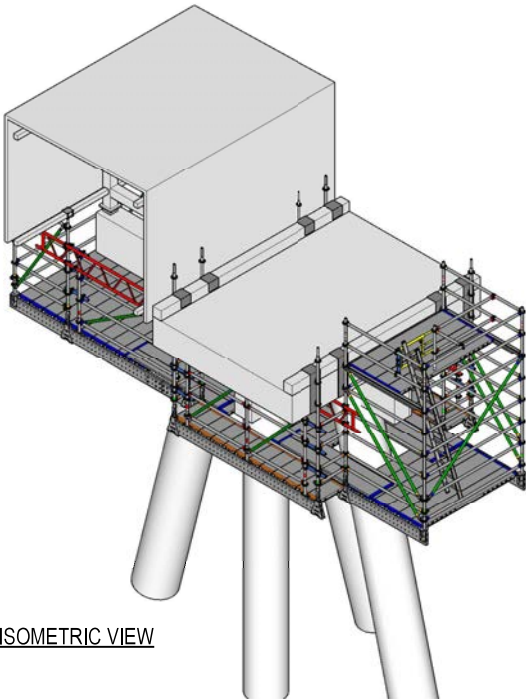
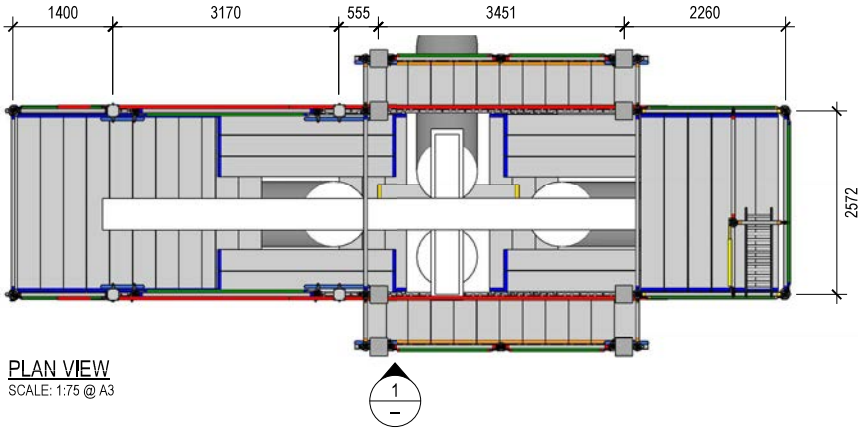
ISOMETRIC VIEW

ALL SCAFFOLD TO BE RINGLOCK

CONCEPT ISSUE				
A	13.05.2025	CONCEPT ISSUE, NOT FOR CONSTRUCTION		M.F. J.H.
REV	DATE	DESCRIPTION	DRAWN CHKD	
CLIENT		DRAWING TITLE		
BENCHMARK		1 PILE ANCHOR BENT		
PROJECT		JOB NO.	DWG.	
LBST TYPICALS		250218	11	
				
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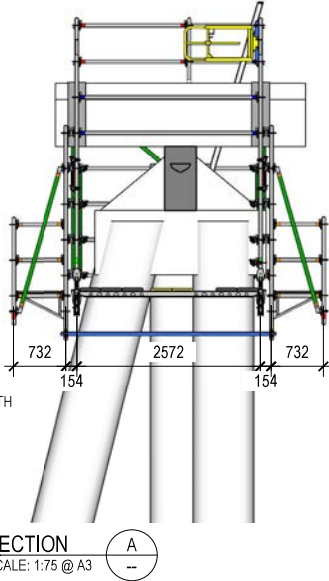
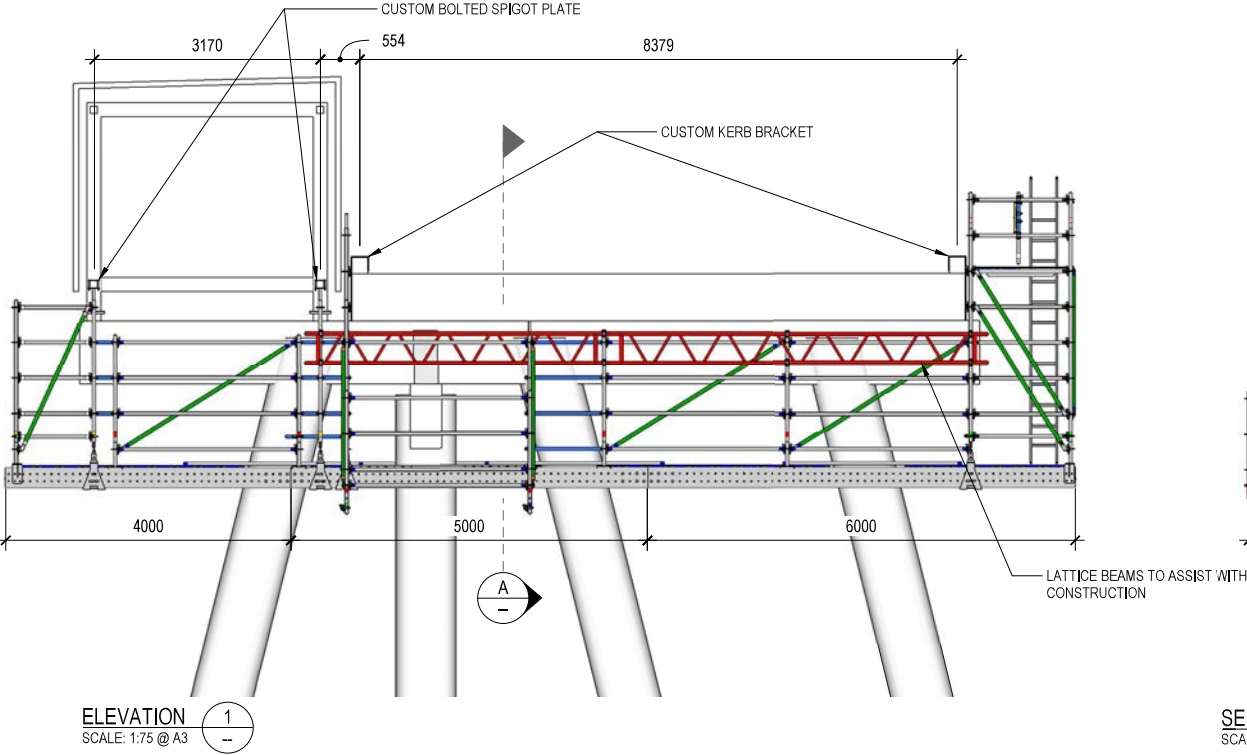
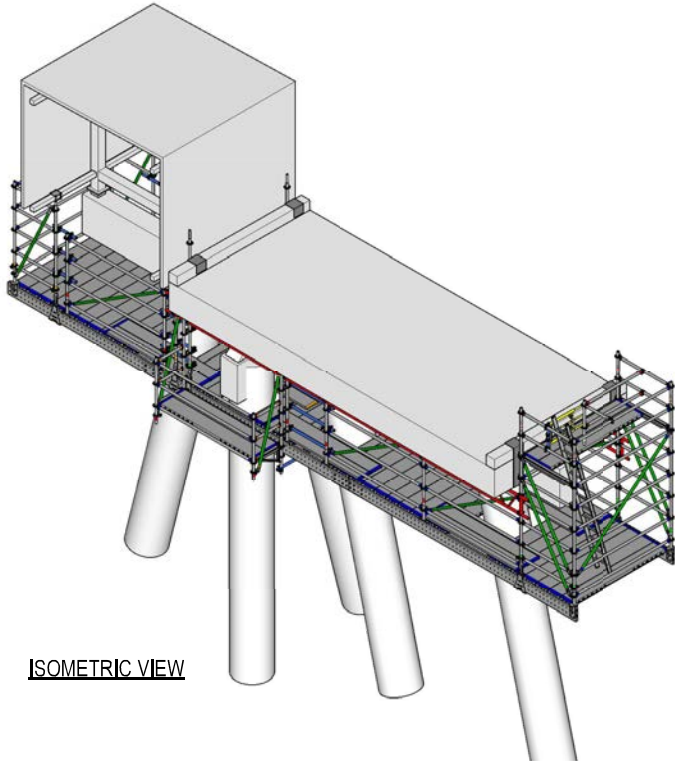
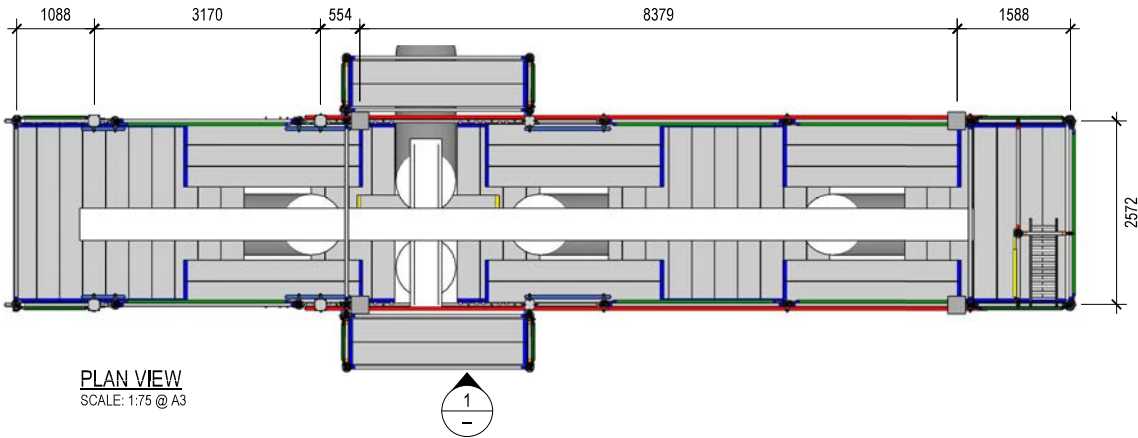


ALL SCAFFOLD TO BE RINGLOCK

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CLIENT		DRAWING TITLE		
BENCHMARK		2 PILE ANCHOR BENT		
PROJECT			JOB NO.	DWG.
LBST TYPICALS			250218	12
				
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ALL SCAFFOLD TO BE RINGLOCK

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CLIENT		DRAWING TITLE		
BENCHMARK		PASSING BAY		
PROJECT		LBST TYPICALS	JOB NO. 250218	DWG. 13
				
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